

Sustainable Energy Use in Dar es Salaam: Current Trends, Future Scenarios, and Policy Options

by

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for the degree of Doctor of Philosophy

Department of Civil and Mineral Engineering
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Abstract

In 2019, Africa accounted for only 5% of global energy demand and 3.7% of energy-related carbon dioxide emissions. However, Africa's rapid urbanization will contribute to rising energy use and emissions, both regionally and globally. Using the case of Dar es Salaam, Tanzania, this thesis offers new insights relevant to the discourse on Africa's evolving energy landscape. The thesis: (1) Estimates possible changes in Dar es Salaam's residential energy use and greenhouse gas (GHG) emissions between 2015 and 2050, (2) Identifies key household and transport-related drivers of energy use and GHG emissions, (3) Assesses variations in energy use at the sub-city (ward) level, i.e., between settlements of differing socio-economic profiles and spatial location in the city, and (4) Examines institutional and societal factors that may constrain low-carbon development in Dar es Salaam.

Three studies are presented to address the four aforementioned thesis aims. The first study – *Modelling Future Patterns of Urbanization, Residential Energy Use and Greenhouse Gas Emissions in Dar es Salaam with the Shared Socio-Economic Pathways* – employs a scenario-framework to scope different urban growth and GHG emissions pathways in Dar es Salaam. The

work demonstrates an approach for projecting GHG emissions in an Africa city context that may be data constrained. The second study – *Does Location Matter? Investigating the Spatial and Socio-Economic Drivers of Residential Energy Use in Dar es Salaam* – shows the differences and clustering of energy use that exist at the ward level, and employs statistical methods to correlate energy use with different socio-economic and spatial characteristics of wards. The final study – *Assessing Institutional and Societal Barriers to Low-Carbon Development in Dar es Salaam* – asserts that processes to implement low-carbon measures (e.g., electrification and public transport projects) would need to engage multiple stakeholders in a collaborative process to leverage the power and mandate of different institutions. Together, these studies seek to inform energy and urban planning policies in Dar es Salaam that (1) enhance synergies between GHG mitigation investments, (2) support implementation strategies that consciously account for local energy use realities and infrastructure access needs, and (3) acknowledge linkages between sustainability, climate change, and socio-economic development strategies.

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I have thought about this moment many times over the course of my PhD research. The moment where I get to reflect on the people who have been a source of unwavering guidance, support, and friendship throughout my research journey. I am finally here. Thinking back to the start. Reflecting on the journey. And completely overwhelmed with gratitude.

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I recall hearing about Dr. Dan Hoornweg's work during my time at the World Bank in Washington, DC. Colleagues at the Global Environment Facility (GEF) lauded him for his contributions to the World Bank's Sustainable Cities Program. When I started the PhD program in 2015, I knew that I wanted to meet him and, if possible, collaborate on research. I am now very grateful for Dan's mentorship and support. It was in an early phone conversation with him that we discussed the scale and challenge of urban growth in Africa. I did not fully know it at the time, but this conversation influenced my decision to shift gears and focus my work on African

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Finally, I would like to thank my incredible family and friends both in Canada and abroad. To my mother, who has a PhD herself, thank you for the many conversations and words of encouragement when I needed them the most. To my husband, Kwame, no words can describe my deep appreciation for you. Thank you for being with me through this journey and helping me to see the lessons and opportunities in every challenge. And last, but not least, to my daughter, Olivia. You are the reason I do this work, in the hope to secure a brighter and more sustainable future for you. This thesis is dedicated to you.

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Nomenclature

BRT – Bus Rapid Transit

GDP – Gross Domestic Product

GHG – Greenhouse Gas

GWP – Global Warming Potential

HDI – Human Development Index

IAM – Integrated Assessment Model

IEA – International Energy Agency

INDC – Intended Nationally Determined Contribution

IPCC – Intergovernmental Panel on Climate Change

LEAP – Long-Range Energy Alternatives Planning Software

LPG – Liquefied Petroleum Gas

LULUCF – Land Use Land-Use Change and Forestry

SDGs – Sustainable Development Goals

SSA – sub-Saharan Africa

SSPs – Shared Socio-Economic Pathways

UN – United Nations

UNFCCC – United Nations Framework Convention on Climate Change

WHO – World Health Organization

Chapter 1

Introduction

1.1 Energy Use in Africa: An Evolving Landscape

As of 2019, economic activities in Africa – particularly the sub-Saharan Africa region – contribute little to the global energy landscape. According to the International Energy Agency (IEA), sub-Saharan Africa accounts for only 5% of global energy demand and 3.7% of energy-related carbon dioxide (CO₂) emissions (IEA, 2019a). However, these emissions are anticipated to rise considerably in coming years. Urbanization will increase the region's energy use and associated greenhouse gas (GHG) emissions (which encompass CO₂ emissions) due to higher energy use from residential activities, electricity generation, transportation and industry (Bai et al., 2018; IEA, 2019a; Nagendra et al., 2018; Steckel et al., 2020). Steckel et al. (2020) also show that carbon-intensive electric power, including new coal-fired capacity in sub-Saharan Africa will contribute significantly to the region's future “carbonization”.

For many cities, GHG emissions are mostly attributed to the energy use of city residents. In 2019, the IEA reported that the residential sector accounted for 64% of sub-Saharan Africa's energy use, compared to only 14%, 15%, and 8% from industry, transport, and other energy-using sectors (IEA, 2019a). Moreover, in cities such as Lagos (Nigeria), Cape Town (South Africa) and Abidjan (Côte d'Ivoire), GHG emissions from residential buildings (not including biomass use) and on-road transportation (including domestic transport) have accounted for the largest share of urban GHG emissions in 2015, i.e., 42% (Lagos), 47% (Cape Town) and 56% (Abidjan) (C40 Cities, 2017). GHG emissions from industry (including manufacturing and commercial/services activities) have been the second largest category, i.e., 32% (Lagos), 43% (Cape Town) and 33% (Abidjan) (remaining GHG emissions have been from solid waste disposal and waste-water treatment). Rising energy use and GHG emissions from residential activities and other key sectors (e.g., industry) are linked to higher population and Gross Domestic Product (GDP) levels anticipated in the next stages of the region's urbanization (Dioha & Kumar, 2020; IEA, 2019a; Steckel et al., 2020; Stone & Wiswedel, 2018).

Of the projected 2.2 billion people who may be added to the global urban population between 2017 and 2050, over half (1.3 billion) are expected to be in Africa (UN, 2018). Figure 1.1 compares the 30 largest cities globally – as estimated by their populations in 2020 (UN, 2018) – with the 15 largest African cities. I rank cities according to their estimated annual urban growth rate, i.e., the average annual rate of change in urban population, between 2020 and 2035 (UN, 2018). Rankings show that the top 9 fastest growing cities between 2020 and 2035 are all African cities (i.e., Dar es Salaam to Abidjan, Figure 1.1). The largest city in the sample, Tokyo (estimated population of 37 million in 2020), is expected to grow at a declining rate (-0.25% annually) between 2020 and 2035. This can be compared to annual urban growth rates exceeding 4% between 2020 and 2035 in Dar es Salaam (6.7 million), Addis Ababa (4.8 million), Mogadishu (2.3 million) and Kinshasa (14.3 million).

Despite Africa's rapid urbanization, the region's cities are largely overlooked in current research on urban energy use and GHG emissions. In the chapter on human settlements, infrastructure and spatial planning in the Intergovernmental Panel on Climate Change's (IPCC) Fifth Assessment Report (AR5), the authors stated that case-studies from African cities are particularly limited, and their review "reflects this limitation in the literature" ((IPCC, 2014), page 951). Similarly, Lamb et al. (2019) developed a database of over 4,000 case-study examples that demonstrated climate mitigation responses in global cities and showcased "city scale reforms" in transportation, building design and urban form. The authors concluded that "cities in world regions with the highest future mitigation relevance" (i.e., Africa) "are systematically under-represented" in the literature ((Lamb et al., 2019), page 279). Africa's high urban growth rates (Figure 1.1) suggests an urgency to initiate more research on urban energy use and GHG emissions in the region as to identify opportunities to establish low-carbon patterns of urban development.

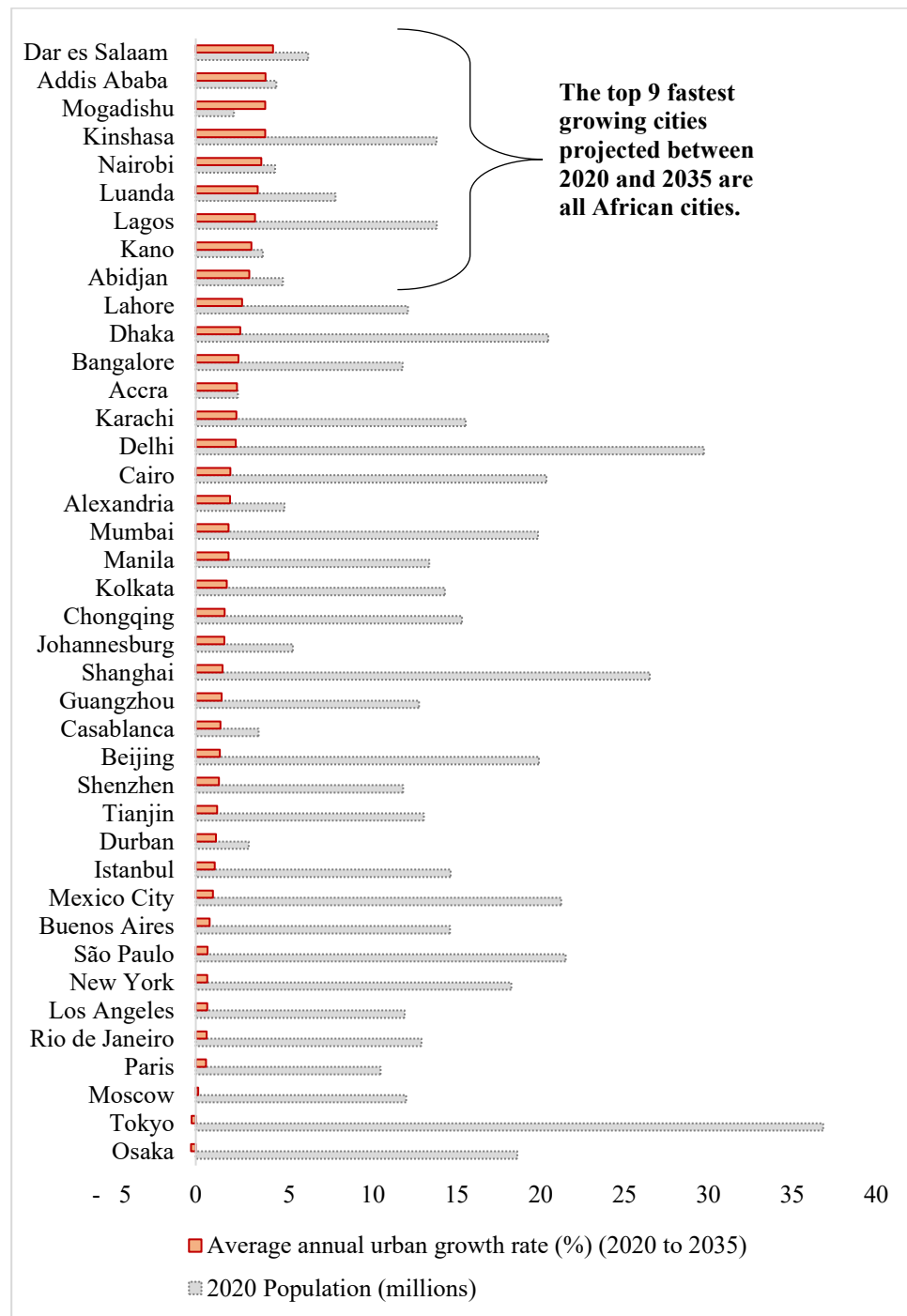


Figure 1.1. City populations (in millions) and estimated average annual urban growth rates between 2020 and 2035. Cities are ordered according to their average annual urban growth rate to show that the top 9 fastest growing cities in the sample are African cities. I developed the figure using data from the United Nations (UN, 2018).

By the end of the 21st Century, Hoornweg and Pope (2017) have asserted that Africa's urban growth could shift the world's largest cities from Asia and Latin America today (based on 2010 data) to Africa. The authors projected that population levels in Lagos (Nigeria), Kinshasa (Congo) and Dar es Salaam (Tanzania) could exceed 70 million (in each individual city) by the year 2100 (Hoornweg & Pope, 2017). From a CO₂ emissions perspective, some studies have shown that Africa's growth could accelerate the region's CO₂ emissions to reach as high as 20% to 23% of global emissions by 2100 (Calvin et al., 2016; Lucas et al., 2015). These changes would represent a five-fold increase from the region's 3.7% contribution (as of 2019, IEA, 2019a), or be equivalent to the combined (2016) emissions of the United States and Canada (IEA, 2017b).

To avoid a potential “lock-in” to such carbon-intensive patterns of urban growth, studies have called on African cities and their local governments to integrate low-carbon and climate-resilience strategies as part of their existing urban planning and development policies, e.g., Bai et al. (2018), Cartwright et al. (2015), Colenbrander et al. (2019), Lwasa et al. (2018), Solecki et al. (2018), and Ürge-Vorsatz et al. (2018). Authors of these studies agree that the changing urban form and economy of African cities may lock-in their future energy use and GHG emissions. Therefore, these next stages of urban development could present an opportunity for African cities to “leapfrog” the carbon-intensive and environmentally destructive development historically experienced among Global North cities (Europe and North America), i.e., leapfrogging could advance low-carbon and climate-resilient urban development.

However, the urbanization of African cities comes with several challenges. To start, urbanization is often coupled with high vulnerability to climate-induced risks such as flooding, droughts, sea level rise, and storm surges (Lwasa et al., 2018). For example, approximately 8% of Dar es Salaam (Tanzania) lies within a low-elevation zone, where an estimated 210,000 people are vulnerable to sea level rise and coastal flooding (Kebede & Nicholls, 2012). In addition, local government authorities (municipalities) are often overburdened with competing development agendas such as addressing the wide-spread urban poverty and informality in cities (over 60% of Africa's urban population lives in informal settlements) (Bawakyillenuo et al., 2018; Lall et al., 2017; Lwasa et al., 2018; UN-Habitat, 2014). Local governments are also limited in their

capacity to spearhead climate change mitigation or adaptation responses at the local level due to governance structures that allocate more power to national or state-led institutions (Dellas et al., 2018; Shemdoe et al., 2015).

To make progress on low-carbon and climate-resilient development, researchers have called on African cities to prioritize the following policy and research actions: (1) implementing climate change mitigation responses to support broader development agendas at the city, national or regional level (Bawakyillenuo et al., 2018; Lwasa et al., 2018), (2) enabling partnerships between local and national authorities during the implementation of climate change mitigation and adaptation initiatives in cities (Bawakyillenuo et al., 2018; Shemdoe et al., 2015), (3) engaging higher level national government institutions and the private sector in financing city level initiatives (Diep et al., 2016; Kithia & Dowling, 2010; Leck & Simon, 2018; Tait & Euston-Brown, 2017), and (4) estimating energy use and GHG emissions trends in African cities and highlighting the “lived” experiences of city residents (IPCC, 2014; Lamb et al., 2019; Nagendra et al., 2018). My work in Chapter 3, Chapter 4 and Chapter 5 uses the case of Dar es Salaam to explore these policy and research priorities.

My work also identifies three gaps in the literature that – if addressed – could inform progress on the policy and research priorities discussed above. Firstly, except for some regional and cross-country comparisons (e.g., Currie et al. (2017), Lucas et al. (2017), Stone & Wiswedel (2018), and van der Zwaan et al. (2018)), few studies have estimated or projected urban energy use or GHG/CO₂ emissions trends for different African cities – possibly, as suggested by Steckel et al. (2020) “owing to the small role they play in current emissions” ((Steckel et al., 2020), page 83). This gap suggests a need for research that examines energy use or emissions trends (e.g., current and future changes) within individual cities, and identifies key sectors (e.g., residential, transportation and industrial) driving changes in energy use/GHG emissions at the city level.

Secondly, studies of Currie et al. (2017), Stone and Wiswedel (2018), and van der Zwaan et al. (2018) estimated energy use and emissions at the aggregate city or regional level and did not account for differences in energy use at the sub-city (settlement, neighborhood or ward) level. Only a handful of studies (e.g., Jagarnath and Thambiran (2018) and Strydom et al. (2020)) have estimated resource (material and energy) use at the sub-city level to identify the specific

neighborhoods and economic activities that have contributed to resource use and GHG emissions in cities. For example, Jagarnath and Thambiran, (2018) estimated neighborhood level GHG emissions for the Durban metropolitan area using spatially disaggregated activity data (e.g., postal codes, ward populations, and firm addresses) collected from national census surveys and sector reports (e.g., transport, industry, residential, and waste sectors). By showing differences in emissions for specific groups and economic activities, the authors have called on city governments to tailor mitigation efforts (e.g., supporting more aggressive GHG mitigation in the highest emissions zones in south, central and north Durban) and address inequalities in infrastructure access among the poorest and lowest-emitting neighborhoods. In Cape Town, Strydom et al. (2020) have likewise suggested that interventions to support sustainable urban energy use should consider the socio-economic profiles of neighborhoods. The authors showed differences in energy use using electricity, gas, charcoal, and wood use data collected from selected low, middle, and high-income neighborhoods in Cape Town (Strydom et al., 2020).

However, studies of Jagarnath and Thambiran (2018) and Strydom et al. (2020) exemplify the larger presence of case-studies from South African cities. There is a general lack of studies from other African cities, which would be critical to informing cross-country comparisons or reviews of energy use/GHG emissions trends in different cities or settlements across the region. Furthermore, for many African cities, implementation strategies to enable progress on low-carbon development at the city level are often poorly articulated. For example, the African Union's Agenda 2063 outlines a strategic framework to promote low-carbon development at the regional level, and calls on governments to deliver "modern, efficient, reliable, cost-effective, renewable and environmentally friendly energy to all African households, businesses, industries and institutions" ((African Union, 2014), page 16). However, the Agenda does not include recommendations for cities to realize the recommended strategies. Furthermore, with the exception of studies by Bawakyillenuo et al. (2018) and Tait and Euston-Brown (2017), current literature offers few examples of city actions to promote low-carbon development at the city level – refer to Chapter 5, for a literature review on the institutional factors that have constrained progress on climate change responses in African cities.

This thesis aims to evaluate energy use trends and discourse around low-carbon development

using the case of Dar es Salaam. In particular, the thesis contributes to the literature gaps discussed above and addresses four research objectives: Objective 1: Estimate current and future residential energy use and GHG emissions in Dar es Salaam (see Section 1.3 for details on this city choice/selection); Objective 2: Identify key drivers of residential energy use and GHG emissions in Dar es Salaam; Objective 3: Assess differences in residential energy use that may exist at the Dar es Salaam sub-city (ward) level; and Objective 4: Assess the institutional and societal factors that may constrain low-carbon development in Dar es Salaam. I focus my work on the residential sector, which, as previously highlighted, contributes to much of the urban energy use and GHG emissions at the regional and city level (though, other sectors, e.g., industry or waste, could be incorporated in future work, see Chapter 6). I use the term “residential energy use” or “residential GHG emissions” throughout the thesis to refer to the sum of energy uses or GHG emissions from domestic household and transportation activities. The following Section (1.2) outlines specific research questions to be addressed for each of the four thesis objectives.

1.2 Thesis Objectives and Questions

Objective 1: Estimate current and future residential energy use and GHG emissions in Dar es Salaam.

- Question 1: What are current levels of residential energy use and GHG emissions in Dar es Salaam, and how might they evolve from 2015 to 2050?
- Question 2: What modelling approaches can be employed to estimate and project residential energy use or GHG emissions when data availability is limited?

Objective 2: Identify key drivers of residential energy use and GHG emissions in Dar es Salaam.

- Question 3: What activities drive residential energy use and GHG emissions in Dar es Salaam in 2015 and in 2050?
- Question 4: What is the influence of household cooking fuel choice, travel mode choice and household wealth on energy use among households in Dar es Salaam?

Objective 3: Assess differences in residential energy use that may exist at the Dar es Salaam sub-city (ward) level.

- Question 5: Is there a statistically significant difference in residential energy use between informal, formal, and mixed ward types in Dar es Salaam?

Objective 4: Assess the institutional and societal factors that may constrain low-carbon development in Dar es Salaam.

- Question 6: How do key informants perceive the enabling environment for low-carbon development Dar es Salaam?
- Question 7: What are possible societal factors that could influence household energy and travel behaviours in Dar es Salaam?

1.3 Outline of Thesis Chapters

The thesis comprises of three studies (presented in Chapters 3 to 5), each written as an individual journal paper. The papers offer new insights to the discourse on Africa's urban energy landscape and address the research gaps discussed previously (Section 1.1). I selected Dar es Salaam (Tanzania) as a case study for two reasons: (1) the city is among the fastest-growing cities in the region, alongside other cities such as Lagos (Nigeria), Addis Ababa (Ethiopia), and Kinshasa (Congo) (see Figure 1.1); and (2) I established strong networks with local researchers, policymakers and universities during an earlier field trip to Dar es Salaam in September 2017, where I presented research at the International Association for People and Environment Studies (IAPS) Symposium hosted by the Institute of Human Settlements Studies (IHSS) at Ardhi University. This early trip culminated in research partnerships being established with IHSS and Ideas in Action Limited (who supported the delivery of the household surveys). Details on these collaborations are provided in Chapter 4 and Chapter 5, and supplementary material appended at the end of these chapters.

I also conducted additional field trip to Lusaka (Zambia) in January 2017 (and later, from November 2018 to February 2019), where I established a partnership with the Center for Urban Research and Planning (CURP) at the University of Zambia. However, resource constraints compromised my ability to deliver a concrete study in Lusaka, though some preliminary findings from the Lusaka work are mentioned in the supplementary material of Chapter 5.

The thesis chapters are described as follows. Full details of the three papers' citations and contributions of the authors are included in Section 1.4.

- **Chapter 2** is a background chapter that outlines a history on the discourse on sustainable development, cities, and their energy use. I also focus on the Africa region, and highlight the region's importance to sustainable development discourse. Later in the chapter, I describe key methods that are relevant to the thesis, i.e., Life Cycle Assessment (LCA), Urban Metabolism (UM), scenario-based energy modelling approaches, data elicitation methods, and statistical analysis.

- **Chapter 3** – *Modelling Future Patterns of Urbanization, Residential Energy Use and Greenhouse Gas Emissions in Dar es Salaam with the Shared Socio-Economic Pathways* – is the first journal paper of the thesis. The work presents the first projection of residential energy use and GHG emissions in an African city, Dar es Salaam. My projections are based on urbanization narratives that are inspired by the Shared Socio-Economic Pathways (SSPs), a scenario framework developed within the climate research community to support integrated analysis of future climate change impacts, vulnerabilities, adaptation, and mitigation (IIASA, 2015). Studies have employed the SSPs to model socio-economic “futures” across a range climate change and environmental megatrends and/or drivers, e.g., population growth (KC & Lutz, 2017), urbanization (Jiang & O’Neill, 2017), energy use (Bauer et al., 2017), air pollution (Rao et al., 2017) and land-use change (Popp et al., 2017). The original SSPs are based on five narratives or “storylines”, each with different consequences for global and regional socio-economic development under increasing climate uncertainty (O’Neill et al., 2017). Only a handful of studies have applied the SSP narratives at the city level (e.g., Kamei et al. (2016); Hoornweg and Pope (2017)), and none for any major African city. Therefore, I present the first known application of the SSPs to Dar es Salaam and demonstrate a method for projecting energy use and GHG emissions in a developing country that may be data constrained. The work also shows the wide uncertainty in long-term GHG emissions projections, while also showing the order of magnitude jump in emissions that can be expected in Dar es Salaam to 2050 (even under optimistic scenarios).
- **Chapter 4** – *Does Location Matter? Investigating the Spatial and Socio-Economic Drivers of Residential Energy Use in Dar es Salaam* – is the second paper. The work “zooms in” to the Dar es Salaam ward to assess residents’ lived experiences. Specifically, the work assesses differences in residential energy use among surveyed wards in Dar es Salaam and elucidates the spatial and socio-economic drivers of energy use. I led fieldwork activities in Dar es Salaam between September and November 2018, and with the support of a 10-person team, administered surveys to 1,363 households in the city. We (the field team) randomly sampled households across eight socio-economically and

geographically diverse informal, mixed, and formal wards to ensure that the final sample reflected the diverse typology of settlements in Dar es Salaam. Among key findings, I show (1) the differences in residential energy use among the surveyed wards, (2) the statistical relationship between residential energy use and ward type, and (3) the influence of spatial and socio-economic factors (e.g., land-use, density, cooking fuel choices, and household wealth, among other variables) on household and transport-related energy use, respectively. For policymakers, I highlight that movement towards higher levels of energy use and a shift toward modern fuels (i.e., electricity and transport oils) could be expected as Dar es Salaam urbanizes and develops socio-economically. Finally, by clustering surveyed wards according to their household and transport-related energy use, I show the need for differentiated approaches to implementing energy policies in Dar es Salaam.

- **Chapter 5** – *Assessing Institutional and Societal Barriers to Low-Carbon Development in Dar es Salaam* – is the final paper. The work interrogates the discourse on low-carbon development in African cities. In the residential sector, low-carbon development could be enabled through policy processes to replace wood fuel use in the home (i.e., “phasing out” charcoal or firewood) with electricity and encouraging energy-efficient public transport use (e.g., Bus or Light Rail systems) (Cartwright et al., 2015; Kennedy et al., 2019, 2018). However, the capacity of African cities to realize low-carbon development may be constrained by various institutional and societal factors. Firstly, local governments are often overburdened with competing development agendas (e.g., rising urban poverty, informality, and inequality) or are severely restricted within multi-level governance structures that allocate more power to national institutions. Secondly, the implementation of low-carbon measures requires a careful consideration of the societal context (e.g., household energy-use choices and preferences) that may constrain the adoption of cleaner or low-carbon fuels (e.g., electricity) in the home. Considering this context, I examine the key institutional and societal factors that may constrain progress on low-carbon development in Dar es Salaam. My data analysis draws on the perspectives and insights of two stakeholder groups: (1) key informants (experts) within

national and local government institutions, donor agencies, academia and the private sector, and (2) surveyed households. I engaged key informants via semi-structured interviews conducted in person (details in the supplementary material appended to Chapter 5) and used the same household survey presented in Chapter 4 to (1) examine the perspectives of surveyed households in Dar es Salaam, and (2) highlight societal aspects that are not included in my previous study (Chapter 4). To ensure higher uptake of low-carbon measures within cities, I recommend that stakeholders address existing institutional barriers to low-carbon development in Dar es Salaam, and at the same time, engage communities during the planning and implementation of low-carbon projects (e.g., electrification projects initiated at the city or national level, or planned phase extensions to the BRT in Dar es Salaam).

- **Chapter 6** presents the main conclusions and contributions of the thesis and reflects on areas of future work.

At a high level, the organization of the thesis, showing linkages across each of the thesis objectives, questions and papers are detailed in Figure 1.2.

Chapter 3 and Chapter 4 have been published, and Chapter 5 is being prepared for publication. Papers have been written in a structure as guided by the peer-reviewed journals, and there may be some limited repetition in background/literature review and methods among the papers.

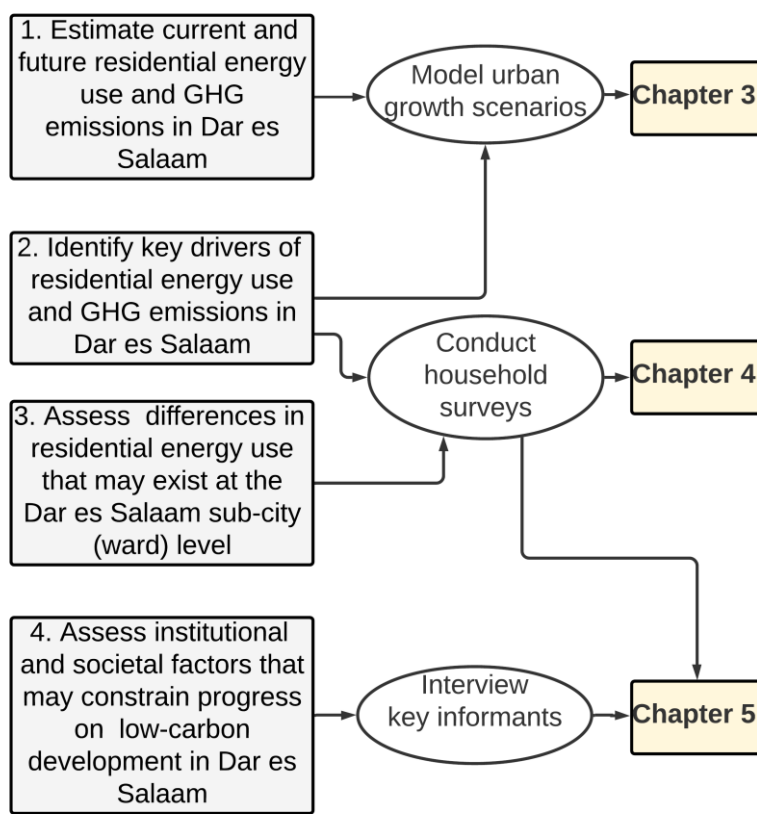


Figure 1.2. Schematic diagram showing linkages between thesis objectives, methods, and main papers/chapters (i.e., Chapters 3, 4, and 5).

1.4 Author Contributions

The citations for the papers related to this thesis (i.e., Chapters 3 to 5) are below along with the contributions of all authors.

Chapter 3:

- Luo, C., Posen, I. D., Hoornweg, D., & MacLean, H. L. (2020). Modelling future patterns of urbanization, residential energy use and greenhouse gas emissions in Dar es Salaam with the Shared Socio-Economic Pathways. *Journal of Cleaner Production*, 254.
<https://doi.org/10.1016/j.jclepro.2020.119998>

As first author, I conceptualized the initial research idea, wrote the first draft, and subsequent drafts, of the paper, and designed the methods and modelling approach that integrated the SSPs within the Long-Range Energy Alternatives Planning (LEAP) software. As co-authors, Professors Daniel Posen (committee member), Daniel Hoornweg (committee member), and Heather L. MacLean (primary supervisor) edited and commented on the various drafts of the paper, assisted with improvements to literature review, methods and modelling approach, and provided overall guidance on the research scope and conclusions. The study was partly funded through a research grant with the Natural Sciences and Engineering Research Council of Canada (NSERC) provided by Professor MacLean.

Chapter 4:

- Luo, C., Posen, I. D. & MacLean, H. L. (2020). Does location matter? Investigating the spatial and socio-economic drivers of residential energy use in Dar es Salaam. *Environ. Res. Lett.* <https://doi.org/10.1088/1748-9326/abd42e>

For this study, I led the fieldwork activities in Dar es Salaam, designed initial and subsequent drafts of the household survey/questionnaire, conducted data cleaning and statistical analysis, and wrote the first draft, and subsequent drafts, of the paper. Co-authors, Professors Daniel Posen, and Heather L. MacLean, assisted in conceptualizing the research objectives and questions, supported data analysis and visualization approaches, and conducted several reviews

of the statistical models and methods. They also commented on and edited the various drafts of the paper and provided overall guidance on the research conclusions and implications for policy. The study was partly funded through a research grant with the Natural Sciences and Engineering Research Council of Canada (NSERC) provided by Professor MacLean. Additional funding for this work was also provided by the International Development Research Centre (IDRC) Canada (Doctoral Research Award); the Institute on Municipal Finance and Governance (IMFG) at the Munk School of Global Affairs at the University of Toronto (Graduate Fellowship); and the International Growth Centre (IGC) (Small Project Fund).

Chapter 5:

- Luo, C., Jean-Baptiste, N., Siame, G., MacLean, H. L (2020). Assessing institutional and societal barriers of low-carbon growth in Dar es Salaam. In preparation for submission to *Cities*.

In this final study, I led the field activities in both Dar es Salaam and Lusaka (finding from Lusaka are presented in the supplementary material of Chapter 5), including outreach and engagement with key informants who were selected for interview in-country (Tanzania and Zambia) and via email or telephone communication. I wrote the first draft of the paper and subsequent drafts following comment and review from co-authors. As co-authors, Dr. Nathalie Jean-Baptiste and Dr. Gilbert Siame facilitated engagement with stakeholders in local and national government and supported planning of workshops in Dar es Salaam and Lusaka (details in Section 1.5). They also commented on and edited drafts of the paper and provided city-specific details and clarifications to the paper, where relevant, given their expertise and knowledge of urban planning issues in Dar es Salaam and Lusaka. Professor MacLean provided overall guidance to the research, which included reviewing and editing drafts of the paper. Like the first two papers, this study was partly funded through a research grant with the Natural Sciences and Engineering Research Council of Canada (NSERC) provided by Professor MacLean. Additional funding for this work was also provided by the International Development Research Centre (IDRC) Canada (Doctoral Research Award); the Institute on Municipal Finance and Governance (IMFG) at the Munk School of Global Affairs at the University of Toronto (Graduate Fellowship); and the International Growth Centre (IGC) (Small Project Fund).

1.5 Note on Fieldwork and Policy Workshops

Studies in Chapter 4 and Chapter 5 employ data from household surveys and key informant interviews collected during fieldwork in Dar es Salaam and Lusaka in 2018/2019. Ethics clearance through the University of Toronto Ethics Review Board (REB) was confirmed prior to the start of these activities in August 2018 (details in Chapter 4: Section 4.11). Household surveys (Chapter 4) were administered by a ten-person field team. The final sample resulted in 1,363 surveys completed across eight formal, informal and mixed wards of Dar es Salaam. Key informant interviews (Chapter 5) included discussions with selected experts across local and national government agencies, academic institutions, the private sector, civil society organizations and donor groups (e.g., the World Bank, African Development Bank, United Nations Development Program, and others) in both Dar es Salaam (Tanzania) and Lusaka (Zambia) (although analysis on field data in Lusaka are not included in the thesis given the limited nature of the work conducted in the city). Twenty-four and sixteen key informants were interviewed in Dar es Salaam and Lusaka, respectively (details in Chapter 5).

Fieldwork also culminated in two policy workshops hosted in partnership with the International Growth Centre (IGC) and local universities (Ardhi University and the University of Zambia) in November 2018 (Dar es Salaam) and February 2019 (Lusaka). Workshops engaged stakeholders in interactive dialogue on strategies for “steering low-carbon growth and sustainable energy use” in key infrastructure sectors such as electricity production, transportation and housing (pictures from both workshops are shown in Appendix B and Appendix C). In attendance was a diverse group of stakeholders from government, academic and private institutions, including representatives from the Dar es Salaam and Lusaka City Councils, sector-based ministries (e.g., energy, transport and local government), electricity and gas companies, civil society and donor groups. Due to research ethics restrictions, workshop outputs are not included in the scope of the work presented in Chapter 5. Overall, discussions created an interest among stakeholders to continue the dialogue and support ongoing policy processes to promote low-carbon development in Dar es Salaam and Lusaka.

1.6 Related Media

Household surveys in Dar es Salaam

- <https://www.youtube.com/watch?v=ySli6u5-QNU&t=146s>

Details of household surveys conducted in Dar es Salaam in 2018. The field team had the opportunity to visit different parts of the city and observe, first-hand, the energy use realities and needs of residents. Funding for the field activities was provided by the International Development Research Centre Canada (IDRC), the International Growth Centre (IGC) (Tanzania Office), and the Natural Sciences and Engineering Research Council of Canada (NSERC).

Policy workshop in Dar es Salaam

- <https://www.youtube.com/watch?v=KAAMys7zyis>

Highlights the outcomes of the “Steering Sustainable Energy Use and Low-Carbon Growth in Dar es Salaam” workshop hosted by the University of Toronto and the International Growth Centre (IGC) on November 7th, 2018. In attendance was a diverse group of stakeholders from government, academic and private institutions, including the Dar es Salaam City Council, the Dar es Salaam Bus Rapid Transit (DART) Agency, and TANESCO (Tanzania’s main electricity supply company). Stakeholders participated in an interactive dialogue on key institutional and financing strategies for low-carbon investments in key sectors such as electricity production, transportation, and housing.

Press

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- Dixon, Keenan (2018). Low-carbon growth in Dar es Salaam | A workshop on governance and finance strategies organized by PhD Candidate Chibulu Luo. *News: Department of Civil & Mineral Engineering*. Retrieved from: <https://civmin.utoronto.ca/low-carbon-growth-in-dar-es-salaam-tanzania-a-workshop-on-governance-and-finance-strategies-organized-by-phd-candidate-chibulu-luo/>

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Chapter 2

Background and Literature Review

This Chapter presents a background and review of relevant literature on (1) sustainable development, cities and their energy use, (2) why African cities are important to discourse on energy use, and (3) methods that are important in the context of the thesis, i.e., common engineering approaches to measuring urban sustainability (life cycle assessment, urban metabolism, and scenario-based energy modelling approaches), and bottom-up data collection methods and elicitation (e.g., qualitative and quantitative survey methods and data analysis).

2.1 Sustainable Development, Cities, and their Energy Use

In their 1987 *Our Common Future* report to the United Nations, the Brundtland Commission defined sustainable development as “development that meets the needs of the present without compromising the ability of future generations to meet their own needs” (Brundtland Commission, 1987). At the time, the Commission called on governments to support “a new era of economic growth” through policies that protect and expand the earth’s environmental resources (Brundtland Commission, 1987). The reality of the 21st Century, however, is that human activities have destabilized the earth’s environmental resource base and constrained sustainable development (Nakicenovic et al., 2016; Rockström et al., 2009; Steffen et al., 2015). Rockström et al. (2009) proclaimed that climate change, land-use change, and biodiversity loss caused by human activities – in what they call the “Anthropocene” – have influenced “irreversible changes” to the earth’s system beyond critical thresholds or “planetary boundaries” (Rockström et al., 2009). In a later report, Nakicenovic et al. (2016) asserted that transgression of these planetary boundaries has been associated with the industrialization and economic growth of Global North regions (i.e., Europe and North America) in the 20th Century. This industrial era accelerated the Anthropocene effect and resulted in “global-scale environmental problems”

((Nakicenovic et al., 2016), page 7). Looking into the 21st Century, the authors affirmed that leapfrogging from the fossil fuel intensive path of these Global North regions is possible in the Global South (i.e., parts of Asia, Latin America, and Africa). They write: “The global North now needs to abruptly and immediately embark on sustainable zero-emissions development pathways while the global South would need to avoid repeating the historical experience of the global North and proceed immediately on a sustainable development pathway” ((Nakicenovic et al., 2016), page 12).

Urban growth is among the key global megatrends that influences sustainable development pathways. Since the time of the Brundtland report, the United Nations and the broader development community have not historically considered policy actions in cities as a priority for sustainable development. Only in the advent of the United Nation’s 2030 Agenda for Sustainable Development in 2015 were cities explicitly recognized and included in the discourse (UN, 2015). Specifically, the 2030 Agenda calls governments to act on 17 Sustainable Development Goals (SDGs) that enable environmental, social, and economic development, globally. Together, the goals are ambitious, and represent a global vision to “shift the world to a sustainable and resilient path”. SDG 11 imagines that cities will be “inclusive, safe, resilient and sustainable” by the year 2030, and is supported by 10 targets that capture this vision. Specifically, “inclusive” cities are realized when access to affordable and basic urban infrastructure is available for “for all” populations (e.g., housing, transport systems, public spaces). “Safe” cities ensure that urban infrastructure is accessible and safe for vulnerable populations, e.g., women, children, the elderly, or persons with disabilities. “Resilient” cities result in minimal economic and environmental losses from climate change and other environmental disasters, especially for the urban poor or communities in “vulnerable situations”. And finally, “sustainable” cities prioritize integrated policies and actions for GHG mitigation and adaptation to climate change (UN, 2015). My thesis work largely focuses on the “inclusive” and “sustainable” dimensions of SDG 11. For example, my policy recommendations call on policymakers to support energy sector and GHG mitigation investments that align with broader socio-economic development goals and the needs of the poorest and vulnerable communities (see Chapters 3 to 5). I do not explicitly incorporate the “safe” and “resilient” dimensions of sustainability in the scope of my work, but my research findings could inform future research on safety and resilience (details in Chapter 6).

Figure 2.1 compares total carbon dioxide (CO₂) emissions from oil, gas and coal combustion for the World and the top CO₂ emitting countries and regions (i.e., OECD member countries, China, the United States and India) relative to the Africa region. In 2017, economic activities in China, the United States and India accounted for half (49%) of total world CO₂ emissions, compared to only 3.6% in Africa (IEA, 2019b). The anticipated build-up of urban infrastructure in developing regions, particularly in Africa, will lead to higher energy use and GHG emissions in these regions during the remainder of the 21st Century (IPCC, 2014). Within countries, cities may drive most future energy demand and GHG emissions in Africa (IPCC, 2014).

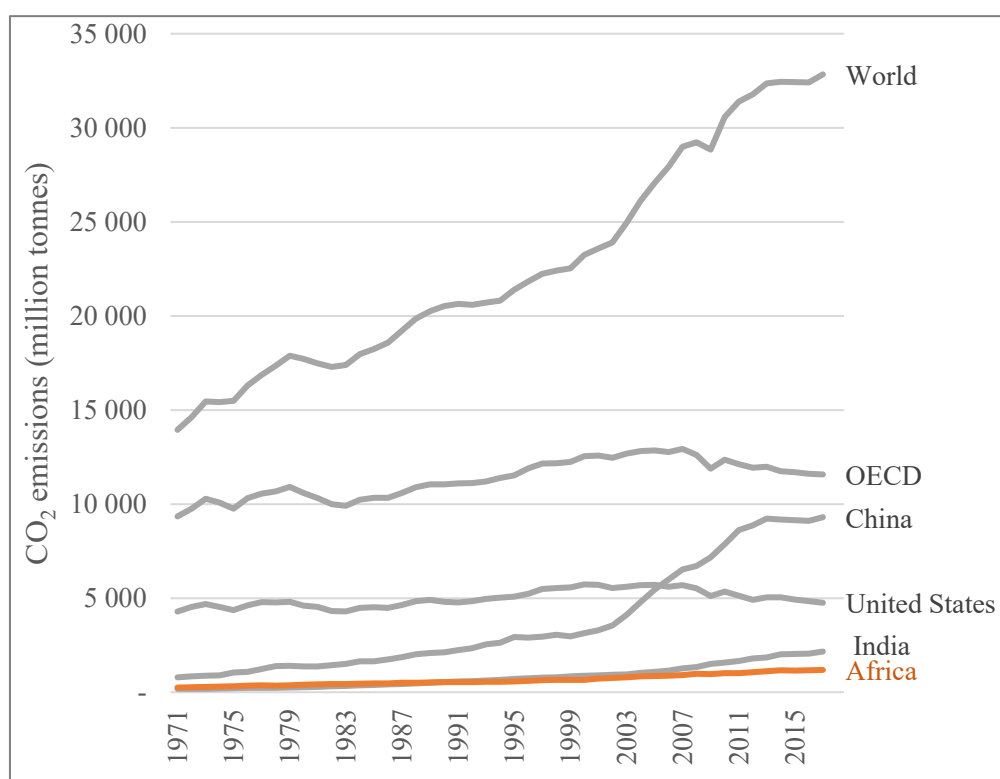


Figure 2.1. Comparing total CO₂ emissions from fuel combustion from 1971 to 2017 for the World, OECD member countries, the top three highest-emitting countries (China, the United States and India) and the Africa region. Historically, Africa's CO₂ emissions have been low, representing only 1.7% of world CO₂ emissions in 1971 and 3.6% in 2017. CO₂ emission are attributed to fuel combustion from the use of coal, oil, and natural gas (IEA, 2019b). I compiled the figure using data from the International Energy Agency (IEA, 2019b).

2.2 Africa's Urban Energy Context

As of 2019, the Africa region accounted for only 5% of global energy demand and 3.7% of energy-related CO₂ emissions (IEA, 2019a, 2019c). African cities will be a key driver of anticipated increases in the region's future contribution to global energy demand and CO₂ emissions (and to levels that may become significant at the global scale) (IPCC, 2014; van der Zwaan et al., 2018). According to the Africa Development Bank, between \$130 and \$170 billion a year in infrastructure investments would be required to support Africa's economic growth (AfDB, 2018). The region's infrastructure demands present a unique opportunity to build more sustainable cities through policies and investments that promote low-carbon and resilient communities (especially benefitting the poor).

According to the Intergovernmental Panel on Climate Change (IPCC), cities and urban areas account for between 67% and 76% of global energy use and 71% and 76% of global CO₂ emissions (based on 2005 data) (IPCC, 2014), mostly due to economic activities in Global North cities and some developed Global South cities (e.g., Chinese and Indian cities). The low urban energy use and CO₂/GHG emissions of African cities is due to their low material and energy use, and limited infrastructure availability and supply. For example, Kennedy et al. (2014) compared the energy use and GHG emissions of 22 global cities across Europe, Asia, North America, and Africa (i.e., Dar es Salaam, the poorest city in their sample based on GDP). Study findings showed that estimated urban GHG emissions in Dar es Salaam (0.56 tCO₂e/capita) represented only 3% of levels reported in the highest emitting U.S. city, Denver (21.54 tCO₂e/capita) (Kennedy et al., 2014). In a later study (Kennedy et al., 2015), the authors estimated that energy and material use was lowest in developing cities in Asia and Africa (e.g., Lagos), arguably because cities are “consuming resources at rates below those that support a basic standard of living for all citizens”. The study by Grubler et al. (2013) shows similar disparities between regions. By quantifying direct final energy use across 225 global cities, which included a sample of 19 African cities, the authors estimated energy use of between 17 GJ/capita (Dar es Salaam) and 44 GJ/capita (Cape Town) for selected African cities, compared to between 74 GJ/capita (London) and 163 GJ/capita (Los Angeles) among European and North American cities.

However, the studies of Kennedy et al. (2014, 2015) and Grubler et al. (2013) did not consider future changes in GHG emissions across their sampled cities. At the regional level, the “Supporting African Municipalities in Sustainable Energy Transitions” project (SAMSET) is among the few regional studies that project changes in urban energy use and GHG emissions in Africa (Stone & Wiswedel, 2018). The study employs data for 47 African countries aggregated into four regions: East Africa, West Africa, Central Africa, and Southern Africa. The authors project, under a Business-as-Usual scenario, that urban energy use in the region doubles by as early as 2025, and quadruples by 2040 (most growth in demand is within the transport and industrial sectors) (Stone & Wiswedel, 2018). In comparison, I estimate a much larger increase in the case of Dar es Salaam – i.e., GHG emissions could rise by 310% to 2300% between 2015 and 2050 due to potentially higher electricity access levels in 2050 and carbon-intensive electricity generation powered by coal and natural gas (see Chapter 3, Table 3.4).

Studies of Calvin et al. (2016), Lucas et al. (2015) and van der Zwaan et al. (2018) have similarly projected changes in GHG emissions at the Africa regional level (i.e., for both rural and urban areas) and identify opportunities to spur low-carbon development in the region. For example, van der Zwaan et al. (2018) employed an integrated assessment model (IAM) to model energy demand and supply pathways for low-carbon development in Africa (including North African countries). Their findings showed that while Africa’s GHG emissions could become substantial at a global scale by 2050, the region could leapfrog fossil-fuel based growth with large-scale use of renewable energy options (van der Zwaan et al., 2018).

Another set of studies has conducted sector-specific or economy-wide energy use or GHG emissions analysis at the national level, e.g., Kemausuor et al. (2015) (future bioenergy use in Ghana), Emodi et al. (2017) (future energy demand in Nigeria, economy-wide) and Mahumane and Mulder (2016) (future energy demand in Mozambique, economy-wide). While other studies, though not projective in their approach, have employed “urban metabolism” (UM) frameworks to quantify current flows of resources (i.e., materials, energy and waste) that shape African cities and their surrounding hinterlands. For example, Currie et al. (2015; 2017) estimated energy and material flows for 120 different African cities. According to the study authors, higher-income cities such as Johannesburg and Casablanca showed higher construction material and fossil fuel

use (e.g., coal, natural gas, and transport oils) due to their larger industry sectors. Lower-income cities such as Dar es Salaam, Luanda (Angola), and Kinshasa (Congo) showed higher biomass use due to their more rural and agricultural-based economies (Currie et al., 2015; Currie & Musango, 2017). Other UM studies conducted in African cities include studies for Cairo (Huzayyin & Salem, 2013; Kennedy et al., 2015), Lagos (Kennedy et al. 2015), and Cape Town (Hoekman & von Blottnitz, 2017). However, none of the above studies accounted for local differences in energy use and access at the settlement level, i.e., to assess which communities, neighborhoods, or income groups account for the most resource (material and energy) use in cities.

In Chapter 4 of this thesis, I show that assessing differences in resource use at the settlement level could inform more differentiated policy actions, e.g., targeting GHG mitigation efforts in high-consuming neighborhoods, and energy access policies/initiatives in low-consuming ones. While my earlier work in Chapter 3 presents the first known estimates of current (i.e., 2015) and future energy use and GHG emissions pathways in Dar es Salaam. The study estimates Dar es Salaam's GHG emissions in the residential sector and highlights the key household and transport-related activities that are influencing rising energy use and GHG emissions in the city.

Furthermore, Chapter 4 and Chapter 5 also examine socio-economic and policy-related questions that are not explicitly considered in Chapter 3, or work by Currie et al. (2015; 2017), Kennedy et al. (2015), and others as discussed above. The work assesses the local reality in terms of how Dar es Salaam residents' access and use fuels (e.g., electricity, charcoal, or gas) and the institutional and societal constraints to communities adopting low-GHG or low-carbon measures. These aspects are important to consider given the unique challenges that African cities face, i.e., where urban growth is occurring in tandem with competing development challenges, and where local governments are often over-burdened and under-capacitated to address resident demands. Therefore, accounting for local differences in energy use and access, and the institutional or societal factors that influence the use of low-carbon measures within communities, may support more effective urban planning and implementation of energy policy visions.

2.3 Background on Dar es Salaam (Tanzania)

With an estimated population of 5.1 million as of 2020, Dar es Salaam is the largest city and economic hub of Tanzania. It is also Africa's fastest growing city (see Figure 1.1 in Chapter 1) and will possibly become one of the world's largest mega-cities by 2050 (Hoornweg & Pope, 2016). Spatially, Dar es Salaam exhibits a monocentric and radial form, with the highest infrastructure and population densities concentrated in the city center, and along the four major arterial roads: to the north along Bagamoyo road, west along Morogoro road, south-west along Nyerere road and south along Kilwa road (Lupala, 2002) (for a detailed map of Dar es Salaam refer to Chapter 3 and 4 – Figure 3.1 and Figure 4.1).

2.3.1 The dual nature of Dar es Salaam's expansion

Spatially, the Dar es Salaam consists of both “formal” and “informal” areas, which reflects the dual nature of the city's expansion. Most residents live in informal areas (over 70%, Limbumba & Ngware (2016)) with most households built on untenured land, and where home ownership is organized at the local level without undergoing the formal legal procedures. The rapid growth of informal areas – particularly in peripheral locations where rural land is being developed and incorporated into the city – has fueled much of the city's urban sprawl and heightened demand for infrastructure services (Andreasen et al., 2017).

Findings from my fieldwork show that widely used definitions of “slum” are poorly suited to the Dar es Salaam context given the variation in socio-economic conditions and building types that one observes throughout the city, and particularly in informal settlements. According to the United Nations Human Settlements Program (UN-HABITAT, 2006), a slum household is defined as “a group of individuals living under the same roof in an urban area who lack one or more of the following: (1) Durable housing of a permanent nature that protects against extreme climate conditions; (2) Sufficient living space which means not more than three people sharing the same room; (3) Easy access to safe water in sufficient amounts at an affordable price; (4) Access to adequate sanitation in the form of a private or public toilet shared by a reasonable

number of people; (5) Security of tenure that prevents forced eviction” ((UNHABIT, 2006), page 1). Based on my observations during my fieldwork in 2018, some settlements in Dar es Salaam would fall under this UN-HABITAT definition, e.g., “Keko”, which was one of my surveyed wards (see Figure 2.5 and Table 4.1). However, it is equally common to observe well-built concrete structures and high-rise buildings in some informal settlements, where residents (who are likely higher income) have access to grid electricity, or piped sewerage and water connections. These differences underscore the grey zone that emerges when defining which settlements constitute as “formal” versus “informal”.

There are only a few settlements in Dar es Salaam that have been developed “formally” with supporting land-use and infrastructure plans. One such area is “Msasani” ward (shown in Figure 2.2) – one of the wealthiest areas of Dar es Salaam and a ward that was included in my survey sample, see Table 4.1. However, given that some informal areas are being regularized (i.e., undergoing redevelopment and infrastructure upgrading to become formal settlements), for the purpose of this thesis, I classify such areas as “mixed settlements” as they include a mix of both informal and formal areas, e.g., “Kimara” and “Kawe” wards would be classified as mixed settlements – they were also included as part of my survey sample, see Table 4.1.



Figure 2.2. Msasani ward (“Oysterbay” segment) – a formal settlement in Dar es Salaam and home to the City’s wealthiest communities. Photo taken by Alice Chibulu Luo.



Figure 2.3. Mwananyamala – an informal settlement in Dar es Salaam that is currently undergoing regularization. I refer to this settlement as “mixed” to reflect these ongoing efforts to upgrade and redevelop certain areas in the ward. Photo taken by Alice Chibulu Luo.

Recognizing the dual (formal versus informal) nature of Dar es Salaam’s urban expansion, my work in Chapter 4 examines the variations in energy use and access among different formal, informal and mixed settlements in the city. I have observed that these differences have not been considered in energy sector-wide policies and strategies for Dar es Salaam, or Tanzania more broadly. However, failing to account for local differences in energy use may lead to generalized energy sector initiatives that do not consider local energy and infrastructure needs (e.g., for electricity or public transportation) in communities.

2.3.2 Climate and Energy Policies in Tanzania and Future Prospects

Tanzania's national energy policies have focused on achieving an energy transition from traditional (wood fuels) to modern fuels (electricity). Tanzania’s policy vision for energy sector development aligns with the country’s *Development Vision 2025* that support socio-economic and industrial growth. According to the Vision, “it is envisioned that Tanzanians will have graduated from a least developed country to a middle-income country by the year 2025 with a

high level of human development. The economy will have been transformed from a low productivity agricultural economy to a semi-industrialized one led by modernized and highly productive agricultural activities which are effectively integrated and buttressed by supportive industrial and service activities in the rural and urban areas” ((Government of Tanzania, 2000), page 2).

Energy and climate change policies, including the national *Power System Master Plan* (Government of Tanzania, 2016), Intended Nationally Determined Contributions (Government of Tanzania, 2015a), and the Sustainable Energy for All *Action Agenda* (Government of Tanzania, 2015b), seek to align with Vision 2025 and view energy sector development as a key driver of economic growth across the country – although, as discussed in Chapter 5, specific implementation strategies have been poorly articulated by relevant stakeholders and institutions. Policies have largely promoted aggressive electrification ambitions, as detailed below, and the need to reduce community dependence on wood fuels. For example, the Action Agenda aims to increase the percentage of the Tanzanian population with access to modern cooking solutions from 16% in 2012 to >35% in 2025 (Government of Tanzania, 2015b).

However, the reality is that wood fuels (or biomass) have continued to dominate Tanzania’s energy mix. As of 2014, wood fuels – namely, charcoal and firewood – have accounted for 90% of final energy use in Tanzania (Government of Tanzania, 2014). Wood fuel use is also common among Dar es Salaam households despite the city’s higher electrification level. As of 2016, an estimated 75% of Dar es Salaam households had access to electricity (compared to the rural and urban averages of 17% and 65%, respectively) (Government of Tanzania, 2016). Wood fuel use is often present in the form of “fuel stacking”, where households use a combination of traditional and modern fuels for household needs. According to Tanzania’s 2017 *Energy Access Situation Report*, most Dar es Salaam households reported using charcoal for cooking (88%), compared to gas (27%), kerosene (22%), firewood (14%), and electricity (1%) (Government of Tanzania, 2017). In comparison, results from my work in Chapter 4 show that between 53% (Keko) and 67% (Mwananyamala) of households reported fuel stacking for cooking, compared to less than 1% of households that used only electricity (Chapter 4 , Section 4.11.6).

Tanzania's Action Agenda (Government of Tanzania, 2018) envisions that over 75% of the national population will be connected to electricity by 2030 (compared to a national electrification level of 33% in 2016, Government of Tanzania, 2016). Relatedly, my sustainability (SSP1) and business as usual (SSP2) scenarios in Chapter 3 project a shift to 100% electrification by 2050. However, as shown from my work, rising electricity access levels may result in higher GHG emissions from electricity generation (Chapter 3), given the largely fossil-fuel intensive generation mix in Tanzania.

As of 2016, Tanzania's electricity generation mix has been dominated by natural gas (59%); hydropower (35%), Heavy Fuel Oil (HFO) (5.7%) and biomass (0.3%) account for the remaining fractions (Government of Tanzania, 2017). The Tanzanian government is exploring other fossil fuels for electrification (i.e., coal) between 2016 and 2040, with the most fossil fuel intensive scenario ("Scenario-3") in the country's 2016 Power System Master Plan projecting a future mix of natural gas (35%), coal (40%), hydropower (20%), and renewables (solar and wind) (5%) in 2040. In comparison, the recommended scenario ("Scenario 2") is equally fossil fuel reliant, i.e., natural gas (40%), hydropower (20%), coal (35%), and solar and wind (5%). (Government of Tanzania, 2016). This means that the carbon-intensity of electrification in Tanzania will drive rising GHG emissions. In Chapter 3, I estimate the GHG intensity to range between 405 and 435 gCO₂e/kWh, compared to the global (IEA) target of 140gCO₂e/kWh (IEA, 2017). Therefore, actions to decarbonize the grid with renewable power (wind, solar or geothermal power) could lead to substantial GHG emissions savings as illustrated in my "aggressive GHG mitigation" scenario in Chapter 3, Section 3.6.7.

2.4 Quantifying Urban Sustainability

Life Cycle Assessment (LCA) and urban metabolism (UM) (described in Sections 2.4.1 and 2.4.2) are widely used engineering methods for measuring urban sustainability, and therefore are relevant background material for this thesis. Together, these methods offer a systems-based approach to quantify urban resource use (i.e., consumption of energy, water and materials in cities) and scoping urban GHG emissions and their environmental impacts.

2.4.1 Life Cycle Assessment (LCA)

Broadly, LCA describes a method for assessing the environmental impacts of a product or service over its entire life cycle, i.e., from the initial gathering of raw material from the earth, until the point at which all residuals are returned to the earth or “cradle-to-grave” (El Haggag, 2005). According to the International Organization for Standardization, there are five components to an LCA (ISO, 2006):

- (1) **Goal:** intended application and reasons for carrying out the study.
- (2) **Scope:** product/system being studied, system boundary, assumptions, and limitations.
- (3) **Inventory analysis:** data collection, calculation, and validation of data (e.g., completeness check, sensitivity analysis, mass balance checks etc.).
- (4) **Impact assessment:** impact categories (e.g., climate change/GHG emissions), indicators (e.g., infrared radiative forcing), characterization factors (e.g., global warming potential for each greenhouse gas).
- (5) **Interpretation of results:** identification of significant issues, valuation of completeness and consistency, conclusions, limitations, and recommendations.

I focus my review specifically on studies that have used LCA to assess the GHG emissions and global warming impacts associated with urban energy and material use, e.g., GHG emissions attributable to the construction and use of urban infrastructure such as roads, housing, transportation and waste systems (see studies by Chavez et al. (2012), Hillman et al. (2011), Hillman and Ramaswami (2010), and Kennedy et al. (2010)). In this context, LCA can be employed to assess urban GHG emissions other environmental aspects (e.g., air pollution or land-use change) for different phases of the infrastructure life cycle: e.g., construction, use/operation, and end-of-life (see Figure 2.2), as well as emissions “embodied” in urban materials (e.g., food, water, or concrete) (Ramaswami et al., 2008).

Most inventories of GHG emissions at the city level focus on the “use/operation” phase of the infrastructure life cycle, where emissions are “scoped” at different spatial scales, e.g., quantifying GHG emissions that occur within the city, or expanding the scope to include emissions from upstream and downstream activities that are attributable to activities within the city boundary (e.g., upstream electricity generation or downstream waste processes) (WBCSD/WRI, 2014). The *Global Protocol for Community-Scale Greenhouse Gas Emission Inventories* (GPC) presents the most widely used standard for GHG emissions reporting for cities. These include “Scope 1” emissions that are produced within the city boundary; “Scope 2” emissions that are from electricity generation at the power plant as a consequence of grid-supplied electricity, heat and cooling within the city boundary; and “Scope 3” emissions that are from upstream infrastructures and supply chains outside of the city boundary (Kennedy et al., 2010; WBCSD/WRI, 2014).

Accounting for Scope 3 emissions (when data are available) can offer a more holistic picture of the GHG implications of urban activities. For example, Chavez et al. (2012) estimated that in-boundary (Scope 1 and 2) GHG emissions accounted for 68% of Delhi's total GHG emissions, i.e., emissions from residential, commercial, industrial and transport activities (including electricity generation). In comparison, the authors also showed that out-of-boundary (Scope 3) GHG emissions contributed the remaining 32% of Delhi's total GHG emissions, i.e., GHG emissions from fuel processing, air travel, cement use, and food production outside the city. A later study by Chen et al. (2016) compared GHG emissions across 25 sectors for selected Chinese and Australian cities. The study authors found that in-boundary and out-of-boundary GHG emissions from construction (including GHG emissions embodied in imports and exports) was the largest contributor to emissions, i.e., representing 21%–24% and 23%–41% of the total carbon footprint in Australian and Chinese cities, respectively.

Finally, only a handful of studies have employed LCA to quantify the GHG emissions in Africa, e.g., see studies for Kampala, (Lwasa, 2017), Durban (Jagarnath & Thambiran, 2018), Cape Town (Kennedy et al., 2009), and sub-Saharan Africa (Godfrey & Xiao, 2015; Stone & Wiswedel, 2018). To my knowledge, my work in Chapter 3 presents the first estimates of GHG emissions in the Dar es Salaam context.

2.4.2 Urban Metabolism (UM)

LCA and UM are related concepts. UM provides a “snapshot” of urban material and energy flows in cities (i.e., in the context of a larger LCA study) but does not show the full environmental implications of such flows (Figure 2.2.). The metabolism of cities, according to Kennedy et al. (2007), is the “the sum total of the technical and socio-economic processes that occur in cities, resulting in growth, production of energy, and elimination of waste”. It is a method for quantifying inter-dependent inputs, outputs, and storage of energy, water, nutrients, materials, and waste in cities (Maranghi et al., 2020).

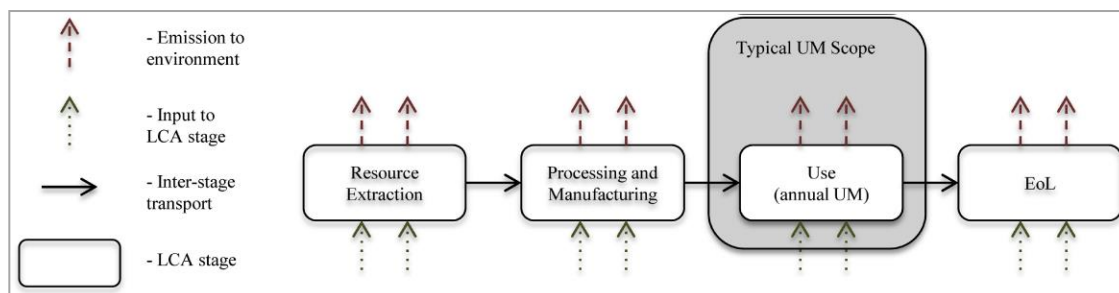


Figure 2.4. The different life cycle stages typically covered in an LCA each with its own associated environmental exchanges in terms of energy and mass requirements and waste and pollutant emissions. The gray box indicates that only the use stage is typically accounted for in traditional UM studies and the up and downstream burdens tend to remain unaccounted for (Note: EoL refers to “End of Life”). Source: (Goldstein et al., 2013), page 3.

Goldstein et al. (2013) outlines the different stages typically covered in an LCA and highlights the “use” stage that is the boundary/scope for most UM analysis (see Figure 2.4). Material Flow Analysis (MFA) is among the most common accounting methods used in UM studies (Goldstein et al., 2013; Pincetl et al., 2012). MFA studies use the principles of mass conservation to quantify the stocks and flows of energy and materials in cities, i.e., $\text{mass in} = \text{mass out} + \text{stock change}$ (Pincetl et al., 2012). However, the MFA approach is unable to estimate environmental burdens (e.g., GHG emissions) associated with resource flows (Pincetl, 2012). To account for these limitations, some studies employ integrated UM/MFA and LCA models to present a more holistic picture of urban energy and material flows, and their environmental impacts, e.g., see studies of García-Guaita et al. (2018) and Goldstein et al. (2013).

Table 2.1. presents a broad overview of example studies that apply LCA and UM approaches to sustainability assessment across global cities. As shown in the Table: (a) there are a few examples of studies from African cities, and (b) energy use/GHG emissions are quantified at the city scale (i.e., based on the study boundary or scope) – there are limited examples of sub-city level to assess the neighborhoods or settlements that may contribute most to urban material and energy use/GHG emissions.

My work in Chapter 3 and Chapter 4 employs an LCA and UM approach to address these gaps in the literature. However, some limitations of my work are that (1) I estimate Scope 1 and 2 residential emissions for Dar es Salaam as upstream (Scope 3) level data was unavailable for most household and transport activities, and (2) I do not assess other environmental impacts associated with resource use, e.g., biodiversity loss, land-use changes, or indoor air pollution associated with household wood fuel use. These are important areas of future work which I highlight in the conclusions (Chapter 6).

Table 2.1. Selected studies that employ Life Cycle Assessment (LCA) and Urban Metabolism (UM) approaches to quantifying aspects of urban sustainability across different global cities.

#	Source	Study Region	Continent	Method	Flow (or resource)	Boundary (and Scope when GHG emissions are estimated)
1.	Kennedy et al. (2009)	Global cities	Global	LCA	GHG emissions	City (Scopes 1, 2, and 3)
2.	Hillman & Ramaswami (2010)	Selection of U.S. cities	North America	LCA	GHG emissions	City (Scopes 1, 2, and 3)
3.	Chavez et al. (2012)	Delhi, India	Asia	LCA	GHG emissions	City (Scopes 1, 2, and 3)
4.	Chen et al. (2016)	Selection of cities in China and Australia	Global	LCA	GHG emissions	City (Scopes 1, 2, and 3)
5.	Dhakal (2009)	Selection of Chinese cities	Asia	LCA	GHG emissions	City (Scope 1)
6.	Lwasa (2017)	Kampala, Uganda	Africa	LCA	GHG emissions	City (Scopes 1 and 2)
7.	Heinonen & Junnila (2011)	Helsinki and Porvoo, Finland	Europe	LCA	GHG emissions	City (Scopes 1, 2, and 3)
8.	Ivanova et al. (2017)	European Union (EU)	Europe	LCA	GHG emissions	EU countries (Scopes 1, 2, and 3)
9.	Jagarnath & Thambiran (2018)	Durban	Africa	LCA	GHG emissions	City (Scopes 1 and 2)
10.	Stone & Wiswedel, (2018)	Selection of African cities	Africa	LCA	GHG emissions	City (Scopes 1 and 2)
11.	Huzayyin & Salem, (2013)	Cairo	Africa	LCA	GHG emissions	City (Scopes 1 and 2)
12.	Pichler et al. (2017)	Global cities	Global	LCA	GHG emissions	City (Scopes 1, 2, and 3)
13.	Fragkias et al. (2013)	U.S. Cities	North America	LCA	GHG emissions	City (Scopes 1 and 2)

#	Source	Study Region	Continent	Method	Flows (or resource)	Boundary (and Scope when GHG emissions are estimated)
14.	Facchini et al. (2017)	Global cities	Global	UM	Material and energy	City
15.	Nagpure et al. (2018)	Selected Indian cities	India	UM	Material, energy and water	City
16.	Hoekman & von Blottnitz (2017)	Cape Town	Africa	UM	Materials, energy and water	City
17.	P. K. Currie et al. (2017)	Cape Town	Africa	UM	Materials, energy and water	City
18.	P. Currie et al. (2015)	Selected African cities	Africa	UM	Materials, energy and water	City
19.	Kennedy et al. (2015)	Global cities	Global	UM	Materials, energy and water	Global
20.	Niza et al. (2009)	Lisbon, Portugal	Europe	UM	Materials and energy	City
21.	Rosado et al. (2016)	Selected Swedish cities	Europe	UM	Materials and energy	City
22.	García-Guaita et al. (2018)	Santiago de Compostela, Spain	Europe	UM + LCA	Materials, energy, water, and GHG emissions	City (Scopes 1, 2, and 3)
23.	Goldstein et al. (2013)	Global cities	Global	UM + LCA	Materials, energy, water, and GHG emissions	City (Scopes 1, 2, and 3)
24.	Quinn & Fernández (2011)	US	North America	UM	Materials and energy	City

2.4.3 Scenario-Based Approaches (Long-Term Energy Modelling)

Recognizing the scale and investments required to support a low-carbon and climate-resilient future, researchers have employed scenario-based models to project future energy use and GHG emissions for different cities, countries, or regions. As described by Iyer & Edmonds (2018), these models are often “based on sophisticated forward-looking energy-economic models that consider a combination of technologies and their attributes together with market-related forces to facilitate decision-making under deep uncertainty” (Iyer & Edmonds, 2018), page 357).

Furthermore, given the large number of variables and processes that underlay most energy scenarios, Iyer & Edmonds (2018) also recommended that researchers (1) acknowledge the limitations of the models and assumptions used, and (2) contextualize their findings by discussing other external factors that may influence future outcomes. In projecting residential energy and GHG emissions in Dar es Salaam (i.e., between 2015 and 2050), I incorporate some of the recommendations proposed by Iyer & Edmond (2018), i.e., that findings from this work provide broad insights to possible energy/GHG emissions futures in Dar es Salaam, but these pathways could change due to underlying uncertainties in model assumptions and other contextual factors not considered in the study (e.g., changes in socio-economic conditions or policy processes) (see Chapter 3).

Only a few studies have projected changes in energy use and/or GHG emissions of African cities (e.g. (Godfrey & Xiao, 2015; SEA, 2015; Stone & Wiswedel, 2018)). For example, Stone and Wiswedel (2018) employed a scenario-based method – using the Stockholm Environment Institute’s Long-Range Energy Alternatives Planning (LEAP) software – to project urban energy use and GHG emissions (from residential, industrial and transport activities) for the entire sub-Saharan Africa region between 2012 and 2040. The authors’ baseline (reference) scenario showed that urban energy use in sub-Saharan Africa increases fourfold by 2040, with GHG emissions rising 280% (Stone & Wiswedel, 2018). LEAP studies are also available for selected Asian and Latin American cities. For example, Huang et al (2019) used LEAP to project GHG emissions in the city of Guangzhou (China). The authors showed that Guangzhou’s GHG emissions could peak by 2023 assuming that current climate mitigation policies are implemented

(Huang et al., 2019). Other LEAP studies are available for São Paulo (Collaço, Dias, et al., 2019; Collaço, Simoes, et al., 2019), Panama (McPherson & Karney, 2014), Bangkok (Phdungsilp, 2010), and various Chinese cities (Fan et al., 2017; Lin et al., 2018; Yang et al., 2017; Zhou et al., 2016)

Outside of LEAP, researchers have employed sector-specific scenario models and frameworks to model changes in energy use/GHG emissions for buildings (e.g., Lin et al. (2017), Li et al. (2019) and Mokhtara et al. (2019)), transportation (e.g., Pongthanaisawan and Sorapipatana (2013), Aggarwal and Jain (2016), Dhar et al. (2017) and Du et al. (2017)) and industry (e.g., Wang et al. (2013) and de Souza et al. (2018)). While, another set of studies have employed Integrated Assessment Models (IAMs) to forecast long-term energy and emissions scenarios, e.g., Riahi et al. (2017), van Sluisveld et al. (2018), Silva Herran et al. (2019) and Wu et al. (2019). Additional details on the above-mentioned literature are presented in Chapter 3.

The study in Chapter 3 employs a LEAP model to estimate current and future changes in Dar es Salaam's residential energy use and GHG emissions (between 2015 and 2050). I chose to use LEAP given its wide use in a range of developing country contexts, i.e., see studies for São Paulo (Collaço, Dias, et al., 2019; Collaço, Simoes, et al., 2019), Panama (McPherson & Karney, 2014), and others mentioned above. The tool has also been used by developing countries across Africa, Asia and Latin America to develop national energy and GHG emissions scenarios to report their Intended Nationally Determined Contributions (INDCs) as part of commitments under the Paris Climate Change Agreement (Heaps, 2016; SEI, 2018).

My modelled scenarios in LEAP are inspired by the Shared Socio-Economic Pathways (SSPs), a scenario-based framework for envisioning future pathways of socio-economic development, i.e., changes in society and ecosystems over a century time scale (to the year 2100) (O'Neill et al., 2017, 2014; Riahi et al., 2017). The climate change research community has combined the SSPs with IAMs to model development pathways across a range of societal and ecosystem drivers, e.g., global population growth (KC & Lutz, 2017), urbanization (Jiang & O'Neill, 2017), energy use (Bauer et al., 2017), air pollution (Rao et al., 2017) and land use change (Popp et al., 2017). The SSPs offer “narratives” or “storylines” that describe different socio-economic outcomes from the above-mentioned drivers (population growth, urbanization, energy use, etc.), which can

later inform policy analysis and future climate change actions (O'Neill et al., 2017). Only a few studies have applied the SSP narratives at the city level, including the study of Kamei et al. (2016) on Tokyo, and the global study of Hoornweg and Pope (2017), although the authors did not quantify future energy use/GHG emissions of cities studied (additional details on this literature is in Chapter 3). To my knowledge, my study in Chapter 3 is the first to employ the SSPs for the purpose of projecting GHG emissions and energy use in Dar es Salaam, or indeed, any major African city.

2.5 Bottom-up (Fieldwork) Approaches to Collecting Data

My work in Chapter 3 employs a scenario-based approach to estimate residential energy use and GHG emissions in Dar es Salaam. However, this early work did not disaggregate energy use/GHG emissions at a smaller spatial scale (e.g., the ward or settlement level), which I note as a research limitation of the work in Chapter 3. To address this research gap, I led a more detailed study based on fieldwork (presented in Chapter 4) to assess differences in residential energy use patterns in a set of wards across Dar es Salaam. The overall aim of this work was to assess differences in energy use at the ward or settlement level with the aim of better informing urban planning and energy policies in Dar es Salaam/Tanzania.

I led fieldwork activities in Dar es Salaam between August and November 2018. Activities were mostly funded through research grants provided by the International Growth Centre (Small Project Fund) and the International Development Research Centre Canada (Doctoral Research Award). We (the field team) employed both quantitative and qualitative approaches to field data collection. Quantitative approaches involved structured interviews with 1,363 households selected across 8 formal, informal and mixed wards in the Dar es Salaam region. Qualitative methods included semi-structured interviews with key informants selected across local and national government, the private sector, civil society and donor groups in Dar es Salaam. A general background on quantitative and qualitative survey research methods is presented in this section and their overall relevance in the context of this thesis. A more detailed account of fieldwork activities is presented in the methods of Chapter 4 and Chapter 5.

2.5.1 Structured Surveys

The structured survey is a commonly applied method in quantitative research and consists of standardized questions and response types (Cheung, 2014; Fontana & Prokos, 2016). Interview formats are controlled and similar for all respondents, where “the interviewer controls the pace of the interview by treating the questionnaire as though it were a theoretical script to be followed in a standardized and straightforward manner” ((Fontana & Prokos, 2016), page 16). Because respondents receive the same set of questions, in the same order and sequence, structured surveys are easily replicable and scalable across a large sample (e.g., over 100 respondents). Depending on research objectives, time and budget, surveys can also be self-administered over the phone or via email (Cheung, 2014). Data collected is often coded and incorporated with statistical analysis (e.g., regression modelling, refer to Chapter 4) or other quantitative methods (e.g., data visualization or GIS).

Government-funded national census or sector-specific surveys (e.g., on electricity access or household fuel choices) are a type of structured survey administered in-person. Other examples, funded through various donor groups, include the US “Demographic Health Surveys” (DHS) (<https://dhsprogram.com/>), the World Bank’s “Living Standards Measurement Study” (LSMS) (<https://worldbank.org/en/programs/lms>), and the United Nation’s “Multiple Indicator Cluster Surveys” (MICS) (<https://mics.unicef.org/>) (UN, 2005). Collected data from these government or donor programs include a diverse set of sector-specific or socio-economic indicators, which are often incorporated in current literature, including studies on urban energy use. For example, Nagpure et al. (2018) estimate the energy material use (e.g., electricity, gas and water use) of low-income households in Indian cities using data from India’s national census report, and various district and city level surveys and master plans (Nagpure et al., 2018). Likewise, Choumert-Nkolo et al. (2017; 2019) use a national dataset published through the World Bank LSMS to assess key socio-economic drivers of household fuel choices in Tanzania. In cases where national or city data is not available, other studies have used “expert elicitation” to estimate, define or map current or future changes in technology, system or environmental processes, e.g., estimating oil sands GHG emissions in Canada (McKellar et al., 2017), defining

a community's infrastructure vulnerability and resilience in disasters (Chang et al., 2014), or mapping energy system and SDG linkages (Fuso Nerini et al., 2018).

In Chapter 4 and Chapter 5, I employ structured surveys to ascertain community level differences in residential energy use among households in Dar es Salaam. The structured format allowed for large sample of surveys to be completed (i.e., over 1,300 households completed the survey) but did not offer scope for respondents or interviewers to delve deeper into specific questions when needed. My research findings could inform future studies that employ unstructured surveys (detailed in section 2.5.2) to understand specific cultural or community aspects or “ways of life” that are not captured using structured formats. Finally, we (the field team) administered the survey in person given the objective to select a socio-economically diverse set of wards, including harder to reach households in informal settlements, e.g., Keko ward, shown in Figure 2.5, is home to several low-income communities in Dar es Salaam.



Figure 2.5. Keko ward. One of the surveyed wards in the fieldwork sample. The ward is home to several low-income communities in Dar es Salaam. Photo taken by Alice Chibulu Luo.

The survey consisted of over 70 questions (shown in Appendix A) and covered a range of topics, including household demographics, energy use, and travel behaviours (details in Chapter 4 and Chapter 5). Field team members participated in a two-day training to ensure proper interpretation of survey questions and to avoid interview biases, e.g., where ineffective or unprofessional delivery of survey questions by the interviewer could influence respondent answers (Fontana & Prokos, 2016). During the training, we facilitated mock interviews among field team members, including live pilot testing of survey questions with households in “Kijitonyama” ward.

2.5.2 Unstructured and Semi-Structured Surveys

Semi-structured and unstructured surveys (or qualitative interviews) stem largely from the social sciences but are widely applied in engineering research. According to Rubin and Rubin (2012), “Qualitative interviews are conversations in which a researcher gently guides a conversational partner in an extended discussion. The researcher elicits depth and detail about the research topic by following up on answers given by the interviewee during the discussion. Unlike survey research, in which the same questions are asked to all participants, in qualitative interviews each conversation is unique, as researchers match their questions to what each interviewee knows and is willing to share” ((Rubin & Rubin, 2012), page 4). Surveys can also occur in the context of participant observation or ethnographic interviews (Fontana & Prokos, 2016), i.e., where information gathered depends on the unique context, culture and setting in which participants are being observed or studied. These methods can lead to a deeper understanding of community culture, language, and way of life (Fontana & Prokos, 2016).

A semi-structured interview is a specific kind unstructured interview where the interviewer relies on an interview guide (i.e., a consistent set of questions or topics), but as noted by Blee & Taylor (2002), retains flexibility “to digress and to probe based on interactions during the interview” ((Blee & Taylor, 2002), page 92). This guided format still provides a “breadth and depth of information” and “the opportunity to discover the respondent’s experience and interpretation of reality”. Rubin & Rubin (2012) highlight the different role of the interviewer during unstructured or semi-structured interviewing, i.e., unstructured interviews are meant to “obtain a general

flavor” respondent perceptions, compared to semi-structured interviews that are guided by a specific set of questions ((Rubin & Rubin, 2012) page 4). Sanchez (2014) also highlights this difference by describing unstructured interviews as “similar to natural conversations” where “researchers ask questions that are largely unscripted”. Relatedly, key informant interviews presented in Chapter 5 are based on semi-structured interviews to ascertain the key barriers to low-carbon development in Dar es Salaam. Criteria for identifying key informants are detailed in Chapter 5. However, broadly, the process involved a consideration of the individual’s prior work, research, expertise and/or present position within the institution they represented. In total, 24 key informants were interviewed in Dar es Salaam (affiliations of interviewed participants are specified in the results section of Chapter 5).

2.6 Concluding Remarks

Urban energy use is integral to the discourse on sustainable development. Research shows that the anticipated growth in energy use among African cities may cause the region’s GHG emissions to become significant at the global level. However, current literature on urban energy use and GHG emissions offers few case studies on African cities. This has translated to calls to action among researchers for studies that (1) quantify energy use and GHG emissions for different African cities (and at different scales, e.g., city-specific or regional), (2) highlight the local reality in terms of how urban residents’ access and use energy, and (3) examine the key institutional and societal barriers of low-carbon development in cities. Papers presented in this thesis (i.e., Chapters 3 to 5) aim to address these research gaps and contribute to the discourse on Africa’s changing energy landscape. Using the case of Dar es Salaam (Tanzania): the first study defines a method for projecting residential energy use and GHG emissions “futures” in the city between 2015 and 2050; the second study shows the heterogeneity in energy use among surveyed formal, informal, and mixed Dar es Salaam wards; and the final study examines the institutional and societal factors that may constrain low-carbon development in the city. Insights from these studies aim to inform more effective urban planning and energy policies that support the energy and socio-economic needs of residents (especially the poor).

2.7 References for Chapter 2

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Chapter 3

Modelling Future Patterns of Urbanization, Residential Energy Use and Greenhouse Gas Emissions in Dar es Salaam with the Shared Socio-Economic Pathways

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3.1 Abstract

This paper presents three scenarios of urban growth, energy use and greenhouse gas (GHG) emissions in Dar es Salaam using narratives that are consistent with the Shared Socio-Economic Pathways (SSPs). I estimate residential energy demand and GHG emissions from 2015 to 2050 for household activities (including upstream electricity generation) and passenger (road) transport (Scopes 1 and 2). I project that by 2050, Dar es Salaam's total residential emissions would increase from 1,400 ktCO₂e (in 2015) up to 25,000 – 33,000 ktCO₂e (SSP1); 11,000 – 19,000 ktCO₂e (SSP2); and 5,700 – 11,000 ktCO₂e (SSP3), with ranges corresponding to different assumptions about household size. This correlates with an increase in per capita emissions from 0.2 tCO₂e in 2015 to 1.5 – 2 tCO₂e (SSP1); 0.7 – 1.3 tCO₂e (SSP2); and 0.5 – 0.9 tCO₂e (SSP3). Higher emissions in SSP1 (the sustainability scenario) are driven by a higher urban population in 2050 and increased energy access and electricity consumption. Through aggressive GHG mitigation policies focused on decarbonization of the electricity sector and road transport, total emissions under SSP1 can be reduced by ~66% in 2050. Study insights aim to

inform policies that identify and capture synergies between low-GHG investments and broader socio-economic development goals in African cities.

3.2 Introduction

How emerging Global South cities – especially in the Africa region – mitigate and adapt to climate change is critical to future sustainability. By the end of the century, over 30 African cities are expected to be among the world’s largest megacities (with populations exceeding 10 million) (Hoornweg and Pope, 2017) compared to two megacities in 2017 (Lagos and Kinshasa) (WorldAtlas, 2017; UN, 2018). Though the region accounts for only 3.7% of global energy-related greenhouse gas (GHG) emissions, rapid urbanization and economic growth will increase future energy demand and GHG emissions (IEA, 2019). The growth of new urban infrastructure, such as power plants, roads, water supply and sewer systems, will push the region’s aggregate material and energy use to much higher levels (Westphal et al., 2017). Urban sprawl, and persistent decline in urban population density, will be an additional driver of energy demand and emissions (Angel et al., 2011). Therefore, steering African cities towards a low-GHG future is critical to energy policy and planning as urban growth will impact global emissions due to the projected expansion of Africa’s population (Godfrey and Xiao, 2015; Calvin et al., 2016). However, literature on the future energy and GHG emissions transitions of African cities is limited to a few studies, e.g., Godfrey and Xiao (2015) and SEA (2015a). This calls for research that investigates different scenarios of urban growth and energy use in African cities, and, specifically, identifies key sectors (e.g., residential, transportation and industrial) driving these changes within individual cities.

There are two main contributions of this paper. To my knowledge, I present the first projections of possible changes in residential energy use and GHG emissions, i.e., from domestic activities, including household and transportation activities, in Dar es Salaam, Tanzania, one of the largest and fastest growing cities in Africa (Hoornweg and Pope, 2017). My analysis highlights the household and transportation drivers that are the primary contributors to future GHG emissions in Dar es Salaam, providing insights for policy makers and urban planners. The projections are to

2050 and use the Shared Socio-Economic Pathways (SSPs) as a guiding narrative. The SSPs were originally established by the climate change research community to facilitate integrated analysis of future climate impacts, vulnerabilities, adaptation, and mitigation (Riahi et al., 2017). There have been only a few applications of the SSPs at the city-level (e.g., Kamei et al. (2016) and Hoornweg and Pope (2017)), and none for the purpose of projecting GHG emissions and energy use in Dar es Salaam or any other major African city.

Second, the paper presents a method for scoping GHG emissions pathways in a relatively data-poor environment, and demonstrates how the SSPs can be used to develop urban growth scenarios. Current urban energy use or GHG emissions studies tend to focus on Global North cities (where data sources and methods are more robust), despite calls to action for research attention and focus on the Global South (especially in the Africa region) (IPCC, 2014; van der Zwaan et al., 2018). The lack of research is further reflected by the few studies that have estimated energy and GHG emissions flows in African cities, e.g., Kampala (Lwasa, 2017), Lagos (Kennedy et al., 2015) and Cape Town (Hoekman and von Blottnitz, 2017). However, these studies do not discuss expected changes in future GHG emissions in the manner presented in this paper. My results show the wide uncertainty in these future projections, while simultaneously demonstrating the order of magnitude jump in emissions that can be expected in Dar es Salaam even under optimistic scenarios.

I focus on the residential sector as it is a large “end-use” sector in the Africa region (IEA, 2014, 2019). Regional estimates indicate that 66% of final energy use occurs in the residential sector (largely due to biomass use), compared to 21% in the industrial, agricultural and services sectors (IEA, 2014). From a fossil fuel perspective (i.e., not including biomass use), residential GHG emissions still account for a large share of urban GHG emissions. For example, in large African cities such as Lagos and Accra, GHG emissions from residential buildings (due to household electricity use for lighting, heating and cooling) were estimated at ~30% (2015) and ~23% (2015), respectively, of total stationery and transport emissions, compared to ~14% and ~5% from industry (i.e., manufacturing and construction) (C40 Cities, 2017). Furthermore, while there is no available estimate of residential GHG emissions in Dar es Salaam (outside of the ones generated within this study), national GHG inventories estimate that electricity production and

transportation (including for residential use) accounted for ~38% of Tanzania's total energy sector emissions (in 2014), compared to ~7% for industry (WRI, 2015). GHG emissions from industry would generally vary on a case-by-case basis or may be linked to industry specific regulations, and therefore emissions projections for industry would scale differently compared to residential emissions. For the above reasons, the focus of this paper is on residential activities, although industrial activities could be incorporated in future work.

To accomplish the contributions outlined above, this paper:

- (1) Estimates the current (2015) emissions in Dar es Salaam and presents narratives (based on the SSPs) that project future changes in GHG emissions from domestic households, including public and private vehicle travel (Scopes 1 and 2), between 2015 and 2050.
- (2) Assesses which household and transportation activities are the primary contributors to emissions to 2050.
- (3) Analyzes how spatial factors such as urban population density influence energy use and GHG emissions.
- (4) Provides actionable urban policy recommendations that can support a low-GHG and sustainable energy transition in Dar es Salaam, and the Africa region more broadly.

3.3 Literature Review: Infrastructure and Energy Transitions in Africa and Other Global South Cities

The African Development Bank estimates the scale of investments required to build Africa's future infrastructure at between \$130 and \$170 billion a year (AfDB, 2018). This infrastructure demand presents a unique opportunity to build more sustainable (and resilient) cities with policies that promote low-GHG and resilient communities (that especially benefit the poor). However, the urbanization of African cities comes with unique challenges. Unlike the transformation in Europe and North American cities, whose urbanization was correlated with industrialization and economic growth (Currie and Musango, 2017), these associations are not evident in the Africa region (Allen, 2014). Rather, urban growth has been predominately

“splintered” and reinforced by socio-economic challenges such as poverty, inequality and vulnerability to climate change (Allen, 2014; Currie and Musango, 2017). Splintered urbanism has heightened inequalities, as basic infrastructure services, such as electricity, water supply and public transportation, are often limited or non-existent for the poorest neighborhoods (Allen, 2014; Currie and Musango, 2017). In this regard, studies find that low levels of infrastructure stock (and urban wealth) in African cities is a key reason for their limited energy use and GHG emissions compared to higher-income cities (Kennedy et al., 2015).

A handful of prior studies have compared electricity use, transportation emissions and/or direct final energy use among global cities (e.g., Schulz (2010); Grubler et al. (2013) and Kennedy et al. (2014)), and report values for Dar es Salaam (0.16 MWh/capita, ~1 tCO₂e/capita and 17GJ/capita) that are far lower than their counterparts in the U.S. (9 – 10 MWh/capita and 4 tCO₂e/capita or greater) or Canada (162 GJ/capita in Toronto). Another set of studies quantify the flows of materials, energy, and waste in cities using urban metabolism frameworks. Metabolism assessments are available for a limited number of African cities, including Lagos (Kennedy et al, 2015) , Kampala (Lwasa, 2017), Durban (Jagarnath and Thambiran, 2018) and Cape Town (Hoekman and von Blottnitz, 2017). Increasing resource access remains a key challenge for these cities. For example, the study by Kennedy et al. (2015; page 5988) conclude that developing cities in Asia and Africa (e.g., Lagos) are “consuming resources at rates below those that support a basic standard of living for all citizens”. This is consistent with other literature that correlates resource use and GDP/capita or Human Development Index (HDI) ratings in African cities (Currie et al., 2015; Currie and Musango, 2017).

Only a few studies have projected energy use and GHG emissions pathways in African cities (e.g., Senatla (2011), Godfrey and Xiao (2015), SEA (2015a) and Stone and Wiswedel (2018)). However, there are several studies in other regions of the Global South, especially Asian and Latin American cities (e.g., McPherson and Karney (2014), Collaço et al. (2019) and Huang et al. (2019)). Emissions pathways are estimated using scenario-based models that aggregate data across different urban sectors. For example, Stone and Wiswedel (2018) used the Stockholm Environment Institute’s Long-Range Energy Alternatives Planning (LEAP) software to assess the scale of GHG emissions growth (from residential, industrial and transport activities) in urban

Africa from 2012 to 2040. The authors' results indicated that urban energy demand in African cities could increase fourfold by 2040, with GHG emissions rising 280%. This would shift the region's share of global emissions from 1% (in 2012) to 4% in 2040. In China, Huang et al. (2019) used LEAP to project peak levels of GHG emissions in the city of Guangzhou. Findings showed that while emissions will peak by 2023 under existing climate mitigation policies, the peak could be moved forward to 2020 with more stringent energy conservation and policies, including (among other interventions): (1) adjusting the energy mix and mode of passenger transport; (2) and replacing coal and oil use with electricity and natural gas in the industrial sector; and (3) enabling large scale-up of renewable energy power. Similar applications of the LEAP model at the city-level are available for São Paulo (Collaço et al., 2019), Panama (McPherson and Karney, 2014), Bangkok (Phdungsilp, 2010), and several Chinese cities (Zhou et al., 2016; Fan et al., 2017; Yang et al., 2017; Lin et al., 2018), among others.

Outside of LEAP, researchers have employed models and frameworks designed for specific sectors, including buildings (e.g., Lin et al. (2017), Li et al. (2019) and Mokhtara et al. (2019)), transportation (e.g., Pongthanaisawan and Sorapipatana (2013), Aggarwal and Jain (2016), Dhar et al. (2017) and Du et al. (2017)) and industry (e.g., Wang et al. (2013) and de Souza et al. (2018)). Other studies have used Integrated Assessment Models (IAMs) to forecast long-term energy and emissions scenarios (e.g., Riahi et al. (2017), van Sluisveld et al. (2018), Silva Herran et al. (2019) and Wu et al. (2019)). IAM literature remains limited in the Africa region, with notable exceptions by Lucas et al. (2015), Calvin et al. (2016), and van der Zwaan et al. (2018). In particular, van der Zwaan et al. (2018) modeled pathways for low-carbon development in Africa (including North African countries) using the "TIAM-ECN" IAM model and simulated the development of energy economies over time. Their findings showed that while Africa's GHG emissions could become substantial at a global scale by 2050, the region could "leapfrog" fossil-fuel based growth with large-scale use of renewable energy options (van der Zwaan et al., 2018).

A final set of studies have coupled IAMs with the SSPs to project a range of socio-economic trends, such as future changes in global population (KC and Lutz, 2017), urbanization (Jiang and O'Neill, 2017), energy use (Bauer et al., 2017) and air pollution (Rao et al., 2017). However, a number of research gaps remain in the IAM and SSP literature. Local or city level data has not

been widely incorporated into models and there is need for additional research at lower geographic scales, i.e., to enable local dynamics to be incorporated into IAMs (Cronin et al., 2018). Currently, studies by Kamei et al. (2016) and Hoornweg and Pope (2017) are among the few studies that translate the SSP narratives to the city level (though, do not use an IAM approach). Kamei et al. (2016) determined long-term socioeconomic scenarios in Tokyo based on a theoretical model and expert interviews, while Hoornweg and Pope (2017) coupled their narratives with regression models to project urbanization trends in the world's largest cities to 2050, 2075 and 2100.

Gaps in modelling approaches remain, and researchers have called for additional studies in developing regions, especially Africa (Cronin et al., 2018; van der Zwaan et al., 2018). This paper contributes to the growing SSP literature and provides the first application of SSPs in Dar es Salaam or Tanzania. The novelty in my approach is embedded in my assumed scenarios and projections. Assessments of urban energy and/or material flows in Kampala (Lwasa, 2017), Lagos (Kennedy et al., 2015), Durban (Jagarnath and Thambiran, 2018), and others aforementioned, did not estimate changes in energy use or GHG emissions over time. Therefore, by estimating current (2015) and possible changes in GHG emissions in Dar es Salaam to the year 2050, my work may derive insights on the emissions pathways of other African cities.

Considering that the IAMs (including the SSPs) have not been applied to city level analysis (Cronin et al., 2018), I couple my SSP narratives with a LEAP modelling approach. The LEAP tool has been widely used to estimate long-term energy use and GHG emissions futures in developing countries. Finally, my work also shows the opportunity for Dar es Salaam to implement policies that support low-GHG urbanization. The urbanization narratives modelled in this paper – SSP1 (Sustainable Growth), SSP2 (BAU Growth), and SSP3 (Fragmented Growth) (described in the Methods) – present distinct urbanization, energy use and GHG emissions futures for Dar es Salaam. The narratives provide a basis for identifying (1) key household and transportation drivers of GHG emissions in Dar es Salaam, and (2) investments that can support future emissions reductions (which could potentially be generalizable to other large African cities).

3.4 Case Study of Dar es Salaam, Tanzania

Dar es Salaam is the largest city and economic hub of Tanzania. In 2015 (the current year considered in this study) Dar es Salaam had an estimated population of 5.1 million (or 1.3 million households, assuming an average household size of four persons per household) (World Bank, 2018). The city is experiencing significant changes in urban form, although it is noted that the city masterplan was last updated in 1979 (Government of Tanzania, 2017a). Structurally, Dar es Salaam exhibits a monocentric and radial urban form, with highest population densities clustered around the city center and along the four major arterial roads, i.e., to the north along Bagamoyo road, north-west along Morogoro road, south-west along Nyerere road and south along Kilwa road (Figure 3.1).

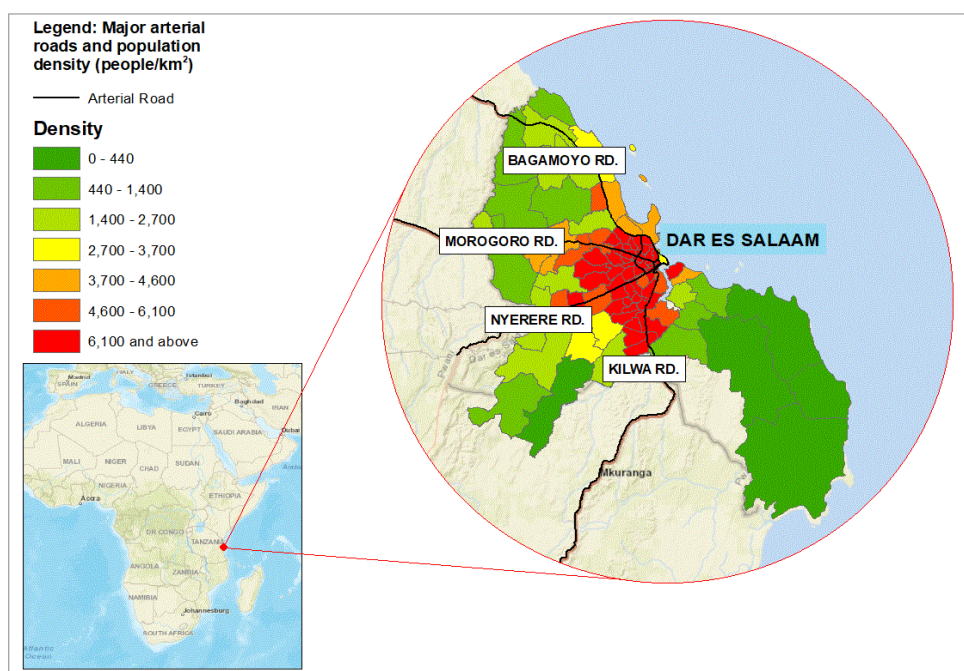


Figure 3.1. Average population densities (by ward) and major arterial roads (Bagamoyo, Kilwa, Morogoro and Nyerere) in Dar es Salaam. I compiled the map in ArcGIS using ward population data from the 2012 National Census Report (Government of Tanzania, 2016b, 2017a).

Generally, energy sector statistics in Tanzania are reported at the national level, including through the National Communications to the United Nations Framework Convention on Climate

Change (UNFCCC) (Government of Tanzania, 2015). An estimated ~75% of Dar es Salaam households have access to electricity (DHS Program, 2016; Government of Tanzania, 2017b). Despite high electrification levels compared to rural areas (Government of Tanzania, 2017b), urban households experience frequent power cuts and fluctuations in voltage that can damage electric appliances (Garside and Wood, 2018). To compensate for electricity shortages, “fuel stacking”, where households use a combination of other fuels such as firewood, charcoal, liquefied petroleum gas (LPG) or kerosene (in addition to electricity) is widespread (Lusambo, 2016). It is estimated that only 2% of Dar es Salaam households use only electricity for cooking and heating needs (DHS Program, 2016).

In the transport sector, approximately 62% of all passenger trips (~81% of vehicle trips) are by small minibuses called “dala-dalas” (Mkalawa and Haixiao, 2014). Other modes include private cars, including taxis (16% of vehicle trips); and motorcycles and tricycles, known locally as “bodas” and “bajajis” (3% of vehicle trips) (Table 3.2) (Mkalawa and Haixiao, 2014). The dala-dala service is widely used by low-income communities given its affordability, though it is often characterized by poor service quality, untrained bus operators and non-adherence to traffic rules and regulations (Nkurunziza et al., 2012). To improve standards of service, the city is implementing a six-phase Bus Rapid Transit (BRT) system, with main corridors operating along the four major arterial roads (Government of Tanzania, 2017a). Phase 1 of the BRT was completed in 2016 and operates along Morogoro road (Figure 3.1), which traverses from Dar es Salaam’s high-income central business district towards middle- and low-income residential areas in the west. Plans to expand the BRT up to six phases are currently underway (World Bank, 2017b). More detail about the BRT implementation is available in the supplementary material appended to the end of this Chapter (Section 3.9.8).

3.5 Methods

I model future pathways of energy use and GHG emissions in Dar es Salaam from 2015 (current year) to 2050 with a focus on the residential sector, including associated public and private road transportation. I include direct (Scope 1) emissions from households (i.e., emissions from the use

of charcoal, firewood, kerosene or liquefied petroleum gas (LPG), and emissions from road travel using private vehicles or public transport modes), as well as upstream (Scope 2) emissions from electricity generation (for household use or electric vehicle charging). We broadly describe these activities as “residential” in the remainder of the paper. I do not account for emissions from fuel production, or from commercial and industrial activities, including air, railway, or marine transport. I also do not include embodied (Scope 3) emissions associated with product manufacture and shipping.

The focus on residential energy use and emissions is due to the large contributions of these activities compared to industrial activities, or other productive sectors. Domestic use of biomass (i.e., charcoal and fuel wood) accounts for over 90% of final energy consumption in Tanzania (Government of Tanzania, 2014a). However, biogenic carbon emissions from biomass combustion, as well as emissions from Land Use Land-Use Change and Forestry (LULUCF) are not included in emissions inventories for the energy sector category. Emissions accounted for in the sector include national electricity (~11%), road transportation (~27%), manufacturing and construction (~7%), and commercial, residential, and agricultural activities (~55%) (WRI, 2015).

All GHG emissions are stated in kilotons of carbon dioxide equivalents (ktCO₂e), which includes CO₂, methane, and nitrous oxide. GHG emissions are calculated using 100-year global warming potentials (GWP) (IPCC, 2013). GWPs and emissions factors for all household and transport fuels are listed in the supplementary information (Section 3.9).

3.5.1 Dar es Salaam’s Urbanization Narratives

My urbanization narratives are inspired by the SSPs which have been developed and modelled by climate change researchers (e.g., Riahi et al. (2017)). The original SSPs are based on five narratives or “storylines”, each with different consequences for global and regional socio-economic development under increasing climate uncertainty (O’Neill et al., 2017). I focus specifically on SSP1 (“Sustainability”), SSP2 (“Business As Usual”) and SSP3 (“Fragmented”)

as they sufficiently illustrate a range of possible futures that encompass results from SSP4 (“Inequality”) and SSP5 (“Fossil Fueled Development”).

The narratives presented in this paper are simplified baseline projections of Dar es Salaam’s future energy use and GHG emissions. Each narrative is distinct and highlights different energy use dynamics and outcomes. I assume no additional climate mitigation actions beyond the baseline narratives (and as outlined in the Methods). Therefore, in Section 3.6.7, I include an additional mitigation scenario that facilitates the examination of aggressive GHG mitigation policies focused on decarbonization of electricity and road transportation, and assesses which activities have the potential to drive the largest emissions reductions to 2050. Table 3.1 describes Dar es Salaam’s urbanization narratives and justifications, as appropriate.

Table 3.1. Dar es Salaam’s Urbanization Narratives inspired by the SSPs: SSP1 (Sustainability), SSP2 (BAU) and SSP3 (Fragmented)

Indicators	SSP1: Sustainability	SSP2: Business-As-Usual (BAU)	SSP3: Fragmented
Population	- Fast initial population growth by 2050. - Lowest peak in population after 2050.	- Moderate population growth, consistent with historic growth trends. - Moderate peak in population after 2050.	- Slow initial population growth. - Highest peak in population after 2050.
Households	100% electrification is realized by 2050 - Net-zero consumption of traditional fossil fuels (i.e., charcoal and wood) by 2050.	100% electrification by 2050. - Households continue to rely on traditional fossil fuels in 2050.	- No change in electrification levels from 2015. - Households continue to rely on traditional fossil fuels in 2050.
Passenger Transport	- Phases 1 to 4 of the BRT are complete by 2050. - BRT ridership accounts for 40% of total passenger trips, similar to reported ridership in Latin American and Chinese cities (WRI, 2018). - Fuel efficiency of light-duty vehicles (LDVs) improves to OECD levels, in line with global targets to 2050 (OECD/IEA, 2017a).	- Phases 1 to 4 of the BRT are complete by 2050. - BRT ridership accounts for 15% of total passenger trips, consistent with existing BRT implementation plans (World Bank, 2017b). - Fuel efficiency of LDVs progresses to the same levels observed in middle- and high-income cities today.	- Phases 1 to 4 of the BRT are complete by 2050. - BRT ridership accounts for 15% of total passenger trips, with future BRT expansion plans halting post-2050. - Fuel efficiency of LDVs progresses to the same levels observed in middle- and high-income cities today.

3.5.2 Modelling Using the LEAP Platform

For each SSP narrative, I use the LEAP modelling platform (Heaps, 2016) to calculate Dar es Salaam’s residential energy use and GHG emissions to 2050. The platform offers a transparent way of structuring complex energy data, projecting different demand and supply scenarios, and integrating factors such as population growth, GDP and policy changes to energy sector analysis (Heaps, 2008, 2016). LEAP has not been employed to model energy use and GHG emissions in Dar es Salaam or Tanzania.

Modelling capabilities include built-in calculations to determine energy use and GHG emissions based on time-varying data points (Heaps, 2008, 2016). The platform's Technology and Energy Database includes GHG emissions data for a range of fuels based on the Intergovernmental Panel on Climate Change (IPCC) guidelines. The supplementary material (Section 3.9.9) provides more detail about the calculation structure within LEAP.

3.5.2.1 Data Sources and Underlying Assumptions (2015 to 2050)

I estimate Dar es Salaam's residential energy use and GHG emissions using the following data and assumptions (see Table 3.2): (1) population, GDP and household size; (2) population density; (3) the GHG intensity of electrification; (4) fuel use at the household level; and (5) fuel use for road transportation. The following sections describe my approach in sourcing data. I also caveat that where data is not available for Dar es Salaam, I draw from national estimates, or proxy data from other Global South cities.

Table 3.2. Key indicators and underlying assumptions for estimating Dar es Salaam’s residential energy use and GHG emissions for SSP1 (Sustainability), SSP2 (BAU), and SSP3 (Fragmented) narratives from 2015 to 2050.

#	Indicator	Unit	Current year – 2015	Data source for current year	SSP1 – 2050	SSP2 – 2050	SSP3 – 2050	Data source for assumptions to 2050
1	Population	million	5.1	(World Bank, 2018)	16	15	12	Equation 3.1
2	GDP/Capita	USD \$	1,100	(IIASA, 2015)	4,700	2,500	1,500	(IIASA, 2015)
3	Household (HH) size	persons/HH	4	(Government of Tanzania, 2014b)	[2 – 4]	[2 – 4]	[2 – 4]	Reduction to 2 persons/HH at the lower bound reflects the lowest HH size observed globally today (UN, 2017)
4	Number of households	million	1.3	(Government of Tanzania, 2014b)	[4 – 12]	[4 – 8]	[3 – 6]	Own calculation
5	Average population density	persons/km ²	3,100	(Government of Tanzania, 2014b)	3,100	3,300	3,500	Downscaled 1km ² population density projections from (Jones and O’Neill, 2016) (Figure 3.3).
6	% change in average urban population density				0%	6%	13%	

#	Indicator	Unit	Current year – 2015	Data source for current year	SSP1 – 2050	SSP2 – 2050	SSP3 – 2050	Data source for assumptions to 2050
7	Electrification level	% of total households	75	(Government of Tanzania, 2017b)	100	100	75	(Government of Tanzania, 2017b)
8	GHG intensity of electricity	gCO ₂ e/kWh	405	Author calculation	405 ³	435 ³	435 ³	Own calculation
9	Electricity use ^{1, 4}	GJ/HH/yr.	5	(IEA, 2014)	46	25	18	Assumption based on SSP narratives for total household energy use (Table 3.3).
10	LPG use ^{1, 4}		4	(Drazu <i>et al.</i> , 2015)	0	16	10	
11	Kerosene use ^{1, 4}		1		0	13	7	
12	Firewood use ^{1, 4}		16	(Drazu <i>et al.</i> , 2015)	0	0	8	
13	Charcoal use ^{1, 4}		21	(SEA, 2015b, 2015a)	0	0	10	
14	Annual VKT per capita	km	870	(Mkalawa and Haixiao, 2014; World Bank, 2017a)	870	860	840	Elasticity between density and VKT (Guerra, 2014)

#	Indicator	Unit	Current year – 2015	Data source for current year	SSP1 – 2050	SSP2 – 2050	SSP3 – 2050	Data source for assumptions to 2050
15	LDV	% of total vehicle trips	16%	(Mkalawa and Haixiao 2014)	12%	15%	15%	Based on assumption that relative change in vehicle trips will mostly shift from dala-dala to BRT as stated in Methods, with small changes in LDV and motorcycle/tricycle use.
17	Dala-dala (standard bus: 40-seater)		81%		55%	67%	67%	
18	Boda or Bajaji (motorcycle or tricycle)		3%		3%	3%	3%	
16	BRT		0% ²	(World Bank, 2017b)	30%	15%	15%	Based on projected completion of BRT Phases 1 to 4 (see (World Bank, 2017b))
19	Electric Vehicles ⁴		0	(IEA, 2017a, 2018)	1%	0.1%	0.1%	(IEA, 2017a, 2018)
20	Fuel use ⁵ (LDV)	litres/100km	12	(World Bank, 2017a)	4.4	7.4	7.4	(IEA, 2014, 2017a)
21	Fuel use ⁵ (BRT)		38	(DART Agency, 2017)	No change.			Own assumption

#	Indicator	Unit	Current year – 2015	Data source for current year	SSP1 – 2050	SSP2 – 2050	SSP3 – 2050	Data source for assumptions to 2050
22	Fuel use ⁵ (dala-dala)		33					
23	Fuel use ⁵ (Boda or Bajaji)		1.8	(IEA/GFEI, 2015)				
Load factor by vehicle mode (from 2015 to 2050): - LDVs – 1.8 passengers/vehicle (World Bank, 2017a) - Dala-dala – 40 passengers/vehicle (DART Agency, 2017) - BRT – 150 passengers/vehicle (DART Agency, 2017) - Boda or Bajaji – 1.2 passengers/vehicle (World Bank, 2017a)								

Table Notes:

¹ Total household energy use remains constant for all future projections, though the relative shares of fuel use change based on the SSP narrative.

² I assume no BRT ridership in 2015. Phase 1 of the BRT was fully operational in May 2016 (DART, 2017).

³ I assume different changes in the generation mix depending on the scenario (Section 3.9.3)

⁴ EV projections are based on current IEA estimates for South Africa (SSP2 and SSP3) and Europe (SSP1).

⁵ Refer to Section 3.9.2 for emissions factors for all fuels used in the LEAP model.

3.5.2.2 Population, GDP, and Household Size

For each SSP narrative, we estimate Dar es Salaam's future population to 2050 as follows:

$$\text{DAR Population}_{\text{Year}} = \text{TP}_{\text{Year}} \times \text{TUP}_{\text{Year}} \times \text{PS}_{\text{Year}} \quad (\text{Equation 3.1})$$

Where “Year” represents the year of prediction, “TP” represents Tanzania's total population (in millions) for the given year, “TUP” represents Tanzania's urban population level (as a percentage) for the given year, and “PS” is the population share of Dar es Salaam (as a percentage of the total urban population) for the given year.

I determine Tanzania's total population (TP) and urban population level (TUP) from the existing population and urbanization projections for the SSPs (Jiang and O'Neill, 2017; KC and Lutz, 2017), which include data from 2010 to 2100. Over the last 20 years, Dar es Salaam has consistently accounted for approximately 30% of the country's total urban population (World Bank, 2018). I assume this share will remain at 30% across all future scenarios (while a rate of 30% may seem low, I expect that this is consistent with the large growth also expected in other Tanzanian cities). Finally, I estimate GDP per capita between 2015 and 2050 by dividing Tanzania's projected GDP, available in the SSP database (IIASA, 2015), by Tanzania's projected total population (TP).

3.5.2.3 Household Size

I estimate the average household size in Dar es Salaam at four persons per household in 2015 (Table 3.2) (DHS Program, 2016). Across all SSPs, Tanzania's total fertility rate (TFR) is projected to fall (Lutz et al., 2014), suggesting that household size will likely decrease in the future. To estimate future changes in household size and impact on household energy use and emissions, I consider two bounding scenarios – (1) as an upper estimate, I assume household size remains constant at four persons per household to 2050; and (2) as a lower estimate, I assume an eventual reduction in household size to 2 persons per household by 2050, consistent with the

lowest household estimates observed globally today (UN, 2017). This also serves the purpose of allowing per capita energy to increase as a function decreasing household size. For example, my assumption that total household energy use remains constant to 2050 (Table 3), implicitly increases per capita energy use with the reduction in household size. Therefore, while I am unable to create a more refined estimate of changes in total household energy use in Dar es Salaam due to data limitations, my modelling explores some possible futures in GHG emissions across a range of estimates (based on both constant and changing household size).

3.5.2.4 Population Density

I project Dar es Salaam's average population density using Jones and O'Neill's (2016) spatial projections which map global and regional changes in urban, rural and total population (based on 1km² grids) from 2010 to 2100. By considering only those grids that fall within Dar es Salaam's administrative boundary, I calculate changes in the city's urban density (i.e., sprawl or concentration) for each processed layer (for SSP1, SSP2 and SSP3).

3.5.2.5 Electricity Generation

Currently, Tanzania's electricity generation mix is dominated by natural gas (59%) (Section 3.9.3); hydropower (35%), Heavy Fuel Oil (HFO) (5.7%) and biomass (0.3%) account for the remaining fractions (Government of Tanzania, 2017c). By 2040, Tanzania aims to expand the generation mix to include coal, solar, wind and geothermal sources (Government of Tanzania, 2016a). According to Tanzania's Intended Nationally Determined Contribution (INDC) (Government of Tanzania, 2015), geothermal potential is estimated at 5GW and hydropower at 4.7GW (though installed capacity is currently 0.6GW (Government of Tanzania, 2016a)). My LEAP model assumes different transformations in the generation mix for each SSP narrative. SSP1 assumes a 10% penetration of renewable energy, consistent with the highest level of renewable energy penetration scenario ("Scenario 6") considered in Tanzania's National Power Plan (Government of Tanzania, 2016a). SSP2 and SSP3 assume a shift in the generation mix to natural gas (40%), hydropower (20%), coal (35%), and 5% penetration of renewable energy (i.e.,

solar and wind sources) by 2050. These advancements are consistent with the preferred scenario envisioned under Tanzania’s National Power Plan (“Scenario 2”) (Government of Tanzania, 2016a).

3.5.2.6 Household Activities

I estimate energy use and GHG emissions associated with fuels used for space and water heating, cooking, lighting, and appliance use within the city (Scope 1), and associated emissions from electricity generation (Scope 2). In 2015, Dar es Salaam’s household electricity use was estimated at 1,250 kWh/household (HH)/yr. (~ 5 GJ/HH/yr.). This is consistent with the World Bank’s “Tier-4” level of electricity access, where households use electricity for lighting and some medium-power appliances (e.g., television, radio, phone charger) (World Bank, 2015). By 2035, Tanzania plans to achieve a national electrification rate of 90% (Government of Tanzania, 2016a). Therefore, my modelling assumes that 100% electrification is realized for SSP1 and SSP2 by 2050. SSP3 assumes no progress is made, with electrification remaining at 75%.

In most households, charcoal or LPG are widely used in combination with electricity. For example, in 2015, 75% of households in Dar es Salaam used electricity and 69% used charcoal (DHS Program, 2016; Government of Tanzania, 2017b), meaning that some households were using both charcoal and electricity for daily needs. Other household fuels include LPG (14%), wood (6%) and kerosene (6%). I implicitly account for these fuel stacking behaviors by calculating the total household energy use (in GJ/HH/yr.) and estimate the relative change in fuel use shares (i.e., charcoal, firewood, LPG and kerosene) for each SSP narrative (Table 3.3). Moreover, all future scenarios assume that total household energy use remains constant, though I change both the household size and the relative energy use shares from the different fuel sources based on the SSP narrative. Finally, although household energy use remains constant, I report results in each scenario for both constant and decreasing household sizes, with the latter implicitly allowing growth in household energy use per person. Refining these projections for household energy use is an important area for future work.

Table 3.3. Modelling assumptions for changes in household energy use for SSP1 (Sustainability), SSP2 (BAU), and SSP3 (Fragmented) narratives.

Scenario	% share of total household energy use in 2015 (current year)	Estimated changes in energy use (by fuel) to 2050
SSP1: Sustainability	Electricity: 11% (5 GJ/HH/yr.) LPG: 9% (4 GJ/HH/yr.) Kerosene: 2% (1 GJ/HH/yr.) Charcoal: 46% (21 GJ/HH/yr.)	Electricity accounts for 100% of total household energy by 2050. Charcoal and wood use phased out by 2030. LPG and kerosene use peak to 35% and 28% of total household energy in 2030 ¹ , followed by a decline and eventual phase out by 2050. Total change in energy use (i.e. from phased out charcoal, LPG, and kerosene) shifts to electricity.
SSP2: BAU	Firewood: 32% (16 GJ/HH/yr.)	Electricity accounts for 100% of total household energy by 2050. Charcoal and wood use halve by 2030 but are entirely phased out by 2050. Total change in energy use (i.e., from phased out charcoal and wood) shifts to electricity, LPG, and kerosene, in equal amounts ² .
SSP3: Fragmented		Electricity accounts for 38% of total household energy by 2050. Charcoal and wood use halve by 2050. Change in total energy use (i.e., from reduced charcoal and wood) shifts to electricity, LPG, and kerosene, in equal amounts ² .

Table Notes:

¹The eventual phase out of charcoal in 2030 shifts total energy use towards electricity, LPG and kerosene between 2015 and 2030. This shift is what drives the peak in LPG and kerosene use in 2030. However, we assume that LPG and kerosene use will eventually decline post-2030 with improved electricity access and economic growth in Dar es Salaam,

²The change in total energy use from charcoal and firewood use is divided by 3 with amounts (in GJ/HH/yr) transferred to electricity, LPG and kerosene (see Table 3.2).

3.5.2.7 Transport Activities

I project future changes in travel demand based on annual vehicle kilometers travelled (VKT) which accounts for city travel by LDVs and public transit, i.e., dala-dalas, “bajajis” (tricycles), “bodas” (motorcycles), and the BRT. For 2015 (the current year), I estimate VKT as a product of the average number of vehicle trips (1.2 trips/person/day (World Bank, 2017a)); average trip distance (20 kilometers (World Bank, 2017a)); mode share; and load factor. Empirical evidence from other developing cities (e.g., Latin America) shows statistically significant correlations between the urban built environment and VKT, e.g., Zegras (2010) and Guerra (2014). To estimate the correlation between VKT and population density, my modelling applies results from Guerra (2014), i.e., using an uncensored latent VKT value that reduces modelling bias associated with different household travel behaviors, a 1% increase in population density is correlated with a 0.03% reduction in VKT (Guerra, 2014). I apply this correlation to my LEAP calculations to estimate the future change in VKT with changes in density for each SSP narrative. All vehicle load factors and fuel consumption estimates are in Table 3.2.

3.5.2.7.1 Electric Vehicles

I anticipate that some penetration of electric vehicles in Dar es Salaam is likely, given the existing policies and plans to increase production of EVs globally (IEA, 2018). However, it is difficult to make reasonable projections for Dar es Salaam to 2050 given the limited data available on the EV market potential in East Africa. Currently, South Africa is the only African country with electric vehicles, representing only 0.1% of passenger vehicle stock (OECD/IEA, 2017b). My SSP2 and SSP3 narratives estimate that Dar es Salaam realizes a similar level of EVs in the LDV fleet by 2050 (Table 3.2); while SSP1 estimates an increase to 1%, similar to levels observed in Europe today (e.g., the Netherlands and Sweden) (IEA, 2018). This seemingly low level of EV penetration is consistent with my assumption that these are baseline projections with no special measures taken toward GHG mitigation beyond the broad narrative of each scenario. This assumption is relaxed in my discussion of aggressive GHG mitigation scenarios in

Section 3.6.7. Finally, I assume electricity consumption of 27 kWh per vehicle-kilometer, consistent with IEA estimates (IEA, 2018).

3.5.2.7.2 The BRT

For all scenarios, I assume that Dar es Salaam completes Phases 1 to 4 of the BRT by 2050, consistent with current implementation plans (Section 3.9.8). Completion of the four phases would result in approximately 900,000 riders per day (World Bank, 2017b), equivalent to 15% of total passenger trips in 2015. SSP2 and SSP3 assume that BRT trips increase to 15% (of all passenger trips), while SSP1 assumes a higher increase to 40%, similar to levels reported in Latin American and Chinese cities (UITP, 2015; WRI, 2018). I estimate BRT fuel consumption at 38 liters/100km (DART Agency, 2017) (Table 3.2), similar to consumption profiles in Latin America and Asian cities, e.g., 33 litres/100km (Jaipur, India) and 40 litres/100km (Quito, Ecuador) (WRI, 2018). I also assume that BRT fuel consumption remains at this level to 2050.

3.5.2.7.3 Minibuses (“dala-dalas”)

I assume no changes in dala-dala fuel consumption to 2050, i.e., consumption remains at 33 litres/100km (DART Agency, 2017), given the current plans to reduce dala-dala use with a shift to BRT (World Bank, 2017b).

3.5.2.7.4 Light Duty Vehicles (LDV)

Fuel consumption estimates for the LDV fleet (~12 litres/100km) are taken from (World Bank, 2017b). Projecting to 2050, SSP1 envisions that LDV fuel consumption improves to 4.4 litres/100km, consistent with IEA targets (IEA, 2017b; OECD/IEA, 2017a). SSP2 and SSP3 assume a less aggressive improvement to 7.4 litres/100km, consistent with projections to 2040 for the Africa region (OECD/IEA, 2014).

3.6 Results and Discussion

3.6.1 Changes in Dar es Salaam's Total Population and Density

Across each of the SSPs, Dar es Salaam is shown to experience substantial population growth between 2015 and 2050. Projections for Dar es Salaam's population to 2050 are based on Equation (3). In all scenarios, Dar es Salaam becomes a megacity by 2050, with the city's population growing to 16 million under SSP1, 15 million under SSP2 and 12 million under SSP3 (illustrated in Figure 3.2). Dar es Salaam experiences the fastest urbanization rate under SSP1, while moderate and slow urbanization occurs under SSP2 and SSP3, respectively. My SSP1 population projection for 2030 (9.2 million in Dar es Salaam) is within 15% of the United Nation's World Urbanization Projections (WUP) estimate for 2030 (~10.7 million) (UN, 2018). In addition, Hoornweg and Pope (2017) extrapolate the WUP dataset to 2100 and project Dar es Salaam's population at 16 million in 2050. This is consistent with my SSP1 and SSP2 estimates.

My urban growth assumptions based on Jiang and O'Neill (2017) who project substantial urban growth in Tanzania across each of the SSPs. Their estimates to 2050 project up to 60% (SSP1), 50% (SSP2) and 30% (SSP3) urbanization in Tanzania (Jiang and O'Neill, 2017), increasing the urban share of Tanzania's population by 7% to 37% between 2015 and mid-century (2050). My calculations show that this is equivalent to absolute population increases of 12 million (SSP1), 11 million (SSP2) and 7.5 million (SSP3) between 2015 and 2050 (Figure 3.2).

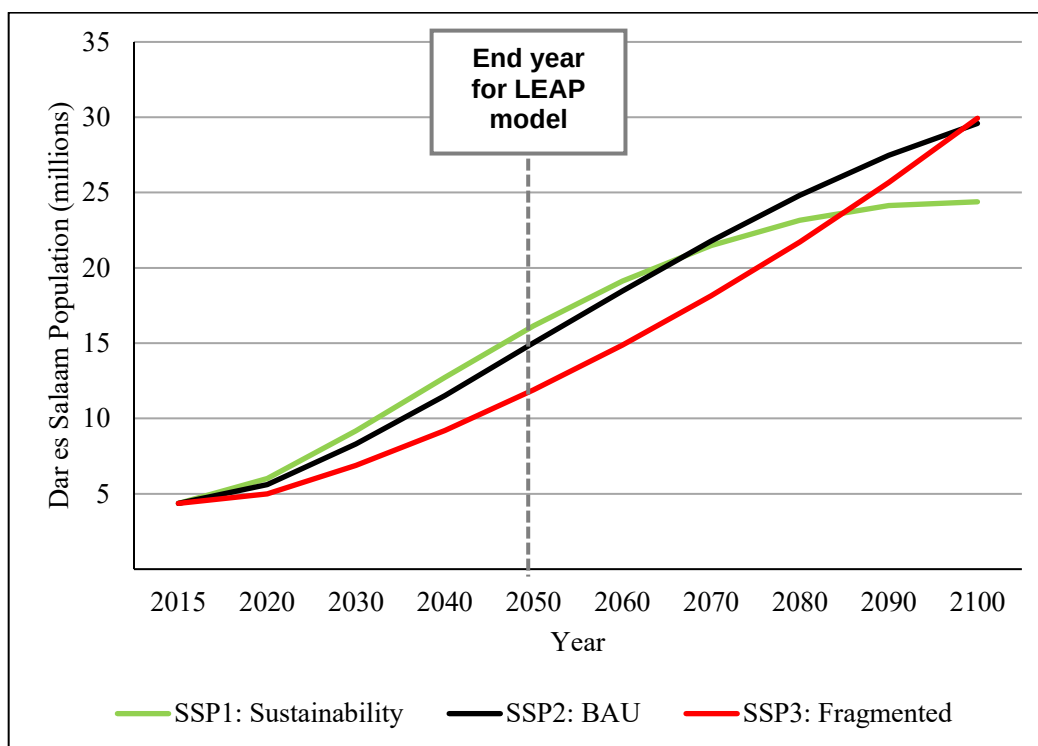


Figure 3.2. Changes in Dar es Salaam's population from 2015 to 2050 for SSP1 (Sustainable), SSP2 (BAU) and SSP3 (Fragmented) narratives. My LEAP model calculates energy use and emissions to the year 2050; though, estimates are extended to 2100 to illustrate the eventual slow-down in Dar es Salaam's population under SSP1. Dar es Salaam's population continues to increase at a higher rate for SSP2 and SSP3.

Dar es Salaam's average population density in 2015 is estimated at 3,100 persons/km² (Government of Tanzania, 2014b). By 2050, I estimate that the city's average population density remains the same for SSP1 (3,100 persons/km²) and increases slightly for SSP2 (3,300 persons/km²) and SSP3 (3,500 persons/km²) (Figure 3.3). My calculations are based on Jones and O'Neill's (2016) "spatially explicit" global population scenarios, which I use to extract the population density projections for Dar es Salaam (see Methods). Given the counter-intuitive nature of the results – i.e., I would expect higher density under SSP1 would be correlated with sustainable resource use (Kennedy et al., 2015) – I caveat that these projections are the only available dataset estimating future population densities based on the SSPs (Gao, 2017) and estimates can be improved with neighborhood level data collection. The maps shown in Figure

3.3 do not illustrate the growth in Dar es Salaam's spatial extent; for example, the likely urban sprawl given the estimated population increases that are projected for each SSP narrative. Therefore, the maps should not be interpreted as accurate projections of density changes of specific neighborhoods. Rather, they provide a baseline assessment of the differences in density change, at the city level, among the three SSP narratives. For example, the sustainability scenario (SSP1) shows higher population densities closer to the city centre and along the four major arterial roads (key development areas for the BRT expansion). While settlement patterns for BAU (SSP2) and fragmented (SSP3) scenarios are more dispersed, i.e., they show higher densities closer to the periphery, particularly in the south-east region of the city (Figure 3.3). Overall, these assumed changes in density for each scenario could inform policies and strategies for future urban infrastructure investment (e.g., the BRT line expansion) in Dar es Salaam.

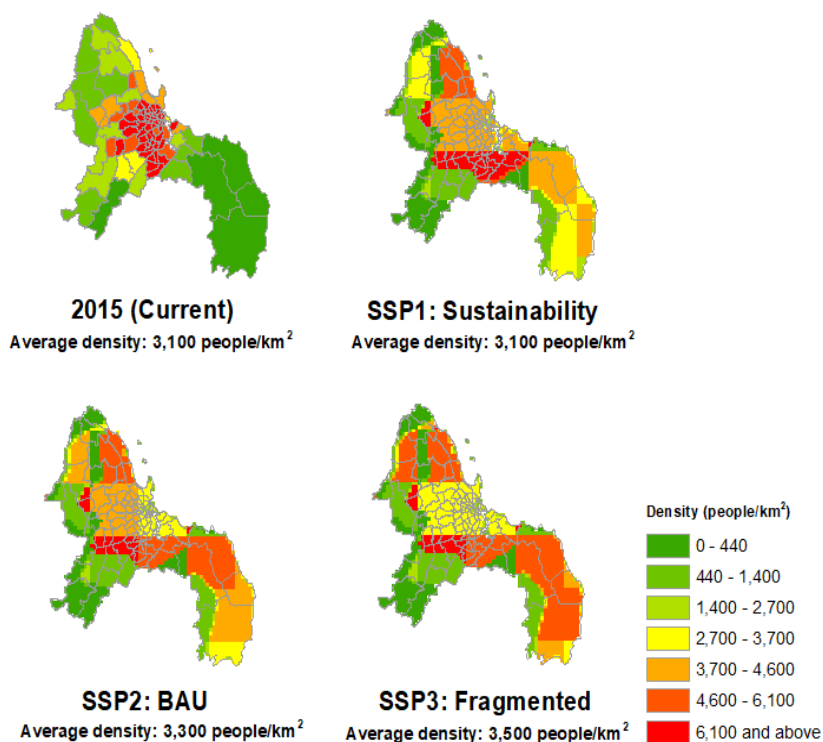


Figure 3.3. Spatial population projections for Dar es Salaam from 2015 to 2050 for SSP1 (Sustainable Growth), SSP2 (BAU Growth) and SSP3 (Fragmented Growth) narratives.

3.6.2 Dar es Salaam's current and future GHG Emissions

Across each of the SSP narratives, population growth is a major driver of rising residential energy use and GHG emissions in Dar es Salaam. In 2015, I estimate total residential emissions, i.e., from domestic households and transport activities, at 1,400 ktCO_{2e} (Table 3.4). In 2014, total energy sector emissions in Tanzania were reported at 22.26 MtCO_{2e} (WRI, 2015). Dar es Salaam accounts for approximately 10% of Tanzania's total population (World Bank, 2018); therefore, I roughly estimate the city's total energy sector emissions at 2,226 ktCO_{2e}. Emissions from domestic households and road transport count for approximately 80% of national energy sector emissions (Government of Tanzania, 2014a), which would scale to approximately 1,780 ktCO_{2e} for Dar es Salaam. Therefore, my estimate of 1,400 ktCO_{2e} for residential sector emissions in 2015 (i.e., resulting from energy uses from domestic household and transport activities) is consistent with the national dataset (within ~18%), as I do not account for energy use in the commercial and industrial sectors.

By 2050, I estimate that Dar es Salaam's total residential emissions will increase to between 25,000 ktCO_{2e} and 33,000 ktCO_{2e} (SSP1); 11,000 ktCO_{2e} and 19,000 ktCO_{2e} (SSP2); and 5,700 ktCO_{2e} and 11,000 ktCO_{2e} (SSP3). This is correlated with an increase in per capita emissions from 0.2 tCO_{2e} in 2015 to between 1.5 tCO_{2e} and 2 tCO_{2e} (SSP1); 0.7 tCO_{2e} and 1.3 tCO_{2e} (SSP2); and 0.4 tCO_{2e} and 0.9 tCO_{2e} (SSP3) (Table 3.4). My estimates represent a 4 to 24-fold increase in emissions to 2050 (relative to 2015), due to the higher urban population in 2050 and increased energy access and electricity consumption. Increased emissions from household electricity use are due to the assumed continued use of fossil fuels for electricity production, consistent with projections under Tanzania's national power plan (Government of Tanzania, 2016a). The Tanzanian government projects that natural gas and coal will continue to dominate Tanzania's electricity mix to 2040, accounting for 40% and 30%-35%, respectively of the mix (Government of Tanzania, 2016a). I apply these projections across each of my scenarios (see Section 3.9.3).

To my knowledge, there are no other projections of residential GHG emissions in individual African cities against which to compare my results. However, a growing number of regional

studies indicate an overall upward trend in GHG emissions due to increased electricity access and economic activity in the region. For example, Calvin et al. (2016) estimate that GHG emissions in the sub-Saharan Africa region will increase by 2.7 % to 3.8% per year from 2005 to 2100 (or by ~122% to ~171% by 2050). The International Energy Agency (IEA) projects slightly lower levels of growth, estimating an ~ 80% increase in GHG emissions in sub-Saharan Africa by 2040 (i.e., from 1,141 Mt CO₂ to 2,051 Mt CO₂ in 2040) under their “Current Policies” scenario (IEA, 2017b). While, van der Zwaan et al. (2018) estimate a 100% (2-fold) increase in GHG emissions in continental Africa (including North Africa) from 2015 to 2050 under their “reference scenario”, and a 30% to 40% increase by assuming (1) a 4% annual increase in the CO₂ price (“TAX” scenario) or (2) a 20% reduction in global emissions by 2050 (“CAP” scenario). In contrast, the results presented in this paper are applicable to the city rather than the regional level (as the above-mentioned regional studies combine both rural and urban data). This partially explains the variation in results, and my substantially higher estimates, given the larger concentration of energy use in cities. Moreover, my emissions scenarios are presented as a range, based on assumptions of household size, with the upper estimate reflecting the lower household size assumption (given that total household energy use is kept constant – see Methods).

3.6.3 Household Emissions

Between 80% and 90% of total residential emissions are due to household electricity use, given that 70% to 75% of the electricity mix is from natural gas and coal to 2050 (Section 3.9.3). The increasing number of households – particularly under SSP1 – is what fundamentally drives emissions from electricity production (assuming that total household energy use remains constant to 2050). Table 2 shows that electrifying all households under SSP1 and SSP2 narratives will be equivalent to electrifying an additional 3 to 11 million households in 2050 (from 1.3 million households in 2015). Moreover, the GHG intensity of electricity generation remains high even under SSP1 (remaining at ~405 gCO₂e/kWh in 2050) (Table 3.2) – a level that well exceeds the IEA target of 254 gCO₂e/kWh by 2060 (IEA, 2017a). Given that the narratives defined in this paper do not assume aggressive GHG mitigation policies – and instead,

offer baseline trajectories to 2050 – I find that the highest GHG emissions are associated with SSP1. Therefore, my findings highlight the opportunity for more aggressive GHG mitigation policies to reduce the GHG intensity of electricity generation (such as integrating renewable sources) to offset future residential emissions increases in Dar es Salaam.

The fact that an SSP3 trajectory results in the lowest residential emissions is largely due to the inequalities in access that are reinforced under this scenario, i.e., no changes in electrification from 2015, and a 25% lower population under SSP3, compared to SSP1. Under SSP1 and SSP2, Dar es Salaam will likely surpass in absolute terms, in 2050, the current (2013 – 2015) GHG emission levels of North American and European cities (C40 Cities, 2017). On a per capita basis, I find that emissions remain low compared to other global cities, assuming that total household energy use remains constant. For example, per capita emissions (from buildings and transportation) in cities such as New York, San Francisco or London (where data is more robust) were estimated at 5.7 tCO₂e/capita (in 2014), 5.5 tCO₂e/capita (in 2015), and 4.5 tCO₂e/capita (in 2013) (C40 Cities, 2017), compared with only 0.5 tCO₂e/capita to 2 tCO₂e/capita across my scenarios (Section 3.9.4).

Finally, I do not account for biogenic carbon emissions from charcoal or wood burning but illustrate biogenic emissions for each scenario in Section 3.9.6, which increase to ~2,500 ktCO₂e – 5,000 ktCO₂e under SSP3 (which assumes a continued reliance on charcoal to 2050). Ultimately, increasing charcoal use under SSP3 may degrade forests in Dar es Salaam's surrounding rural areas, given that the city already consumes nearly 70% of all charcoal produced in Tanzania, which impacts an estimated 2.8 million hectares of forests (~8.5% of Tanzania's total forest cover) (Msuya et al., 2011). The use of wood fuels (i.e., charcoal and firewood) is also linked to premature mortality and morbidity from indoor air pollution (WHO, 2012). Globally, the World Health Organization (WHO) estimates that over four million premature deaths were attributed to household air pollution from the traditional use of biomass fuels for daily cooking activities in 2012 (WHO, 2012).

3.6.4 Transport Emissions

Road transport is a smaller driver of total residential emissions compared to household emissions. Overall, total emissions from transport increase from 490 ktCO₂e (in 2015) to 600 ktCO₂e (SSP1); 900 ktCO₂e (SSP2); and 700 ktCO₂e (SSP3) in 2050 (Table 3.4). I find that annual VKT per capita does not change substantially across any of the narratives (Table 3.2), with the highest drop (only ~3%) in VKT per capita, projected under SSP3, which is due to the slightly higher population density assumed under this narrative. In addition, although population increases by three to four times by 2050, transportation emissions in all scenarios increase much more slowly. This is due primarily to improving fuel economy and changes in mode share (responsible for a 20% - 60% drop in per capita transportation emissions relative to 2015). Across all scenarios, emissions from LDV travel (with minimal ridesharing) dominate; accounting for over 80% of transport emissions (Table 3.4).

Table 3.4. Total residential emissions from household and transport activities in Dar es Salaam by activity. Results for SSP1 (Sustainability), SSP2 (BAU) and SSP3 (Fragmented) narratives for 2030 and 2050.

		Current Year – 2015	SSP1 – 2030	SSP2 – 2030	SSP3 – 2030	SSP1 – 2050	SSP2 – 2050	SSP3 – 2050
HOUSEHOLD								
▪ Electricity use	ktCO ₂ e	700	[6,400 - 7,100] ¹	[3,200 - 4,100]	[1,900 – 2,400]	[24,000 – 32,000]	[9,000 – 17,000]	[4,500 – 8,900]
▪ LPG use		60	[300 – 390]	[230 – 290]	[130 – 170]	-	[700 – 1,300]	[330 – 650]
▪ Kerosene use		10	[170 – 210]	[110 – 140]	[50 – 60]	-	[400 – 700]	[200 – 300]
▪ Charcoal use ²		120	-	[90 – 120]	[80 – 140]	-	-	[130 – 260]
▪ Wood use ²		20	20	20	20	-	-	[20 – 50]
TOTAL (Household)	ktCO₂e	910	[6,700 - 7,500]	[3,700 – 4,700]	[2,200 - 2,800]	[24,000 - 32,000]	[10,000 - 18,000]	[5,000 - 10,000]
TRANSPORT								
▪ LDV use	ktCO ₂ e	440	560	600	500	500	800	600
▪ Dala-dala use		20	40	40	30	50	60	40
▪ Bajaji or Boda use		30	50	40	40	80	80	50
▪ BRT use		-	0.4	0.2	0.1	2.0	0.7	0.3
TOTAL (Transport)	ktCO₂e	490	700	700	600	600	900	700

3.6.5 Correlation Between Total Residential Emissions, GDP and Population

By plotting population and total residential emissions on a logarithmic scale, I find that population is positively and linearly correlated with GHG emissions for all SSPs. For example, my findings show a 1% increase in total population is correlated with a 2.2% to 2.4% increase in total residential emissions for SSP1, compared to an increase of 1.7% to 2.1% for SSP2 and 1.5% to 2.2% for SSP3. Dar es Salaam's population growth is projected to result in a super-linear scaling relationship for all SSP narratives, with emissions growing at 150% to 240% faster rates than population to 2050. While some studies have shown a linear (Fragkias et al., 2013) and sub-linear (Kennedy et al., 2015) scaling relationship between city population and emissions, these correlations have been weakest in low-GDP cities (including African cities) given their low levels of access to basic infrastructure services such as electricity (Kennedy et al., 2015).

Urban growth in low-GDP cities such as Dar es Salaam requires that resource use increases to a threshold that supports sustainable living standards for residents. My results show that emissions in Dar es Salaam increase super-linearly due to improved energy access and electricity-use, and the likely high GHG-intensity of new electricity sources to 2050 (Table 3.3). Furthermore, the large growth in emissions is influenced by the potential drop in household size and assumption that traditional sources being phased out (wood and charcoal) would result in low emissions reductions due to the exclusion of biogenic CO₂ emissions from the emissions accounting.

SSP1 is associated with the highest level of economic growth (IIASA, 2015). Projections show that Tanzania is expected to experience a nearly eight-fold increase in GDP under SSP1, from USD 49 billion in 2015 to USD 400 billion in 2050, while under SSP2 and SSP3, GDP is expected to increase to USD 260 billion and USD 177 billion, respectively. These estimates are available in the SSP database (IIASA, 2015). Therefore, plotting Dar es Salaam's annual residential emissions per capita against the projected GDP per capita (using a logarithmic scale) reveals a weak (sub-linear) correlation between GDP and emissions. For example, a 1% increase in GDP per capita is correlated with an increase of 0.07% to 0.1% for SSP1 and SSP2, and 0% to 0.1% for SSP3 (Section 3.9.5). As my model does not explicitly account for the likely rise in

demand for household energy services and transportation in response to growing GDP, these correlations are (a) likely underestimated, and (b) not explicitly causal (though potentially linked via the SSP narratives).

3.6.6 Comparing Dar es Salaam's Emissions Projections with other Global South Cities

A limited number of studies project changes in residential GHG emissions in individual African cities, or at the regional level. The studies reveal an overall increasing trend in GHG emissions, though at much lower rates than projected in my paper. Like the current study, some of the studies find that electricity-based emissions play a dominant role in emissions increases (Table 5). However, accounting methods vary among the studies, where electricity emissions are calculated separately or included within a larger energy sector. For example, in their “BASE” scenario, Senatla (2011) show that electricity generation contributes more than 95% of Gauteng’s residential sector GHG emissions between 2007 and 2030. Regional projections by Stone and Wiswedel (2018) estimate a 240% increase in total urban emissions between 2012 and 2040, with transport and industry (including electricity use from industry) being the largest contributors. Similarly, studies in other regions of the Global South (i.e., Asia and Latin America) show that transportation and industry drive GHG emissions given their more advanced levels of socio-economic development. Table 3.5 compares my results with those of other studies in the literature to (1) demonstrate the large difference between my results and example results from other regions, and (2) further illustrate the need for additional GHG emissions studies in large African cities such as Dar es Salaam.

Table 3.5. Comparing GHG emissions results and main drivers of GHG emissions for selected cities or regions in the Global South.

Study	City/Region	Scope	Projection timeline	Percentage change in GHG emissions from starting year	Main driver of GHG emissions
This paper	Da es Salaam, Tanzania	Residential sector emissions	2015 – 2050	310% - 2300%	Electricity (Electricity use increases from 5GJ/HH/yr. in 2015 to 18 – 46GJ/HH/yr. in 2050 – see Table 3.2)
Senatla (2011)	Gauteng, South Africa	Residential sector emissions	2007 – 2030	~100% ¹	Electricity
Stone and Wiswedel (2018)	Sub-Saharan Africa	Total urban emissions ²	2012 - 2040	240% ¹	Transport and Industry
Godfrey and Xiao (2015)	Sub-Saharan Africa	Total urban emissions ²	2012 – 2030	61% ¹	Variable based on city income categorization (i.e., middle-income or least developed city)
Collaço et al. (2019)	São Paulo, Brazil	Total urban emissions ²	2014 – 2030	43%	Transport
Huang et al. (2019)	Guangzhou, China	Total urban emissions ²	2010 – 2030	~20%	Industry and Transport

Table Notes:

¹Projections are based on business-as-usual or baseline scenarios mentioned in each study.

²Total urban emissions refer to emissions in all urban sectors, including industrial, commercial, residential and transportation. Though, studies may use other categories in their accounting approach.

3.6.7 Implementing Aggressive GHG Mitigation Policies under SSP1

Of all regions in Africa, East Africa has the highest renewable energy potential (Lucas et al., 2017). Estimates project that Tanzania can realize the following grid mix under an SSP1 narrative by 2040: 12% hydropower, 30% solar, 21% wind, and 14% geothermal (leaving 23% for natural gas and coal combined) (Lucas et al., 2017). These estimates are consistent with regional models for electricity generation in East Africa and reflect the more rapid development of renewables (wind and solar) in rural areas. The different electricity generation scenarios are detailed in Section 3.9.3. I include an additional narrative (based on SSP1 data and assumptions; see Table 3.3) to test the impact of aggressive decarbonization of electricity, combined with low-GHG investments in transportation. Actions examined are as follows.

- 70% of the electricity generation to be from solar, wind and geothermal sources by 2050 (Lucas et al., 2017).
- The BRT system carries ~50% of all passenger trips.
- 60% of the LDV fleet is electrified by 2050, consistent with global trends (IEA, 2017a).

As shown in Section 3.9.3, generating 70% of electricity from renewable sources in 2050 would reduce the GHG intensity of the grid to ~129 gCO₂e/kWh, compared to 405 gCO₂e/kWh under SSP1 (Table 3.3). By 2050, total residential emissions would increase to 7,400 ktCO₂e – 11,000 ktCO₂e, which is ~66% lower than under my original sustainability narrative (SSP1), though still far higher than current (2015) emissions. Total residential emissions for this aggressive GHG mitigation narrative, are compared with those of the other SSP narratives in Section 3.9.7.

3.7 Research Limitations and Areas of Future Work

There are important areas of future work that are not explicitly considered in my modelling. First, the assumption that household energy use remains constant is an important limitation. This assumption is expected to underestimate demand for energy in a developing economy such as Dar es Salaam. Thus, my scenarios are likely conservative, even though they show an order of magnitude increase in GHG emissions by 2050 (ranging from 4 to 24 times the 2015 level, as

detailed in the results and conclusions). Second, if vehicle manufacturers fulfill commitments to scale up production of EVs or hydrogen fuel cell vehicles in the coming decades (IEA, 2018), and these become more broadly affordable, Tanzania may see growth in EVs by 2050 beyond the estimates assumed in my model (Table 3.2). Also, improvements in road infrastructure and public transit (with the BRT expansion) may result in induced or latent travel demand similar to trends observed in European and North American cities (Cervero, 2002; Noland and Lem, 2002), which will impact transport-related emissions. Third, my estimates exclude Scope 3 or upstream emissions from infrastructure supply chains, which could also contribute substantially to projected GHG emissions. For example, research conducted in Delhi, India estimated that up to 32% of the city's emissions was due to out-of-boundary (Scope 3) activities such as fuel processing, air travel, cement use, and food production (Chavez et al., 2012). Fourth, biogenic emissions from charcoal use are considered as carbon neutral, consistent with IPCC guidelines. However, biogenic emissions would nearly double (assuming HH size reduces to two persons per household by 2050) under SSP3 (Section 3.9.6), influencing land degradation and public health outcomes (due to indoor air pollution). Finally, as noted in my introduction and methods, future work could also incorporate emissions from other sectors, especially industry, which are expected to contribute substantially to future energy demand in the Africa region (IEA, 2019).

3.8 Conclusions and Implications for Energy Policy

In this paper, I:

- Provide the first projection of residential energy use and GHG emissions in Dar es Salaam and demonstrate the use of the SSPs at the city scale.
- Analyze the key drivers of residential energy use and GHG emissions in a large SSA city, Dar es Salaam, offering new insights for the Africa region.
- Demonstrate a method for projecting emissions in a data-poor environment.
- Show the wide uncertainty in these future projections, while also demonstrating the order of magnitude jump in emissions that can be expected in Dar es Salaam to 2050.

Key results are summarized as follows:

- **Dar es Salaam is projected to experience a 4 to 24-fold increase in residential GHG emissions by 2050.** Though Dar es Salaam's current (2015) emissions of 1,400 ktCO₂e (~ 0.2 tCO₂e/capita) are low compared to the emissions of other global cities (see S5), emissions are expected to increase to between 5,700 ktCO₂e (~ 0.5 tCO₂e/capita) and 33,000 ktCO₂e (~ 2 tCO₂e/capita by 2050). The upper estimate is as high as the recorded emissions of Global North cities such as New York, San Francisco, and London, among others.
- **Electricity access (e.g., for lighting, heating and cooling) is the largest driver of residential emissions to 2050.** Assuming that total household energy use remains constant to 2050, with the relative shares of fuel use changing for each SSP narrative (Table 3.3), I estimate that GHG emissions from electricity production (due to improved electrification and access to services) will be a major driver of future residential emissions in Dar es Salaam, i.e., accounting for between 80% and 90% of total residential emissions. This is largely due to continued reliance on fossil fuels for electricity generation. Even under SSP1 (the sustainability scenario), I project that fossil fuels will account for a dominant portion of Tanzania's electricity mix, i.e., 40% and 30% from natural gas and coal, respectively, compared to 20% and 10% from hydro and other renewables (i.e., wind and solar) (Section 3.9.3).
- **Across all scenarios, Dar es Salaam's residential emissions increase super-linearly with population size, mainly due to household electricity use.** The high GHG intensity of electricity – which remains at 405 gCO₂e/kWh for SSP1 and SSP2 – results in a 6 to 35-fold increase in total household emissions relative to 2015.
- **The sustainability scenario (SSP1) has the highest residential emissions due to increased household and transportation energy services.** This suggests a particularly acute need to promote low-GHG development in Dar es Salaam to reduce any tension between social and environmental goals.
- **Dar es Salaam's current low emissions provides an opportunity to design a low-GHG future. This will hinge on the implementation of low-GHG investments (namely, the decarbonization of electricity production) during these next stages of**

urban growth. As shown in my aggressive GHG mitigation scenario (Section 3.6.7), decarbonizing Tanzania's electricity grid through the use of renewable energy sources such as solar, wind and geothermal could reduce the city's total residential emissions by up to 66% by 2050 (SSP1). However, realizing this pathway will hinge on the development of urban policies and financing for aggressive GHG mitigation during these next stages of urban growth.

Lastly, though not explicitly explored in this paper, realizing a low-GHG transition in Dar es Salaam requires the consideration of the city's broader socio-economic development goals. Policies need to leverage synergies between energy sector investments, i.e., financing to decarbonize electricity with renewable technologies or scale-up public transport with the BRT network, and socio-economic development objectives at the city and national level. For example, given that Dar es Salaam is growing amidst other socio-economic challenges, including urban inequality, poverty and climate change, policy actions would require cross-sectoral collaboration between key stakeholders, government agencies, infrastructure service providers and the private sector to identify co-benefits between low-GHG investments and priorities in key sectors. This will be critical for ensuring that low-GHG investments improve the living standards of marginalized groups and that they benefit from the transition.

3.9 Supplementary Material

The supplementary material includes supporting methods, calculations and background data that are important in the context of this chapter (Chapter 3). Tables and figures are presented with the letter “S” in their caption title, and are referenced throughout the main body of this chapter, where relevant.

3.9.1 Global Warming Potentials for Major GHGs

Table S 3.1. 100-year Global Warming Potentials (GWP) for major GHGs. The values represented in the table are from the IPCC Fifth Assessment Report (AR5) (IPCC, 2013) and are included within the LEAP software. Note that the GWP values for non-CO₂ gases do not include climate-carbon feedbacks (climate-carbon feedbacks in response to the reference gas CO₂ are always included) (IPCC, 2013).

Effect	GWP (100 yr.: CO _{2e})
Carbon Dioxide (CO ₂)	1
Carbon Dioxide (CO ₂) (Biogenic)	0
Methane (CH ₄)	28
Nitrous Oxide (N ₂ O)	265

3.9.2 GHG Emissions Factors

Table S 3.2. Assumed emissions factors (for all major GHGs) from residential fuel uses in Dar es Salaam. Emission factors are taken from the 2006 IPCC Guidelines and are included within LEAP's Technology and Environment Database (TED) (Beers et al., 2006). Information can also be sourced from the IPCC Emissions Factor Database (EFDB) (<https://www.ipcc-nggip.iges.or.jp/EFDB/main.php>).

	CO₂	CO₂ (Biogenic)	Methane	Nitrous Oxide
	tons/TJ	tons/TJ	kg/TJ	kg/TJ
Natural Gas	72.92	-	5	0.6
Coal	92.64	-	1	1.4
LPG	72.92	-	10	0.6
Kerosene	72.55	-	10	0.6
Firewood	-	109.56	300	4
Charcoal	-	109.56	200	1
Gasoline	69.3		10	0.6
Diesel	74.1		10	0.6
Solar	-	-	-	-
Wind	-	-	-	-
Hydropower	-	-	-	-

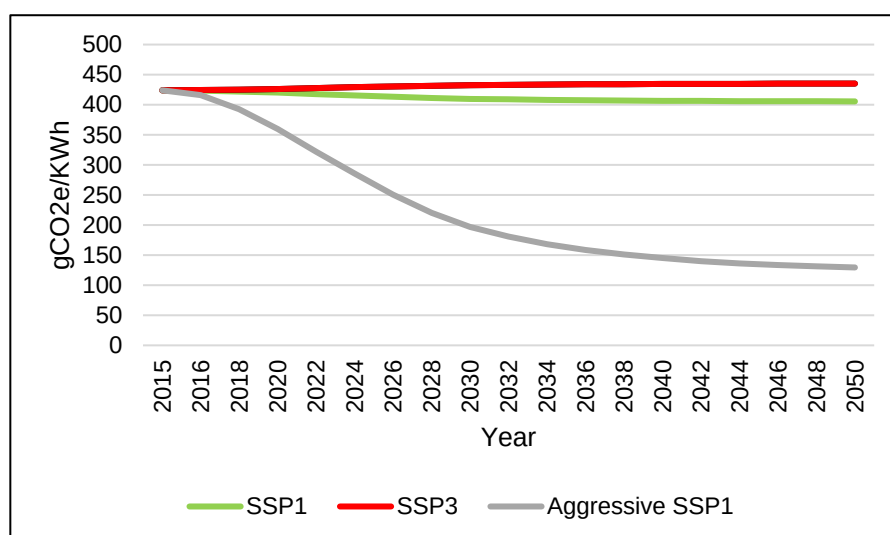
3.9.3 Electric Power Development Scenarios

Table S 3.3. Estimated electric power development scenarios for SSP1 (Sustainable Growth), SSP2 (BAU Growth) and SSP3 (Fragmented Growth) narratives for Dar es Salaam. Changes in the generation mix under SSP2 and SSP3 are based on the preferred option (“Scenario 2”) under Tanzania’s Power System Master Plan (Government of Tanzania, 2016b). Changes for SSP1 are consistent with the most aggressive scenario for renewable energy integration under the Plan (“Scenario 6”).

Fuel Source	2015 (baseline¹)	SSP1 (2050)	SSP2 (2050)	SSP3 (2050)	Aggressive Climate Action (2050)
Natural Gas	59%	40%	40%	40%	13%
Hydropower	35%	20%	20%	20%	12%
Heavy Fuel Oil	5.7%	0%	0%	0%	0%
Biomass	0.3%	0%	0%	0%	0%
Coal	0%	30%	35%	35%	10%
Solar PV	0%	5%	2.5%	2.5%	30%
Wind	0%	5%	2.5%	2.5%	21%
Geothermal	0%	0%	0%	0%	14%
Total	100%	100%	100%	100%	100%

¹Baseline year estimates for Tanzania’s generation mix are extracted from the Energy and Water Utilities Regulatory Authority (EWURA) Annual Report (for the year ended 30th June 2017) (see: <http://www.ewura.go.tz/wp-content/uploads/2018/03/Annual-Report-for-the-Year-Ended-30th-June-2017.pdf>)

Table S 3.4. GHG intensity (gCO₂e/kWh) of Dar es Salaam’s electricity grid for SSP1 (Sustainable Growth), SSP2 (BAU Growth) and SSP3 (Fragmented Growth) narratives. Note that the GHG intensity of electrification is the same for SSP2 and SSP3 (refer to Table 2 in Methods in the manuscript). Aggressive actions to decarbonize Tanzania’s electricity grid with renewable energy sources would reduce the GHG intensity of electrification by ~70% relative to SSP2 and SSP3 (i.e., to ~129 gCO₂e/kWh) in 2050.



3.9.4 Comparing Dar es Salaam's Projected Emissions with Global Cities

Table S 3.5. Total (per capita) residential emissions compared to equivalent emissions in other global cities (stationary energy and transport only). Data taken from C40 Cities GHG emissions interactive dashboard (C40 Cities, 2017). Range in emissions for SSP1 (Sustainable Growth), SSP2 (BAU Growth) and SSP3 (Fragmented Growth) corresponds to LEAP model assumptions on household size.

City	Year	Per Capita GHG Emissions (tCO ₂ e/capita)
Dar es Salaam (SSP1)	2050	[1.5 – 2]
Dar es Salaam (SSP2)		[0.7 – 1.3]
Dar es Salaam (SSP3)		[0.5 – 0.9]
New York City	2014	5.7
Bangkok	2013	5.4
London	2013	4.5
Johannesburg	2015	4.8
Durban	2013	6.5
Lagos	2013	0.9
Cape Town	2015	4.7
Chennai	2015	2.6
Tshwane	2014	5.5
Toronto	2013	5.8
Rio de Janeiro	2012	2.3
Bogotá	2012	1.5
Amman	2014	2.1
San Francisco	2015	5.5

3.9.5 Influence of GDP on Dar es Salaam's Emissions

Table S 3.6. Elasticity with respect to GDP per capita and total residential emissions per capita for SSP1 (Sustainable Growth), SSP2 (BAU Growth) and SSP3 (Fragmented Growth) narratives in Dar es Salaam.

	LN (Total Residential Emissions/capita)		
	SSP1 (Sustainable Growth)	SSP2 (BAU Growth)	SSP3 (Fragmented Growth)
LN (GDP/capita)	[0.07 – 0.1]	[0.07 -0.1]	[0 – 0.1]

3.9.6 Dar es Salaam's Projected Biogenic Emissions to 2050

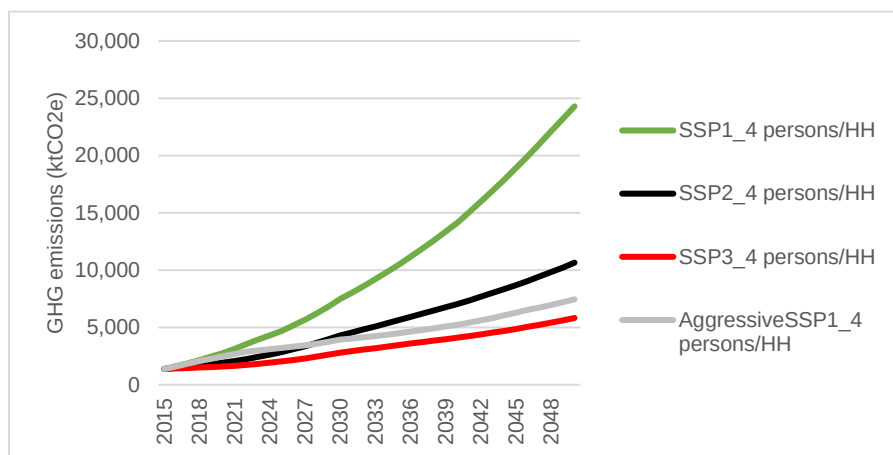
Table S 3.7. LEAP model¹ results showing biogenic CO₂ emissions from charcoal and wood burning for all SSP narratives, SSP1 (Sustainable Growth), SSP2 (BAU Growth) and SSP3 (Fragmented Growth) from 2015 to 2050.

SCENARIO	EFFECTS	2015	2016	2017	2018	2019	2020	2030	2040	2050
SSP1 (Sustainability)	CO ₂ (Biogenic) emissions assuming 4 persons/HH	2,300	2,200	2,100	2,000	1,900	1,800	-	-	-
	CO ₂ (Biogenic) emissions assuming 2 persons/HH	2,300	2,300	2,200	2,100	2,000	1,900	-	-	-
SSP2 (BAU)	CO ₂ (Biogenic) emissions assuming 4 persons/HH	2,300	2,200	2,200	2,200	2,100	2,100	1,800	1,200	-
	CO ₂ (Biogenic) emissions assuming 2 persons/HH	2,300	2,300	2,300	2,300	2,300	2,300	2,300	1,900	-
SSP3 (Fragmented)	CO ₂ (Biogenic) emissions assuming 4 persons/HH	2,300	2,200	2,200	2,200	2,100	2,000	2,200	2,500	2,500
	CO ₂ (Biogenic) emissions assuming 2 persons/HH	2,300	2,300	2,300	2,300	2,300	2,300	2,800	3,800	5,000

¹LEAP does not include biogenic CO₂ emissions from charcoal and wood burning. Consistent with IPCC guidelines, bioenergy is considered as “carbon neutral” given the assumption that emissions from these sources are already accounted for in the Agriculture, Forestry and Other Land-Use (AFOLU) sector (IPCC, 2015). However, the table presents the scale of biogenic CO₂ emissions increase for each SSP narrative to offer broader context on the additional emissions that would arise from charcoal and wood burning in Dar es Salaam. (Note: Values are rounded to 2 significant figures.)

3.9.7 Assuming “Aggressive GHG Mitigation” Under SSP1

[4 persons/HH]



[2 persons/HH]

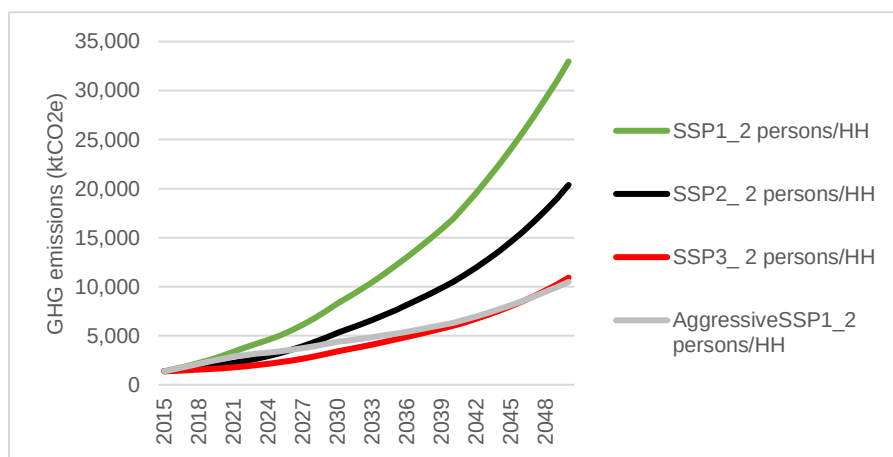


Figure S 3.1. Emissions from aggressive GHG mitigation policies under SSP1. I find that this scenario would result in up to 70% reduction in emissions relative to the initial SSP1 baseline narrative. The values presented in the two figures below show the variation in emissions, based on the different assumptions for household size.

3.9.8 Dar es Salaam's BRT

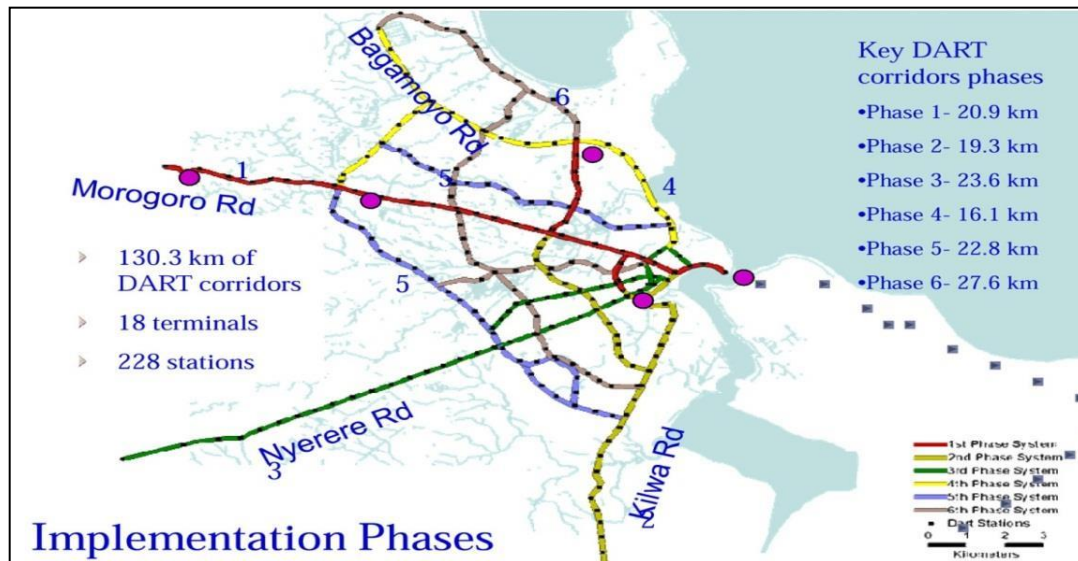


Figure S 3.2. Implementation Phases of Dar es Salaam's BRT. Source of image: (Government of Tanzania, n.d.)

The BRT, also known as the Dar es Salaam Bus Rapid Transit (DART) system, was developed to reduce overall congestion and improve public transport conditions for city residents. Overall, Phase 1 (currently in operation as of 2016) is accessible along ~21 kilometers of road, with 27 bus stations and five terminals (Government of Tanzania, 2017a; IGC, 2018). Plans to expand the BRT corridor up to six phases are currently underway, with main lines running along the city's major roads (Morogoro, Bagamoyo, Nyerere and Kilwa). This would increase BRT accessibility to the majority of residents. By the end of phase 6, the BRT is expected to cover a total network of ~133km in BRT corridors, 18 terminals and 228 stations (Government of Tanzania, 2017a).

3.9.9 Quantifying GHG emissions in LEAP

Total energy consumption of end-use sectors (i.e. household or passenger transport) correspond with different activity levels and energy intensities. Total GHG emissions are then calculated from these end-uses based on the following equation (Heaps, 2008; Ouedraogo, 2017).

$$CEE = \sum_i \sum_j \sum_k AL_{k,j,i} \times EI_{k,j,i} \times EF_{k,j,i}$$

where **CEE** is the total GHG emissions from end-uses; **AL** is the activity level; **EI** is the energy intensity; **EF** is the GHG emissions factor of fuel type **k** through device or vehicle **j** in sector **i**; **k** is the fuel type; **i** is the sector; and **j** is the end device or vehicle.

The activity level for the household sector is the total number of households in Dar es Salaam within a given year, whereas in the transport sector, the activity level is the total annual passenger kilometers travelled. The activity level also corresponds with a series of individual activities (typically defined by a set of shares or penetrations) and their end-uses. For example, an individual activity defined by my model would be the share of electrified households in Dar es Salaam who use LPG for cooking needs. Similarly, another activity is the “share of non-electrified households who use LPG for cooking”.

Energy intensities (i.e. energy consumption per device or vehicle) for households are calculated based on fuels used for both cooking or lighting needs i.e. electricity, charcoal, LPG, wood and kerosene. While, in transportation, energy intensities correspond with vehicle fuel type (i.e. diesel and gasoline) by mode i.e. private vehicle, mini-bus (“dala-dala”), motor-cycle or tricycle (“boda” or “bajaji”), or the BRT.

Finally, given that I also account for Scope 2 emissions from household activities (which include energy uses from electricity generation) GHG emissions from energy conversion (i.e. at the power plant) is shown in the following equation (Heaps, 2008; Ouedraogo, 2017).

$$CET = \sum_q \sum_t \sum_p ETP_{q,t} \times e_{p,q,t} \times EF_{p,q,t}$$

where **CET** is the total GHG emissions from processing and conversion; **ETP** is the energy transformation product; **e** is the energy consumption of fuel type **p** to produce unit secondary fuel type **t** in equipment **q**; **EF** is the emission factor of producing secondary energy **t** from primary energy **p** through equipment **q**.

Therefore, total GHG emissions are calculated based on emissions from end uses and energy conversion (i.e., the sum of the two above-mentioned equations).

3.10 References for Chapter 3

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Chapter 4

Does Location Matter? Investigating the Spatial and Socio-Economic Drivers of Residential Energy Use in Dar es Salaam

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4.1 Abstract

Africa is set to become a key contributor to global energy demand. Urban growth and the energy use of city residents will drive much of the region's changing energy picture. However, few studies have assessed residential energy use among African cities, and the heterogeneity in energy use at the sub-city scale. I use the case of Dar es Salaam, which is among Africa's fastest-growing cities, and to my knowledge, present the first disaggregated estimates of residential energy use at the ward level. I show three main findings. First, I find a statistically significant difference in mean residential energy use among the surveyed wards, which group into four clusters representing distinct levels of household and transport-related energy use. These results show that mean residential energy use (the sum of household and transport-related energy use) is not always correlated with the socio-economic or spatial characteristics of wards – e.g., Msasani (high-income, formal ward) showed similar residential energy use as Keko (low-income, informal ward). Second, I show differences in energy use and fuel switching that occur between low-income and high-income wards: wood fuel (i.e., charcoal) is a majority contributor to

residential energy use in low-income wards (Buguruni, Keko and Manzese), compared to gas, electricity and transport oils in high-income wards (Msasani and Kawe). Finally, regression models indicate that ward density has a statistically significant effect on transport-related energy use, while fuel stacking and proxies for household wealth have a statistically significant effect on household-related energy use. To conclude, I recommend that policymakers account for ward level differences in residential energy use when crafting energy sector strategies for Dar es Salaam (e.g., electrification, energy-efficient cooking, or public transportation initiatives). Policymakers may also anticipate possible convergence towards higher levels of energy use and a shift towards modern fuels, as wards develop socio-economically over time.

4.2 Introduction

Current literature presents few examples of the energy use of African cities at the sub-city scale (i.e., the settlement or ward level). The Africa region accounts for only 5% of global energy demand (IEA, 2014), and 3.7% of (2018) global energy-related greenhouse gas (GHG) emissions (IEA, 2019a). However, urbanization and economic activity in the region could increase future energy use and GHG emissions to reach as high as 20% to 23% of global emissions in 2100 (Calvin et al., 2016; Lucas et al., 2015), equivalent to the current (2016) emissions of the United States and Canada (IEA, 2017b). Despite these staggering trends, most regional studies only quantify energy and/or material use (e.g., water and waste consumption) at the aggregated city scale (e.g., (Currie & Musango, 2017; Currie et al., 2015; Hoekman & von Blottnitz, 2017; Olaniyan et al., 2018)), and generally, studies have drawn two main conclusions. First, that estimated national or city-level electricity use is strikingly low vis-à-vis other developed countries, e.g., between 90 and 135kWh per household per month in Nigeria (in 2017) (Olaniyan et al., 2018), compared to 909 kWh per household per month in the United States (in 2018) (EIA, 2018). Second, there are wide disparities in energy use within and between African cities, e.g., between rural and urban areas (Olaniyan et al., 2018), or high-GDP and low-GDP cities (Currie et al., 2015). Relatedly, some studies that have disaggregated energy use at the sub-city scale (e.g., (Hughes-Cromwick, 1985; Chowdhury et al., 2019; Smit et al., 2019; Strydom et al.,

2020)) highlight the inequalities in infrastructure access for different population groups, and the low energy use of informal settlements (which are sometimes the most deprived settlements though constitute most urban residential land-use). However, how energy use varies across different settlement types (e.g., between formal or informal settlements) is largely unknown, and could result in policy measures that do not consider the local reality, i.e., the heterogeneity that may exist between settlements (and populations) of differing socio-economic status, or spatial location within an individual city.

This study broadly addresses these existing data gaps. Specifically, I elucidate the spatial and socio-economic drivers of residential energy use that could be considered in the advancement of energy policies in Dar es Salaam, which is among the fastest growing cities in Africa, alongside Lagos, Lusaka and Accra, among others (UN, 2018). If policymakers understand the heterogeneity in energy uses that exists between populations, this could inform different approaches to implementing policy visions at the settlement level (e.g., in formal versus informal settlements). Current literature on urban infrastructure in Africa has shown that centralized structures of services delivery (e.g., via state-led utilities) often do not result in equal access to services for the poor. For example, studies have highlighted inequalities in the delivery of water and sanitation services in Dar es Salaam (Monstadt and Schramm, 2017) and Kampala (Uganda) (Lawhon et al., 2018), or electrification in Accra (Ghana) (Silver, 2015), Cotonou (Benin) and Ibadan (Nigeria) (Rateau and Jaglin, 2020). Authors of these studies have also highlighted the broader societal and urban governance factors that influence service delivery and the need to integrate “hybrid” or “decentralized” solutions to ensure more equitable infrastructure configurations, e.g., engaging informal service providers or local communities in the provision of services. However, authors of these studies – particularly energy-specific studies of Silver (2015) and Rateau and Jaglin (2020) – did not quantify differences in electricity or energy use among communities (e.g., kilowatt hours (kWh) of electricity, or gigajoules of energy per household) as I do in this chapter. Specifically, I show that considering the intra-community differences in energy access could support the implementation of differentiated energy sector strategies that account for local community needs and their different lived experiences.

I assume that rates of urbanization and economic growth projected for Dar es Salaam (UN, 2018) could play a role in driving future energy convergence, as historically observed in China (Fan et al., 2017) and India (Chaturvedi et al., 2014), for example. Therefore, by quantifying energy uses at the settlement level, policymakers could anticipate future changes and differences in demand, e.g., if lower consuming settlements catch up to the per capita rates of higher consuming ones. Similar trends could unfold at the aggregated city level in Dar es Salaam, and the sub-city (ward) data presented in this study could predicate possible changes at the settlement level. For example, a more rapid increase in energy use could be expected among low-income wards that become more developed over time, compared to a gradual and slower increase in energy use among high-income wards whose consumption is evolving, e.g., from wood fuels (charcoal and firewood) to modern fuels (electricity or transport oils).

My study focuses on residential energy use, which accounts for most of Tanzania's energy demand (70% as of 2017, (IEA, 2019c)). Specifically, I focus on (1) total residential energy use, (2) cooking fuels, and (3) transport energy. There are several factors that are likely to drive large changes in these specific categories. To begin, Tanzania has an ambitious policy vision to develop their energy sector, including access to reliable, affordable, and efficient energy for all (Government of Tanzania, 2015a, 2015b). Improved access will drive higher residential energy use and GHG emissions (Luo et al., 2020), which would require clear understanding of ward level differences to manage these possible changes. The country's Action Agenda (Government of Tanzania, 2015b) promotes universal access to modern cooking solutions by 2030, i.e., replacing wood fuels (charcoal and firewood) with electricity or liquefied petroleum gas ("gas") to promote the sustainable use of wood fuels and alleviate their associated health burdens (e.g., from indoor air pollution). In 2018, wood fuels accounted for 93% of energy used for cooking by Tanzanian households. This can be compared to only 5% from gas, 1% from kerosene and 1% from electricity (IEA, 2019a).

In the transport sector, rising private vehicle travel and urban wealth will contribute to higher transport energy in Dar es Salaam (Chapter 3). As of 2015, the local "dala-dala" minibus accounted for most passenger trips given its high use among the city's low-income population (though is characterized by poor service quality). Improvements to the public transport network

are currently underway, largely through the city's Bus Rapid Transit (BRT) system, to attract broader use from different population segments (Government of Tanzania, 2017a).

To realize my stated research goals, I examine:

- (1) Differences in residential energy use (i.e., the combined energy use from household and transport-related activities) between selected formal, informal and mixed wards of Dar es Salaam.
- (2) The statistical relationship between: (1) ward type and residential energy use; (2) cooking fuel choice and household-related energy use; (3) public transport use and transport-related energy use; and (4) other spatial and socio-economic factors (e.g., land-use, ward density and wealth) and household-related and transport-related energy use, respectively.

This work builds on prior studies (e.g., (Lee, 2013; Sun et al., 2014; Mensah and Adu, 2015; Sakah et al., 2019)) that have similarly examined the statistical relationship between household energy use and/or cooking fuel choices and their socio-economic and spatial profiles (e.g., expenditure, income, education, location and infrastructure access) in other countries. For example, Lee (2013) examined the effect of household expenditures, location (rural or urban area), education and water access levels on household electricity, kerosene, and wood fuel use in Uganda. In Ghana, Mensah and Adu employed similar methods as Lee (2013) but did not include electricity use or water access as variables in their analysis (Mensah and Adu, 2015).

In the case of Dar es Salaam, available estimates for residential energy use (e.g., electricity, gas, or charcoal use) (e.g. (Government of Tanzania, 2017b), (NBS, 2013)) are mostly aggregated at the national or city level. For example, electricity use estimates for Dar es Salaam in the 2017 *Energy Access Situation Report* (Government of Tanzania, 2017b) encompass the city's five major districts (data at the ward level are not provided). To my knowledge, studies have not disaggregated energy uses across different settlements in Dar es Salaam or other Tanzanian cities, though data are available at the settlement level for other African cities, e.g., electricity use patterns in Johannesburg (South Africa) and Ndola (Zambia) (Roy Chowdhury et al., 2019), and household energy use in Nairobi (Kenya) (Hughes-Cromwick, 1985). Therefore, my current study offers possible insights to the different energy behaviours of Dar es Salaam residents, and

the spatial and socio-economic drivers that could be considered when implementing energy policies in the city.

4.3 Methods

4.3.1 Study Region

Dar es Salaam is the largest city and one of 31 administrative regions in Tanzania. It is a port city, located along Africa's eastern coast, and a major hub for international trade. The metropolitan area, i.e., the "Dar es Salaam region", has a population of 6.1 million (as of 2018), which accounts for 32% of Tanzania's total urban population (World Bank, 2018). The city's population is estimated to more than double in size by the year 2050, i.e., between 15 and 16 million in 2050 (Chapter 3). At the sub-city scale, the Dar es Salaam region is sub-divided into 90 wards (NBS, 2013), situated within larger districts (of which there are five: Ilala, Kinondoni, Temeke, Ubungo and Kigamboni). Wards consist of "informal", "formal" or "mixed" settlements, containing various low, middle and high-income households. The term "informal" refers to the nature of land tenure where land and home ownership is organized privately between individuals without regulation by local or national government (Kironde, 2000; Lupala, 2002). Such activities are regulated by government in formal settlements. Mixed settlements contain both formal and informal areas within the same sub-city boundary.

Among other socio-economic factors, I consider the influence of the city's Bus Rapid Transit (BRT) system that began operation (i.e., phase 1) in 2016. The system has received widespread recognition globally (e.g., (ITDP, 2017; World Bank, 2017)) given the few examples of successful BRT operations in African cities (not including Johannesburg and Cape Town). The phase 1 line operates along 21km of road, carries 160,000 passengers a day on average (ITDP, 2017), and traverses the north-west segment of the city, i.e., Kimara to Kivukoni along Morogoro Road, shown in Figure 4.1. The completed system is expected to cover 137 kilometers of road, to be built in 6 sequential phases; as of June 2020, construction and tendering for phases 2 and 3 (44 km) is underway (DART, 2020).

4.3.2 Methods Overview

At a high level, I:

- Conducted fieldwork (with the support of a field team) to collect household level energy use data across eight wards in Dar es Salaam.
- Estimated mean residential energy use (i.e., in GJ/HH/year), including disaggregated household- and transport-related energy use, in the surveyed wards.
- Employed Analysis of Variance (ANOVA) to test the effect of ward type and cooking fuel choice on mean residential and household-related energy use, respectively.
- Employed multivariate Ordinary Least Squares (OLS) and Tobit regressions (coupled with Principal Component Analysis) to model the statistical relationship between cooking fuel choice, public transport use, and other spatial and socio-economic factors on household-related and transport-related energy use, respectively.

4.3.3 Description of Fieldwork

Fieldwork was conducted in Dar es Salaam between August and November 2018. A prior fieldtrip was organized in 2017 for early pilot testing of the survey/questionnaire. I recruited a 10-person field team through in-person interviews in August 2018 (more details in the supplementary material, Section 4.11.1). In total, surveys were completed for 1,363 households across a socio-economically and spatially diverse set of wards (8 wards in total, which represented 9% of all wards in Dar es Salaam). Ethics clearances received for the study are outlined in the supplementary material, Section 4.11.2. Table 4.1 summarizes key socio-economic, spatial, and sampling data for the surveyed wards. Of the total number of households surveyed, most were in informal and mixed settlements to reflect the higher proportion of the population in Dar es Salaam living in these types of settlements.

Table 4.1. Socio-economic, spatial characteristics and sampling data for surveyed wards in the Dar es Salaam region.

Ward ¹	Type	Density (persons/km) ⁵	Economic bracket ⁶	Mean distance of surveyed households to closest BRT stop (km) ⁷	Mean distance of surveyed households to city center (km) ⁷	Total number of households surveyed
Msasani	Formal ²	4,402	High-income	3.4	7.3	94
Sinza	Formal	12,151	Middle-income	1.1	8.8	111
Buguruni	Informal ³	20,460	Low-income	3.1	6.0	242
Keko	Informal	24,179	Low-income	1.6	3.5	113
Manzese	Informal	38,496	Low-income	0.2	7.8	235
Kawe	Mixed ⁴	4,336	High-income	7.9	13.9	178
Kimara	Mixed ¹	5,569	Middle-income	1.7	14.9	221
Mwananyamala	Mixed ¹	20,409	Low-income	0.6	5.8	169

Table Notes:

¹ Pilot tests were also conducted in Kijitonyama ward, not included in this table.

² A formal settlement contains housing and land obtained through national and local government (Kironde, 2000).

³ An informal settlement contains housing and land obtained individual (informal) means (Kironde, 2000).

⁴ A mixed ward contains both formal and informal settlements, with some settlements currently being formalized.

⁵ Ward density was determined by the authors in ArcGIS using population data provided in (NBS, 2013).

⁶ At the time of the survey, ward level income data was not available for Dar es Salaam. Due to this, income categories were determined based on anecdotal evidence from the fieldwork, which drew on the perspectives of local researchers and officials with demonstrated knowledge of ward level socio-economic differences.

⁷ Mean distances were calculated in ArcGIS based on locational (GPS) coordinates of each surveyed household.

4.4 Sample Design and Survey Method

Households were selected through stratified sampling where the total number of households surveyed in each ward correlated with the total ward population, i.e., more populous wards accounted for a larger portion of the sample. Households were also selected based on their distance (i.e., near or far) from the city center (“Kivukoni”), and the closest stop along the BRT line (Table 4.1). Figure 4.1 presents the spatial distribution of the surveyed wards relative to the city center and the BRT line.

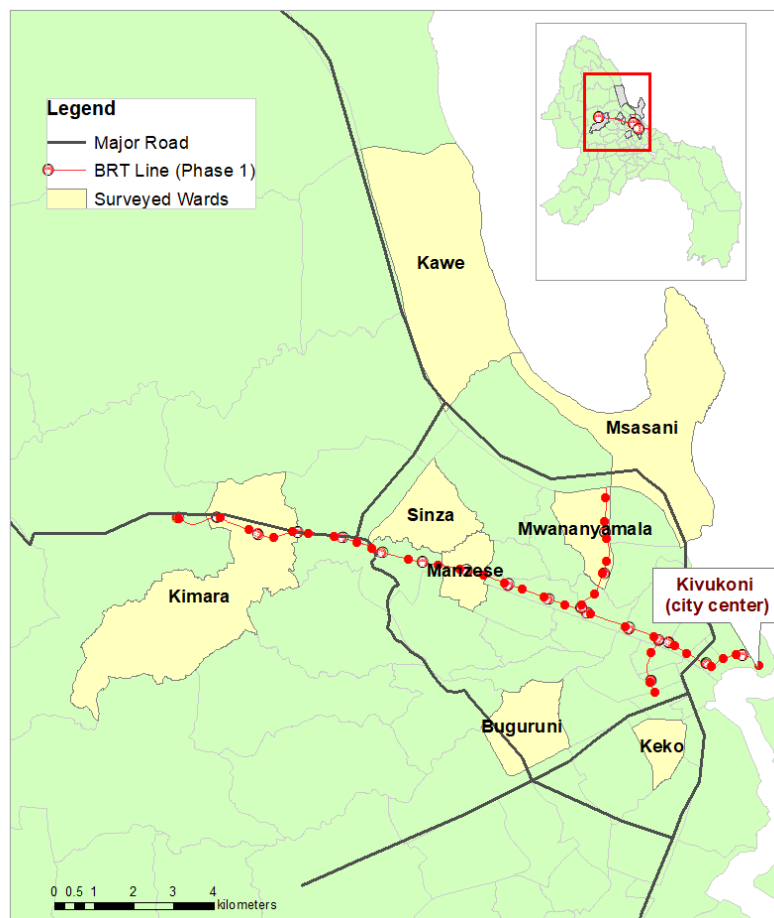


Figure 4.1. Map showing surveyed wards in the Dar es Salaam region relative to the city center and BRT line (Phase 1).

Households were randomly sampled via door-to-door street canvassing. Street Chairmen, known locally as “Mwenyekiti wa Mtaa” (who were elected and trusted community leaders), introduced

the field team to interview participants prior to each interview. Interview participants included consenting adults (household members) with the capacity to respond to the survey questions. Only consenting individuals who could report on household member composition, energy needs and activities, and weekly commuting patterns, were selected.

The survey was administered digitally via android tablets that were configured with locational features (GPS) for storing geo-spatial data at the ward level, i.e., the location of each surveyed household, street names, and travel destinations. Survey questions were sectioned as follows (see Appendix A for full question set):

- Demographic and household information, including the number of household members, education levels and living arrangements (e.g., single, or shared household).
- Locational data, e.g., household location, trip routes and destinations.
- Fuel consumption estimates, e.g., charcoal, gas and electricity use.
- Cooking fuel choices e.g., with charcoal, gas and/or electric stove.
- Electric appliance ownership e.g., washing machine, refrigerator, television, among other regular household appliances.
- Travel mode choices e.g., private vehicle (including Uber or Taxi) and/or public modes such as the BRT, “dala-dala” (local minibus), “boda-boda” (motorcycle), “bajaji” (tricycle). The dala-dala, boda and bajaji operate within an informal network where transit routes and use fares are not institutionalized at the city level (Government of Tanzania, 2017a).

To mitigate possible risks associated with identification, each household was assigned a random household identification number (names and genders of participants were not collected). Only participants that completed the informed consent were interviewed. Finally, if a household was unoccupied at the time of a visit or if a member declined to participate in the survey, then the next closest household was surveyed. In total, 1,679 households were visited, with results as follows: 1,363 (81%) completed the survey, 210 (13%) were not at home, and 106 (6%) refused to participate.

4.5 Approach to Quantifying Residential Energy Use

I use the term “residential energy use” to refer to the sum of energy uses from both household- and transport-related activities. Energy use was estimated based on fuel use and travel behaviors reported by surveyed households.

For household-related activities, this included:

- Kilowatt hours of electricity used per household per year (kWh/HH/year) based on pre-paid electricity receipts. In Tanzania, households can purchase pre-paid electricity units to provide electricity for daily, weekly, or monthly needs.
- Kilograms of charcoal used for cooking per household per year (kg/HH/year) based on number of bag(s) used per day, week, or month (in kg).
- Kilograms of gas used for cooking per household per year (kg/HH/year) based on number of cylinder(s) used per day, week, or month (in kg).

I broadly describe household charcoal and firewood use as “wood fuel” use. However, I do not include firewood use in my estimates for residential energy use. Early pilot tests (in 2017) showed that interview participants were unable to estimate their consumption (i.e., in kg) of kerosene and firewood for cooking. My previous study, which aggregated energy uses at the city level for the year 2015 (Chapter 4) concluded that kerosene accounted for only 2% of Dar es Salaam’s residential energy use. Similarly, the International Energy Agency (IEA) estimates that kerosene accounted for only 1% of energy use at the national (Tanzania) level for 2018 (IEA, 2019a). My prior work reported higher firewood use (32%), but based on this survey, only 8% reported using firewood for cooking (even though households did not report on specific amounts, i.e., kg of firewood). For these reasons, I concluded that excluding both kerosene and firewood from my final estimates would not have a major impact on my results. However, their omission is an important limitation that could be considered in future studies in Dar es Salaam and other African cities.

For transport-related activities, interviewed participants reported on the travel behavior of each household member during an average working day during the week. This included reporting on

“start” and “end” points for each trip (including transfer points), travel modes used, and common routes taken (e.g., names of major roads). The trip data was later converted to latitude and longitude coordinates based on locational (GPS) data collected from the survey. Travel routes and trip distances for each household member were estimated and digitized in ArcGIS by the authors and the support of a GIS consultant. Note, I did not incorporate questions on vehicle fuel use (e.g., diesel or gasoline) and efficiency in the survey as respondents were unable to answer these questions during pilot tests. Vehicle fuel use and efficiencies (by travel mode) were determined based on World Bank (2016) data.

The supplementary material (Section 4.11.3) outlines equations used to estimate energy use from household- and transport-related activities (in GJ/HH/year), respectively; as well as the fuel conversion factors applied in this study.

4.6 Statistical Methods

4.6.1 Analysis of Variance

One-way Analysis of Variance (ANOVA) tests were applied to determine whether there was a statistically significant effect of (1) ward type on mean residential energy use and (2) cooking fuel choice on household-related energy use. Where there was a statistically significant effect, post-hoc tests, i.e., Fisher’s Least Significant Differences (LSD) test, were applied to determine which sample groups were statistically different.

4.6.2 Principal Component Analysis

Due to privacy concerns raised during initial pilot tests, the field team was unable to collect data on household income (i.e., average earnings per household). Instead, I relied on other indicators (or proxies) for wealth, based on data collected from the survey (Table 4.2). I employed Principal Component Analysis (PCA) to reduce these indicators to a smaller set of principal components (or “PCs”), while still retaining much of the variation in the original variables (Jolliffe, 2002). Landgraf’s “Logistic PCA” package in R was employed given the binary nature of the data, i.e.,

households responded “yes” or “no” to questions related to household appliance ownership or education. The two PCs (i.e., PC1 and PC2) explained 80.3% of the deviance in the original model ((Jolliffe, 2002) recommends a cut-off of between 70% and 90%).

Table 4.2. Principal Component (PC) loadings on variables associated with household wealth. Household responses were binary in nature, where households indicated yes (coded as 1), or “no” (coded as 0) in their responses. PC loadings are compared with the proportion of the surveyed households that responded yes, for the entire sample, and for households in Msasani and Kawe (the two high-income wards).

Variable	PC1	PC2	Proportion (%) of all surveyed households	Proportion (%) of surveyed households in high-income wards (Msasani and Kawe) ¹
1. Air-conditioning unit	0.51	-0.35	11%	53%
2. Refrigerator	0.45	0.21	60%	94%
3. Washing machine	0.32	0.19	9%	41%
4. Electric stove	0.24	0.46	12%	46%
5. Television	0.16	0.57	80%	94%
6. Private vehicle	0.41	0.08	25%	56%
7. Tertiary education	0.41	-0.51	39%	84%
Cumulative percentage (%) of total deviance explained	80.3%			

Table Notes:

¹Full summary statistics for other wards in the supplementary material, Section 4.11.6.

I interpret the PC loadings, shown in Table 4.2, in two ways. Firstly, PC1 provides a general picture of household wealth, and households could be sub-divided based on their position along the PC1 dimension. High loadings on PC1 (e.g., owning an air-conditioning unit, refrigerator) could indicate wealthier households (Table 4.2). For example, households in Msasani and Kawe (high-income wards) reported higher levels of appliance use compared to levels in all surveyed wards and had higher PC1 scores on average (boxplots showing PC scores by ward are shown in Section 4.11.4). I am unable to clearly interpret PC2; note that it may not be strictly related to wealth, given that Msasani and Kawe do not have higher PC2 scores compared to other wards (Table 4.2). However, high PC2 loadings for convenience appliances (e.g., television, electric stove) could indicate a household's predisposition for these common technologies.

4.6.3 Multivariate Regression Models

I employed multivariate Ordinary Least Squares (OLS) and Tobit (or censored) regressions to model the influence of socio-economic and spatial variables on household-related and transport-related energy use. The supplementary material (Section 4.11.5) provides a description of data variables used in both models. Similar to the study of Lee (2013), I conducted OLS and Tobit regressions to model variable relationships. Household PC1 and PC2 scores were also applied as independent proxy variables in an attempt to control for household wealth, which resulted in improved regression results, i.e., coefficients for PC1 and PC2 showed a statistically significant and positive correlation with household-related and transport-related energy use (details in Table 4.4 and Table 4.5). Comparative results with the seven original variables applied as independent proxy variables are shown in the supplementary material (Section 4.11.9), where, in some cases, there was no discernable relationship between the original variables and household-related or transport-related energy use, respectively.

The Tobit model was left-censored at zero, which allowed for the regression to inherently account for data clustering at zero (as some households reported zero household-related or transport-related energy use i.e., they did not cook at home or use motorized transport). The OLS model did not account for this data clustering (i.e., the regression incorporated households with

zero values within the regression), which may have influenced biases in results. By applying both models, I account for these model differences and comparatively assess findings.

4.7 Results and Discussion

4.7.1 Differences in Residential Energy Use across the Surveyed Wards

Boxplots presented in Figure 4.2 show the variation in energy use at the ward level from all activities (i.e., residential energy use) and separated household-related and transport-related activities, respectively. Across all surveyed wards, I estimate mean residential energy use (Plot A) at 38 GJ/HH, ranging from approximately 30 GJ/HH at the lower-bound (Buguruni, Kimarra, Manzese and Sinza) to 50 GJ/HH at the upper-bound (Kawe, Keko and Msasani). Disaggregated data summaries for all wards are indicated in the supplementary material, Section 4.11.6. Other notable information is that: (1) all wards have some households with near-zero residential energy use; (2) all wards have substantial overlap in their ranges; (3) high-income wards (Msasani and Kawe) have larger interquartile ranges, where the upper quartile is even more skewed compared to other wards and relative to their medians; and (4) all wards exhibit a right skew, i.e., means are greater than medians, and there are several high-end outliers.

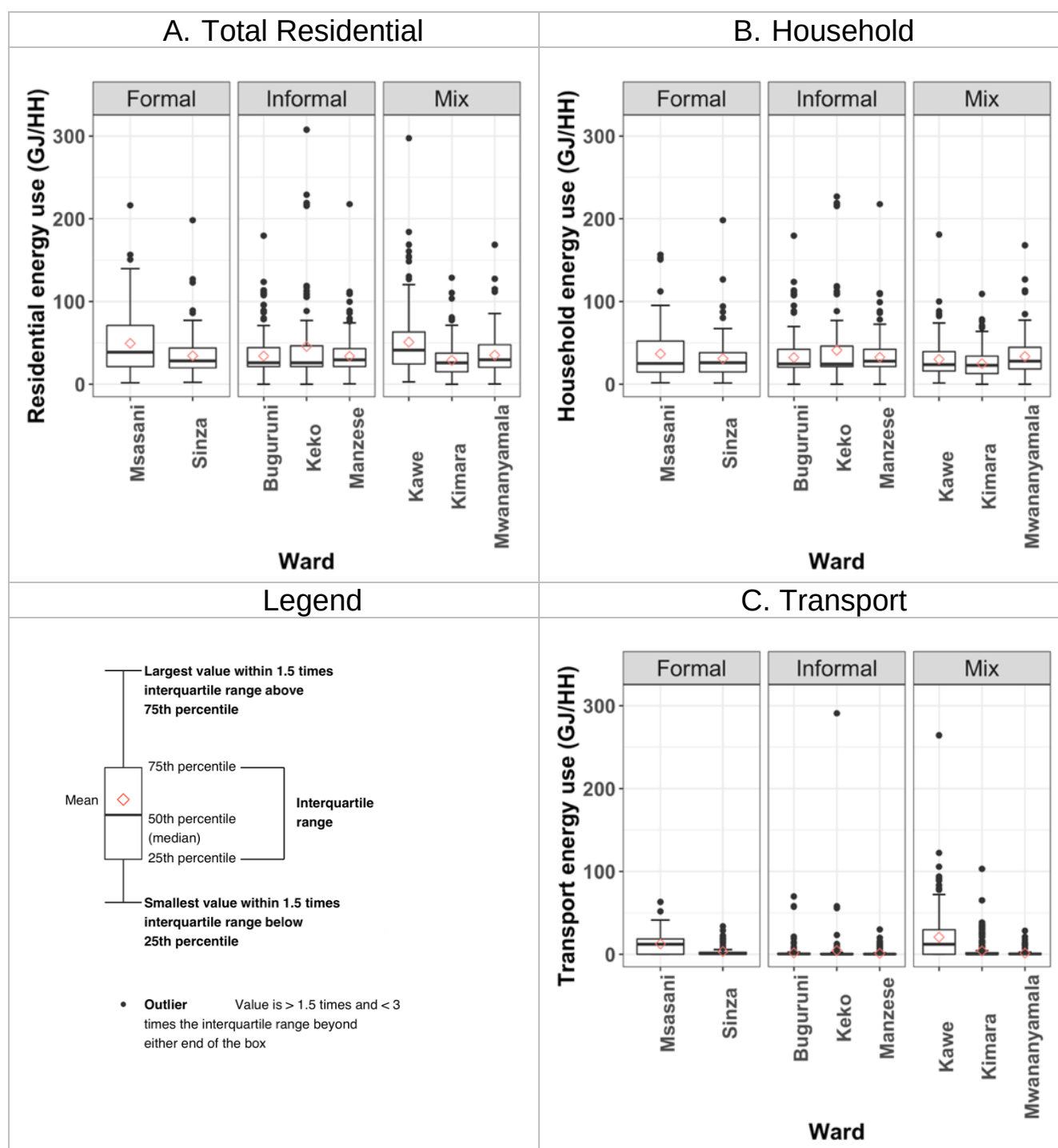


Figure 4.2. Variation in energy use across the sampled wards. Residential energy use from both household-related and transport-related activities (A). Energy use from household-related activities (B). Energy use from transport-related activities (C).

Mean household-related energy use ranges between 20 GJ/HH (Kimara) and 40 GJ/HH (Kawe and Keko) across the surveyed wards (data ranges are wide and right-skewed, i.e., mean values are higher than medians, and there are several high-end outliers) (Figure 4.2, Plot B). My estimates are higher than averages reported for Global South cities in Asia (e.g., 21 GJ/HH in Jakarta (Surahman & Kubota, 2018) and 9.6 GJ/HH in Nepal (Shahi et al., 2020)), but lower than those in North America (e.g., 105 GJ/HH in Canada (Statistics Canada, 2011), and 101 GJ/HH in the United States (Nakagami et al., 2008)). Notably, however, I find that some households use energy at rates comparable to or exceeding the averages in North America (see outliers in Figure 2, Plot B). Across the surveyed wards, between 1% (Kimara) and 10% (Kawe) of households already exceed the average household energy use in the United States.

Data ranges for transport-related energy use are narrower (Figure 4.2, Plot C), due to several households having an absence of motorized travel (i.e., by private vehicle or public transport). Excluding Kawe and Msasani, I estimate near-zero values for mean and median transport-related energy use (and zero interquartile ranges). However, wide disparities are present within and between wards (shown as outliers, Figure 2, Plot C). Similar trends exist when the cumulative share of transport-related energy use is plotted against the cumulative share of all surveyed households (supplementary material, Section 4.11.11), where consumption is unequally distributed towards a few (possibly wealthier) households.

4.7.3 Effect of Ward Type on Residential Energy Use

ANOVA test results show a statistically significant effect of ward type on mean residential energy use [$F(7, 1355) = 9.4, p < 0.01$], i.e., there is a statistically significant difference in mean residential energy use between wards. I cluster wards into four categories of residential energy use (“high”, “medium-high”, “medium”, and “medium-low” consumers, shown in Figure 3) based on results from post-hoc tests (i.e., Fisher’s LSD, which confirmed where the statistical differences occurred between groups). In some cases, I find no statistically significant difference in mean residential energy use between formal and informal wards, or high-income and low-income wards (illustrated in Figure 4.3), e.g., Msasani (high-income, formal ward) and Keko (low-income, informal ward) both have a “medium-high” level of energy use relative to other wards.

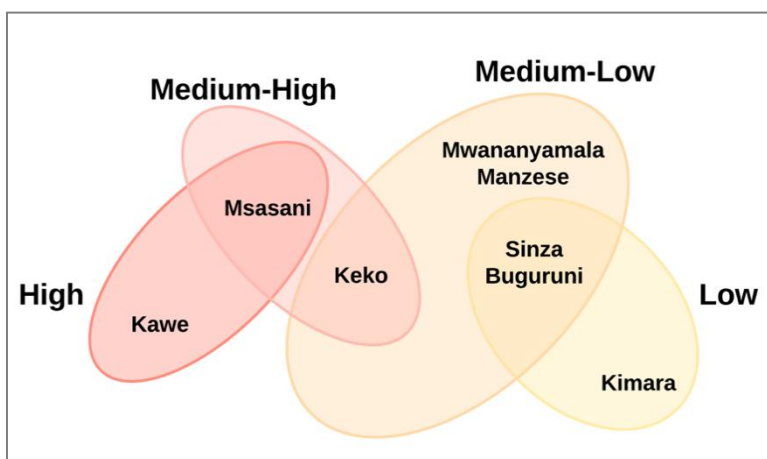


Figure 4.3. Results from post-hoc tests (i.e., Fisher’s LSD) showing ward level differences in residential energy use (at a significance level of 5%). Wards are ordered according to their mean residential energy use i.e., high, medium-high, medium-low and low residential energy use. Wards shown in the same circle or intersection have no significant differences between them.

Figure 4.4 illustrates the relative shares of total residential energy use by fuel category, i.e., electricity, gas, and transport oils (gasoline and diesel), which vary based on ward type. For example, in low-income wards (Buguruni, Keko and Manzese), charcoal accounts for most residential energy use (over 80%). In high-income wards (Msasani and Kawe), modern fuels (electricity, gas and transport oils) account for larger shares (over 50%). My findings are

consistent with other studies in the Africa region that have similarly shown higher wood fuel use among low-income households and higher gas/electricity use among high-income households. For example, the historic study of Hughes-Cromwick (1985) disaggregated urban household energy use (by income group) among surveyed households in Nairobi (Kenya). The study found that wood fuels (charcoal) accounted for most residential energy use among low-income households, i.e., 89% of households in the lowest income group used wood fuels for cooking compared to 31% in the highest income group (Hughes-Cromwick, 1985). More recent studies for South Africa (Bohlmann and Inglesi-Lotz, 2018), Uganda (Katutsi et al., 2020) and Zambia (Mulenga et al., 2019) similarly found considerable wood fuel use among surveyed low-income and high-income households. However, distinct from this current study, the authors of these studies did not include transport energy in their estimate for residential energy use, which means that the energy use estimates from previous studies of Hughes-Cromwick (1985) and others are likely underestimated.

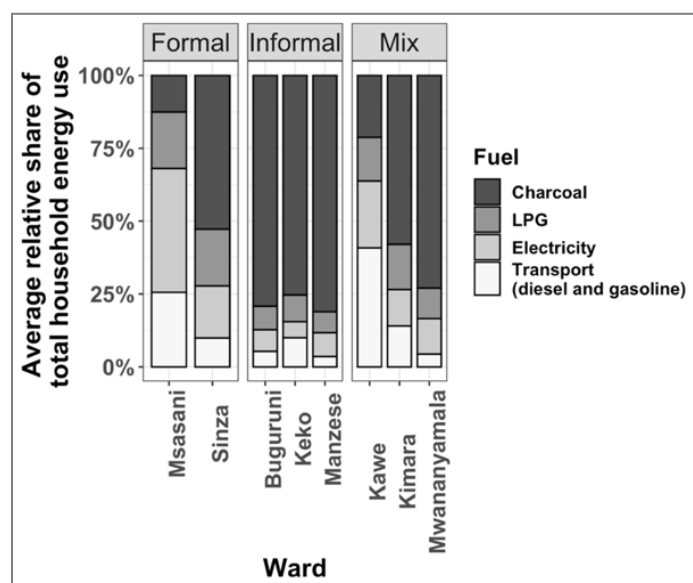


Figure 4.4. Average relative shares of total household energy use by type of fuel in the Dar es Salaam surveyed wards.

In transportation, I find that private vehicle travel accounts for most transport-related energy use, which could indicate underlying disparities in ward-level transport options. This is further illustrated in Figure 4.5, which shows the relative share of transport-related energy use by mode

(Figure 4.5, Plot A) compared to the total number of households that reported the use of a private vehicle (Figure 4.5, Plot B). Disparities are most striking in informal (low-income) wards, where the few households that used a private vehicle (5% or less) contribute 50% to 80% of transport-related energy use. Furthermore, energy use in transportation is the most unequally distributed across surveyed households. Adapted Lorenz curves showing inequalities in energy use yield the highest inequality measure (Gini coefficients) for transport-related energy use (see supplementary material, Section 4.11.11). For example, I estimate a Gini coefficient (G_e) of 0.87 for transport-related energy use (where $G_e = 1$ indicates the highest level of inequality), compared to 0.38 for household-related energy use (Section 4.11.11). Comparing my results with other global regions, I find similar differences: e.g., 0.407 for China (total household energy use, (Wu et al., 2017)), 0.87 for Kenya (electricity use, (Jacobson et al., 2005)), and 0.14 and 0.28 for Canada (electricity and gas use, (Mirnezami, 2014)).



Figure 4.5. Average relative shares of total transport energy use among surveyed wards in the Dar es Salaam region according to transport mode (A). Proportion of households that reported the use of a private vehicle among the surveyed wards (B).

4.7.4 Effect of Cooking Fuel Choice and other Socio-Economic and Spatial Factors on Household-Related Energy Use

I find that most surveyed household do not use electricity for cooking despite relatively high electrification levels at the ward level, i.e., between 75% (Buguruni) and 100% (Msasani and Sinza) of surveyed wards were electrified (supplementary material, Section 4.11.6), but only 9% of households reported using electricity as a cooking fuel. The remaining 81% used a combination of gas, wood fuels (charcoal) and kerosene as a cooking fuel (Table 4.3). Other studies in the Africa region have likewise shown limited electricity use compared to wood fuels and gas among households in rural and urban areas, e.g., Mozambique (Castán Broto et al., 2020), Ghana (Mensah and Adu, 2015), Uganda (Katutsi et al., 2020), and South Africa (Bohlmann and Inglesi-Lotz, 2018). Though a more recent study for Lagos (Edomah and Ndulue, 2020) has shown that increases in electricity use for cooking could be anticipated in the near term (i.e., 2020 onwards) due to the coronavirus (COVID-19) pandemic and mandatory stay-at-home lockdowns.

I also find that electrification levels in Buguruni (75%) are consistent with averages for the Dar es Salaam region reported in Tanzania's energy access report (i.e., 75% in 2016), which is much higher than the national and rural average of 36% and 16% in 2016 (Government of Tanzania, 2017b). However, variation in ward level energy use is not considered in the energy access report – data are generalized for the entire city region. For example, higher electrification levels in other wards (e.g., Msasani) are not considered in reported data, or the variation in electricity sources (e.g., grid-supplied electricity, community generators or solar panels). In the supplementary material (Section 4.11.8.), I show the different sources of electricity among my surveyed households, i.e., from a low of 72% (Buguruni) to a high of 100% (Msasani and Sinza) from grid-supplied electricity, from a low of 0% (Buguruni) to a high of 25% (Msasani) from community or private generators, and from a low of 0% (Mwananyamala) to a high of 5% (Kawe) from solar panels. Note that these ranges do not add to 100% as some households reported multiple sources of electricity supply.

Across all surveyed wards, 62% of households fuel stacked, though, with some variation at the ward level (see the supplementary material, Section 4.11.6, for data summaries by ward).

Households that fuel stack (i.e., use two or more fuels for cooking needs) are correlated with higher household-related energy use. ANOVA tests found a statistically significant effect of cooking fuel choice on mean household-related energy use [$F(11,1349) = 59.05$, $p < 0.01$]. Post-hoc (LSD) tests that map the effect of cooking fuel choice on mean household-related energy use show five groupings of households (A to E) (see Table 4.3). However, note that results in Table 4.3 should not be over-interpreted given the different levels of reporting from households, e.g., in the highest group (A), only one household reported fuel-stacking with electricity, charcoal and “other” (i.e., either kerosene or firewood). Considering these limitations, I find that households that fuel stack with three fuels (6% of all households surveyed) generally appear to be the highest energy users (Table 4.3, group A). Except for charcoal, using a single fuel (e.g., only gas, or only electricity) is associated with lower household-related energy use.

Results from OLS and Tobit regressions correlate fuel stacking with 35% higher household-related energy use (i.e., taking the exponent of the coefficient, ~ 0.3 , shown in OLS and Tobit models – Table 4.4). Fuel stacking with electricity, charcoal and kerosene/firewood (3% of households surveyed) is correlated with the highest household-related energy use (11 and 13 times more, based on estimated coefficients: 2.5 and 2.7 for OLS and Tobit models, respectively) relative to households that use only electricity (reference case). Finally, changes along PC1 and PC2, which I interpret as wealth and possibly propensity to invest in common appliances, respectively, show a significant and positive correlation with household-related energy use. My findings are consistent with other studies that likewise found positive and significant correlations relating both income and fuel stacking to household energy use and/or fuel choices in Tanzania, e.g., (Choumert-Nkolo et al., 2019; D’Agostino et al., 2015).

Table 4.3. Results from LSD test results showing differences in mean household-related energy use based on household cooking fuel choice among surveyed households in the Dar es Salaam region. ‘A’ represents the high energy using groups, and E the lowest. Surveyed households grouped with the same letter indicate no statistically significant difference between group means.

Cooking fuel choice ¹	Dependent variable: log(Household) ²	Ordered groupings ³	Number (and %) of households surveyed ⁵
Electricity, charcoal, and other ⁴	4.7	A	1 (0.1%)
Electricity, gas and charcoal	3.62	AB	43 (3.2%)
Charcoal and electricity	3.59	ABC	2 (0.1%)
Gas, charcoal and other ⁴	3.47	ABC	37 (2.7%)
Only charcoal	3.42	BC	305 (22.4%)
Gas and charcoal	3.41	C	549 (40.3%)
Gas and electricity	3.35	C	73 (5.4%)
Charcoal and other ⁴	3.3	C	133 (9.8%)
Only gas	2.51	D	193 (14.2%)
Gas and other ⁴	2.23	D	8 (0.6%)
Only electricity	1.37	DE	1 (0.1%)
Only other ⁴	0.64	E	16 (1.2%)
Significance (ANOVA)	p<0.01 ***		

Table Notes:

¹ My dataset included no household reporting of the following cooking fuel choice combinations: (1) electricity, gas and other (i.e., either kerosene or firewood), (2) charcoal and other, and (3) electricity and other.

² Household-related energy use estimates were log transformed to reduce skewness. Note, household-related energy use is estimated based on the sum of energy uses from household charcoal, gas and electricity use.

³ Variables with one or more of the same letters are not statistically different from each other.

⁴ “Other” refers to either kerosene or firewood. Note, these fuels are not included in my final estimate for household-related energy use.

⁵ Total households surveyed sum to 1,363. Break-down by ward and additional summary statistics in the supplementary material, Section 4.11.6 and Section 4.11.8.

Table 4.4. Multivariate OLS and Tobit regression results showing the statistical relationship between cooking fuel choice, selected spatial and socio-economic variables, and household-related energy use across the surveyed households in the Dar es Salaam region.

		Dependent variable: Log(Household Energy Use)			
Independent variable:		OLS (1)	OLS (2)	Tobit (1)	Tobit (2)
<u>Spatial</u>					
	Informal settlement	-0.023 (0.065)	0.058 (0.076)	-0.025 (0.065)	0.057 (0.076)
	Mixed settlement	-0.089* (0.051)	-0.034 (0.059)	-0.090* (0.051)	-0.034 (0.060)
	Log(Ward density)	0.086** (0.036)	0.149*** (0.042)	0.086** (0.036)	0.150*** (0.042)
<u>Socio-economic</u>					
	Log(Household members)	0.214*** (0.031)	0.362*** (0.035)	0.215*** (0.032)	0.363*** (0.036)
	PC1	0.018*** (0.002)	0.010*** (0.002)	0.018*** (0.002)	0.010*** (0.002)
	PC2	0.007*** (0.002)	0.013*** (0.003)	0.007*** (0.002)	0.013*** (0.003)
<u>Cooking fuel choice</u>					
	Only charcoal	2.512*** (0.142)		2.669*** (0.150)	
	Only gas	1.508*** (0.147)		1.666*** (0.155)	
	Gas and electricity	1.904*** (0.169)		2.058*** (0.175)	
	Gas and charcoal	2.413*** (0.141)		2.571*** (0.149)	
	Charcoal and electricity	2.243*** (0.448)		2.398*** (0.451)	
	Gas and other	1.475*** (0.251)		1.635*** (0.256)	
	Charcoal and other	2.416*** (0.147)		2.574*** (0.155)	
	Electricity, gas and charcoal	2.241*** (0.174)		2.396*** (0.181)	
	Gas, charcoal and other	2.504*** (0.170)		2.662*** (0.177)	
	Electricity, charcoal and other	3.961*** (0.612)		4.119*** (0.616)	

Independent variable:	Dependent variable: Log(Household Energy Use)			
	OLS (1)	OLS (2)	Tobit (1)	Tobit (2)
Fuel stack when cooking		0.326*** (0.040)		0.328*** (0.040)
Constant	0.067 (0.357)	1.129*** (0.378)	-0.094 (0.362)	1.116*** (0.381)
Observations	1,363	1,363	1,363 (Censored:14)	1,363 (Censored: 14)
Pseudo R ² (McFadden)			0.805	0.769
R ²	0.418	0.193		
Adjusted R ²	0.411	0.188		

Table Notes:

- *p<0.1; **p<0.05; ***p<0.01.
- Standard error denoted by values in parentheses.
- Reference variables not included in table: Formal settlement; Only electricity (for cooking).
- “Other” refers to either kerosene or firewood. Note that these fuels are not included in my final estimate for household-related energy use.
- See supplementary material for: (1) supporting correlation matrix of all data variables (Section 4.11.6), and (2) Full-set of results from OLS and Tobit regressions, i.e., including original proxy variables (education and appliance ownership) applied as predictors (Section 4.11.9).

4.7.5 Effect of Public Transport Use and other Socio-Economic and Spatial Factors on Transport-Related Energy Use

I find a positive and significant correlation between transport-related energy use and the PC1 dimension for wealth (Table 4.5). Msasani and Kawe (high-income wards) contribute the highest transport-related energy use (consistent with their high PC1 scores). Furthermore, using public transport as part of a household’s commuting activities (i.e., either by BRT, dala-dala, bajaji or boda) is correlated with 13% lower transport-related energy use (for the OLS model. Results were indiscernible for the Tobit model). Public transport use (particularly use of the dala-dala) is higher in informal wards, e.g., 93% of households in Buguruni reported using public transport as part of their weekly commuting, compared to 67% of households in Msasani (supplementary material, Section 4.11.6). To interpret these differences in resident travel behavior, we conducted multinomial logit regressions to assess spatial and socio-economic factors affecting travel mode choice (see supplementary material, Section 4.11.10). Findings suggest that household wealth

(assumed based on coefficients along the PC1 dimension of wealth) is significantly and positively correlated with households' choice to use private vehicles (including taxi or Uber rides) relative to the dala-dala. Moreover, even though the BRT is perceived to be more expensive than the dala-dala (see Chapter 5), distance to the city center (Kivukoni) is a key predictor of increased BRT use. Households that are further away from the city center are 17 times more likely to use the BRT relative to the dala-dala (taking the exponent of the coefficient, 2.9. See supplementary material, Section 4.11.10).

Relatedly, distance from the city center is correlated with higher transport-related energy use, e.g., a 100% increase in distance between residence location and the city center is correlated with 2% and 3% higher transport-related energy (Table 4.5). I also find a negative and significant relationship between ward density and transport-related energy use, where a 100% increase in density is correlated with 19% or 16% lower transport-related energy use. These findings are consistent with other literature that associates higher urban densities with lower transport-related energy use, e.g., (Creutzig et al., 2015; IPCC, 2014; VandeWeghe & Kennedy, 2007). However, in the case of Dar es Salaam, higher densities may not directly indicate a sustainable or energy-efficient transport system. Considering the unique context in which African cities are urbanizing, where urban growth is coupled with other challenges, including urban poverty and informality, this negative correlation may be due to other hidden socio-economic factors (e.g., financial constraints that prohibit the use of motorized transport) not explicitly considered in this study. Details on these aspects are beyond the scope of this research and would be an important area of future work.

Table 4.5. Multivariate OLS and Tobit regression results showing the statistical relationship between public transport use, selected spatial and socio-economic variables, and transport-related energy use across the surveyed households in the Dar es Salaam region.

		Dependent variable: Log(Transport Energy Use)	
Independent variables:		OLS (1)	Tobit (1)
<u>Spatial</u>			
	Informal settlement	0.316*** (0.103)	0.402*** (0.145)
	Mixed settlement	0.141* (0.085)	0.148 (0.118)
	Log(Ward density)	-0.197*** (0.064)	-0.177* (0.091)
	Log(Distance to City Center)	0.028 (0.088)	0.078 (0.124)
<u>Socio-economic</u>			
	Log(Household members)	0.119** (0.048)	0.184*** (0.069)
	PC1	0.043*** (0.003)	0.053*** (0.004)
	PC2	-0.007* (0.004)	-0.005 (0.005)
	Use public transport	-0.130* (0.074)	0.054 (0.107)
	Constant	2.851*** (0.682)	2.053** (0.961)
	Observations	1,363	1,363 (Censored: 441)
	Pseudo R-squared (McFadden)		0.703
	R ²	0.326	
	Adjusted R ²	0.322	

Table Notes:

- *p<0.1; **p<0.05; ***p<0.01
- Standard error denoted by values in parentheses.
- Reference variables not included in table: Formal settlement.
- See supplementary material for: (1) supporting correlation matrix of all data variables (Section 4.11.6), and (2) Full-set of results from OLS and Tobit regressions, i.e., including original proxy variables (education and appliance ownership) applied as predictors (Section 4.11.9).

4.8 Policy Considerations

The results presented in this study are suggestive and causality is not assumed. Considering this context, three main findings are clear: (1) there is a statistically significant difference in mean residential energy use among Dar es Salaam wards, where differences are between four distinct clusters of household and transport-related energy use (which I illustrate in Figure 4.6); (2) a shift in fuel consumption, i.e., from traditional (charcoal) to modern fuels (electricity/gas/transport oils), is evident from low-income to high-income wards; and (3) consumption of charcoal (typically present in fuel stacking) and private vehicle use are major drivers of household and transport-related energy use, respectively.

Figure 4.6 shows mean household-related energy use versus transport-related energy use across the surveyed wards, where I identify four clusters of wards, which represent different levels of household-related and transport-related energy use, respectively : (A) Low-Low (Kimara); (B) Medium-Low (Sinza, Buguruni, Manzese and Mwananyamala); (C) High-Low (Keko); and (D) High-High (Msasani and Kawe – the high income wards). This clustering suggests that a possible transition in energy use may occur as wards develop from A to D (Figure 4.6), where convergence towards higher energy use could be expected as more wards consume at levels similar to Kawe/Msasani in the future (and where a switch to more modern fuels could be expected). I also find that fuel stacking plays a critical role in rising energy demand and could be considered in the context of ongoing policy measures promoting electrification and energy efficiency. For example, 100% electricity use, or fuel stacking with electricity and gas, could be encouraged as wards develop socio-economically (e.g., towards D, Figure 4.6) to promote a gradual phase-out of charcoal use. Whereas, 100% gas use, or fuel stacking with gas and charcoal (which is associated with lower energy use than using only charcoal, Table 4.4), could be considered for wards “in transition” (e.g., A to C, Figure 4.6). The use of energy-efficient charcoal and gas stoves, which are already stated as a key consideration in Tanzania’s energy policies, e.g., (Government of Tanzania, 2015b), could be promoted alongside these differentiated interventions.

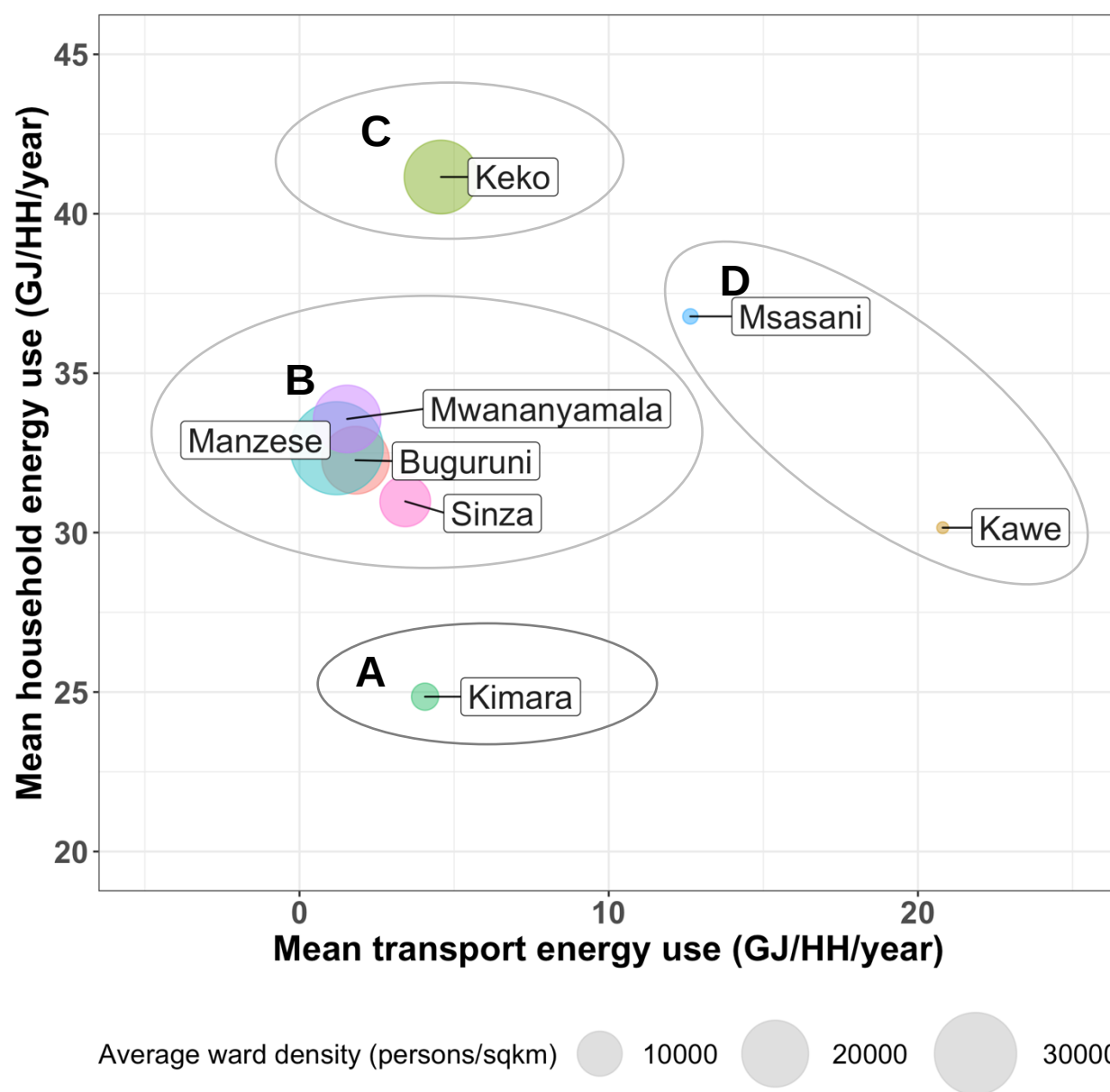


Figure 4.6. Differences in household and transport-related energy use across surveyed wards in Dar es Salaam. The size of bubble indicates average ward density. Surveyed wards map along four major typologies of energy use (based on their mean household-related and transport-related energy use, respectively): (A) Low-Low (Kimara); (B) Medium-Low (Sinza, Buguruni, Manzese and Mwananyamala); (C) High-Low (Keko); and (D) High-High (Msasani and Kawe).

In transportation, public transport use, particularly BRT use, could be encouraged in all wards, but more so in wards such as Msasani/Kawe (cluster D) with high levels of transport related energy use due to their higher private vehicle travel (Figure 4.6). Therefore, policymakers would need to balance investments in public transport and the BRT service across a wide range of settlements in the city: (a) to improve access/affordability (especially for informal, low-income wards with limited travel options) and (b) to control transport-related energy use (especially in wealthier wards due to their higher private vehicle use). These efforts could feed into ongoing urban planning policies (e.g., Government of Tanzania (2017a)) that already envision an energy-efficient and accessible transport system in Dar es Salaam.

4.9 Study Limitations

I highlight three study limitations linked to my methods and results. Firstly, as stated in the methods, I was unable to collect data on firewood and kerosene use (households were unable to estimate their consumption for these specific categories during pilot surveys in Kijitonyama ward). Therefore, my values for household-related energy use may be under-estimated in some cases. I also did not include weekend or leisure travel in my transportation estimates, even though these may have an effect on transport-related energy use and associated GHG emissions. For example, in their review paper, Czepkiewicz et al. (2018) found significant positive correlations between urban density and GHG emissions in transportation when long-distance travel is included in the analysis (i.e., from car weekend trips and international flights).

Secondly, my structured survey did not allow for further analysis of contextual factors (e.g., cultural, or local beliefs) that may influence residential energy use. Therefore, future work could incorporate qualitative methods, e.g., participant observation or ethnographic approaches, as to confirm or validate the data we collected in the structured survey. Except for electricity data that was collected via household electricity bills, charcoal, gas, and transport data were reliant on self-reported data, and surveyed households may have under or over-estimated their energy use or travel behaviors.

Finally, future work may broaden the scope of analysis by (1) tracking changes in residential energy use over time (e.g., to validate our assumptions of energy convergence), (2) conducting similar studies in other cities in Tanzania or African countries to compare research findings, (3) estimating GHG implications of energy use and other environmental impacts (e.g., health burdens associated with air pollution), including geographic mapping to identify settlements or populations that contribute most to urban GHG emissions, and (4) engaging with policymakers to develop strategies to translate our data into meaningful policy actions.

4.10 Concluding Remarks

To my knowledge, I present the first estimates of disaggregated ward level residential energy use for Dar es Salaam. I find a statistically significant difference in energy use among the surveyed wards, which can be grouped into four clusters, each representing distinct levels of residential (household and transport-related) energy use. These findings suggest the need for differentiated approaches to implementing energy policies at the sub-city scale. In addition, I highlight that movement towards higher levels of energy use could be expected with continued urban growth in Dar es Salaam (and possibly, other African cities), e.g., towards Msasani/Kawe levels, or those of outliers shown in Figure 4.2. At the same time, fuel switching towards more modern fuels (especially for cooking, e.g., electricity and gas) could be expected with rising wealth, fuel stacking and travel distances, among other socio-economic and spatial factors, contributing to higher residential energy use. Finally, my calculations show the different household and transport-related drivers of residential energy use at the ward level. However, other contextual factors (i.e., culture, society and urban governance structures) that influence infrastructure service delivery would need to be considered alongside differentiated policy interventions.

4.11 Supplementary Material

The supplementary material includes supporting methods, calculations and background data that are important in the context of this chapter (Chapter 4). Tables and figures are presented with the letter “S” in their caption title, and are referenced throughout the main body of this chapter, where relevant.

4.11.1 Recruitment of Field Team

The field team was recruited following interviews conducted in August 2018. The field team consisted of ten members: Alice Chibulu Luo (Lead Researcher), six graduate students from Ardhi University, and three experienced staff (including a Field Supervisor). The experienced staff were from Ideas in Action Ltd., a local research company, recruited under the study to support field coordination (see: <https://www.iact.co.tz/>). Each member signed a contract that detailed terms of the field work, remuneration and training requirements. In early September 2018, members participated in a 2-day training, which included pilot testing of the questionnaire in Kijitonyama ward.

4.11.2 Research Ethics Approvals and Local Clearances

The study received research approvals/clearances from institutions listed below.

(1) **The Human Sciences Research Ethics Board (REB) at the University of Toronto.**

Approval granted on November 11th, 2017. Protocol Reference #: 34987.

(2) **The Tanzania Commission for Science and Technology (COSTECH) in Dar es**

Salaam. Approval granted on May 7th, 2018. Research Permit #: 2018-327-NA-2018-133.

(3) **Various regional, district and ward level institutions.** Approvals were granted in

August and September 2018. Ideas in Action Ltd. supported the authors in following-up with local government authorities. This included submitting research introduction letters

and the COSTECH approval (item 2, noted above) to the Regional Administrative Secretary, within the office of the Regional Commissioner, who granted research clearance in the Dar es Salaam region. Subsequent approvals were obtained from district and ward level authorities in Kinondoni, Ubungo, Ilala and Temeke districts, and Msasani, Sinza, Manzese, Keko, Buguruni, Mwananyamala, Kawe and Kimara wards.

4.11.3 Quantifying Energy Use

I use the term “residential energy use” to refer to the sum of energy uses from both household and transport-related activities. I determined energy use from household activities based on reported household electricity, charcoal, and gas use, and from transport activities based on the commuting behavior of household members during an average weekday (i.e., Monday to Friday). Results from pilot tests in Kijitonyama ward showed that household reporting on weekday trips showed a more consistent picture of household travel behavior, and therefore, my calculations do not consider possible variations due to weekend travel. Interviewed participants were questioned on “start” and “end” locations of average weekday trips (including transfer stops), travel modes used, and common routes taken, e.g., names of major roads. I digitized and quantified travel routes and travel distances for each trip using the trip data for each household (Mr. Kevin Onjiko, Research Consultant, was recruited to support me with this work).

Equations (1) and (2) show my approach to estimating annual energy use from household and transport-related activities, respectively, and Table S 4.1 summarizes the fuel conversion factors applied in the study.

- (1) **Equation 1:** Annual household-related energy use (in GJ/HH/year) from charcoal and gas and electricity use in the home.

$$HE = EDC \times CF \times 365$$

$$CF = 0.0036 \text{ GJ/kWh (for electricity)}$$

$$CF = 0.0295 \text{ GJ/kg (for charcoal)}$$

$$CF = 0.0473 \text{ GJ/kg (for gas)}$$

Where, for each fuel, HE is the estimated household-related energy use in GJ/HH/year. EDC is the estimated daily consumption in kg (for charcoal and gas) or daily kWh for electricity. CF is the fuel conversion factor, which varies according to fuel type (using estimates shown in Table S 4.1). All values are multiplied by 365 to convert from daily to annual use.

- (2) **Equation 2:** For each household member, annual transport-related energy use (in GJ/passenger/year) from private and public vehicle travel by household members. Note that the final reported value in GJ/HH/year is the sum of all household members' transport-related energy use for each surveyed household.

$$TE = EI \times TD \times D \times CF \times 365$$

$$EI = \frac{1}{(FE \times LF)}$$

$$CF = 0.0443 \text{ GJ/kg (for gasoline)}$$

$$CF = 0.043 \text{ GJ/kg (for diesel)}$$

Where, for each mode, TE is the estimated transport-related energy use in GJ/passenger/year, EI is the energy intensity (liters/passenger-kilometer), which is the inverse product of FE , the fuel economy (vehicle-kilometer/liter) (shown in Table S 1, Notes) and LF , the load factor (passengers/vehicle) taken from (World Bank, 2016) and (DART, 2017) (shown in Table S 4.1, Notes). TD is the daily vehicle travel distance (kilometers) estimated from data digitization in ArcGIS. D is the vehicle fuel density (in kilograms/liter) (Engineering ToolBox, 2003). CF is the fuel conversion factor (based on estimates shown in Table S 1). I assume the following fuel uses within the vehicle fleet taken from (World Bank, 2016) and (Chapter 3): BRT: diesel (100%); bajaji and boda-boda: gasoline (100%); dala-dala: gasoline (80%) and diesel (20%); and private vehicle: gasoline (90%) and diesel (10%). All values are multiplied by 365 to estimate annual use.

Table S 4.1. Energy conversion factors applied in this study. Data available from (United Nations, 2017). Energy content is determined based on the net calorific value of each fuel.

Fuel	Standard Net Calorific Value (Gigajoules per kg, excluding electricity)
Electricity (kWh)	0.0036
Charcoal	0.0295
Liquefied Petroleum Gas (“gas”)	0.0473
Motor – gasoline ^{1,2}	0.0443
Motor – diesel ^{1,2}	0.043
<u>Table Notes:</u> ¹Assumed Vehicle Fuel Economy (FE) (Equation 2) taken from (World Bank, 2016): • Boda-boda or Bajaji – 50 vehicle-kilometers/liter • BRT – 7.41 vehicle-kilometers/liter • Private vehicle– 7.6 vehicle-kilometers/liter • Dala-dala – 7.41 vehicle-kilometers/liter ²Assumed load factors (LF) by vehicle mode taken from (World Bank, 2016) and (DART, 2017) • Boda-boda or Bajaji – 1.2 passengers/vehicle • BRT – 150 passengers/vehicle • Private vehicle– 1.8 passengers/vehicle • dala-dala – 40 passengers/vehicle	

4.11.4 Principal Component Scores by Ward

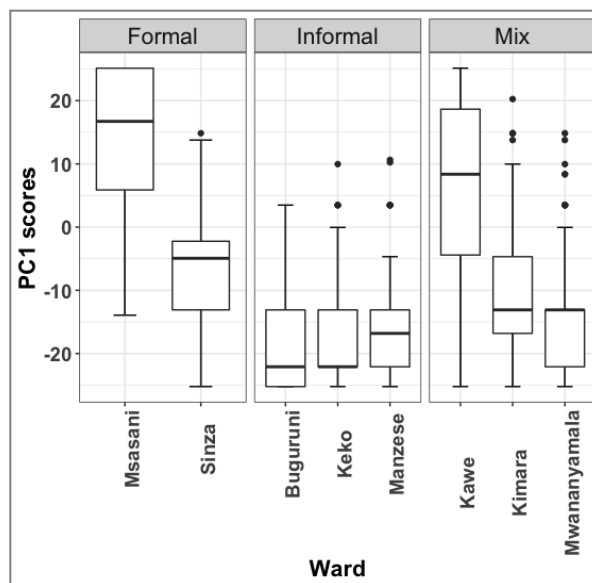


Figure S 4.1. Variation in PC1 scores by ward (which indicate household wealth). High PC1 scores in Kawe and Msasani are indicated in the boxplots, supporting my assumption that these wards are generally presumed as high-income by Dar es Salaam locals.

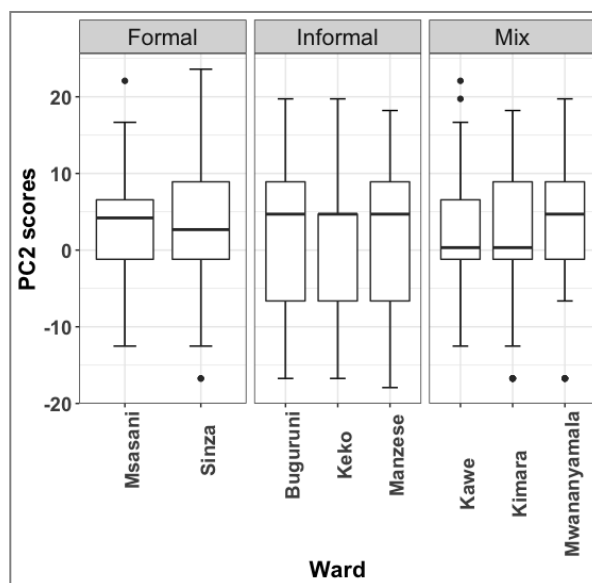


Figure S 4.2. Variation in PC2 scores by ward (which indicate possible propensity for common electric appliances). Box plot ranges appear similar, indicating no clear difference in PC2 scores across the surveyed wards.

4.11.5 Description of Variables Used in OLS and Tobit Regressions

Table S 4.2. Description of variables used in OLS and Tobit models showing the influence of cooking fuel choice¹, public transport use and other socio-economic and spatial variables on household related and transport-related energy use, respectively.

Variable	Description as applied in this study
Dependent variables	
(Note: variables are log transformed, where relevant, to reduce skewness)	
Log(Residential Energy)	The natural log of residential energy use (in GJ/HH), which is the sum of both household-related and transport-related energy among surveyed households.
Log(Household Energy)	The natural log of household-related energy use (in GJ/HH), which is the sum of energy use from charcoal, gas and electricity use among surveyed households.
Log(Transport Energy)	The natural log of transport-related energy use (in GJ/HH) which is the sum of energy use from private vehicle, dala-dala, bajaji and BRT travel among surveyed households.
Independent variables	
<u>Spatial</u>	
Informal settlement	Settlement contains housing and land obtained through informal means i.e., through individuals (Kironde, 2000).
Formal settlement	Settlement contains housing and land obtained through national and local government (Kironde, 2000).
Mixed settlement	Settlement contains both formal and informal settlements, with some areas currently being formalized.

Log(Ward Density)	Natural log of surveyed ward density, measured as people per square kilometer based on 2012 census data (NBS, 2013).
Distance to Kivukoni	Average distance of surveyed wards to the city center (Kivukoni) which is also the last stop south-east of the BRT line (Phase 1).

Socio-economic

Log(Household Members)	Natural log of total household members (both adults and children) within an individual household surveyed.
PC1	First Principal Component dimension.
PC2	Second Principal Component dimension.
Use public transport	Household uses public transport, which encompasses one or more of the following travel modes: the BRT, boda-boda, bajaji or dala-dala

Cooking fuel choice

Only gas	Household uses only gas for cooking.
Only charcoal	Household uses only charcoal for cooking.
Only electricity	Household uses only electricity for cooking
Charcoal and electricity	Household uses charcoal and electricity for cooking.
Gas and electricity	Household uses gas and electricity for cooking.
Gas and charcoal	Household uses gas and charcoal for cooking.
Gas and other	Household uses gas and other fuel (i.e., either kerosene or fuelwood) for cooking.
Charcoal and other	Household uses charcoal and other fuel (i.e., either kerosene or fuelwood) for cooking
Electricity, gas and charcoal	Household uses electricity, gas and charcoal for cooking.

Electricity, charcoal and other	Household uses electricity, charcoal and other fuel (i.e., either kerosene or wood) for cooking.
Gas, charcoal and other	Household uses gas, charcoal and other fuel (i.e., either kerosene or wood) for cooking.
Fuel stacking	Household fuel stacks (i.e., uses 2 or more fuels for cooking).

Table Notes:

¹ Due to an absence of reporting by households, we excluded following cooking fuel choice combinations from my analysis: (1) gas, electricity and other (i.e., either kerosene or wood), (2) charcoal and other, and (3) electricity and other.

4.11.6 Summary of Data Variables Relevant to the Study

Table S 4.3. Full summary statistics for key variables relevant to this study. Surveys were conducted across a socio-economically and spatially diverse set of wards (8 in total, shown in alphabetical order), which represented a total sample size of 1,363 households.

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Number of household members									
Mean (SD)	5 (3)	5 (2)	5 (3)	5 (2)	6 (3)	5 (2)	5 (3)	5 (2)	5 (3)
Median [Min, Max]	5 [1, 20]	5 [1, 10]	4 [1, 10]	5 [1, 10]	5 [1, 20]	5 [1, 10]	5 [1, 10]	5 [1, 10]	5 [1, 20]
Number of household rooms									
Mean (SD)	5 (3)	8 (4)	5 (3)	6 (3)	5 (4)	8 (4)	5 (3)	5 (3)	6 (4)
Median [Min, Max]	4 [1, 20]	8 [1, 30]	5 [1, 10]	6 [1, 30]	4 [1, 40]	7 [1, 20]	4 [1, 20]	5 [0, 10]	5 [0, 40]
Primary education									
No	179 (74.0%)	171 (96.1%)	82 (72.6%)	202 (91.4%)	174 (74.0%)	92 (97.9%)	142 (84.0%)	105 (94.6%)	1147 (84.2%)
Yes	63 (26.0%)	7 (3.9%)	31 (27.4%)	19 (8.6%)	61 (26.0%)	2 (2.1%)	27 (16.0%)	6 (5.4%)	216 (15.8%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Secondary education									
No	110 (45.5%)	157 (88.2%)	65 (57.5%)	138 (62.4%)	110 (46.8%)	87 (92.6%)	72 (42.6%)	73 (65.8%)	812 (59.6%)
Yes	132 (54.5%)	21 (11.8%)	48 (42.5%)	83 (37.6%)	125 (53.2%)	7 (7.4%)	97 (57.4%)	38 (34.2%)	551 (40.4%)
Tertiary education									
No	207 (85.5%)	35 (19.7%)	86 (76.1%)	119 (53.8%)	198 (84.3%)	12 (12.8%)	131 (77.5%)	48 (43.2%)	836 (61.3%)
Yes	35 (14.5%)	143 (80.3%)	27 (23.9%)	102 (46.2%)	37 (15.7%)	82 (87.2%)	38 (22.5%)	63 (56.8%)	527 (38.7%)
Vocational training									
No	231 (95.5%)	171 (96.1%)	106 (93.8%)	204 (92.3%)	223 (94.9%)	91 (96.8%)	162 (95.9%)	107 (96.4%)	1295 (95.0%)
Yes	11 (4.5%)	7 (3.9%)	7 (6.2%)	17 (7.7%)	12 (5.1%)	3 (3.2%)	7 (4.1%)	4 (3.6%)	68 (5.0%)
Multi-story house									
No	241 (99.6%)	125 (70.2%)	112(99.1%)	218 (98.6%)	234 (99.6%)	72 (76.6%)	166 (98.2%)	110 (99.1%)	1278 (93.8%)
Yes	1 (0.4%)	53 (29.8%)	1 (0.9%)	3 (1.4%)	1 (0.4%)	22 (23.4%)	3 (1.8%)	1 (0.9%)	85 (6.2%)
Single-story house									
No	7 (2.9%)	67 (37.6%)	4 (3.5%)	14 (6.3%)	8 (3.4%)	57 (60.6%)	8 (4.7%)	3 (2.7%)	168 (12.3%)
Yes	235 (97.1%)	111 (62.4%)	109 (96.5%)	207 (93.7%)	227 (96.6%)	37 (39.4%)	161 (95.3%)	108 (97.3%)	1195 (87.7%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Electrified									
No	61 (25.2%)	8 (4.5%)	25 (22.1%)	17 (7.7%)	32 (13.6%)	0 (0%)	11 (6.5%)	0 (0%)	154 (11.3%)
Yes	181 (74.8%)	170 (95.5%)	88 (77.9%)	204 (92.3%)	203 (86.4%)	94 (100%)	158 (93.5%)	111 (100%)	1209 (88.7%)
Grid Connection									
No	67 (25.2%)	8 (4.5%)	26 (23.0%)	18 (8.1%)	32 (13.6%)	0 (0%)	11 (6.5%)	0 (0%)	162 (11.9%)
Yes	175 (74.8%)	170 (95.5%)	87 (77.0%)	203 (91.9%)	203 (86.4%)	94 (100%)	158 (93.5%)	111 (100%)	1201 (88.1%)
Community generator									
No	242 (100%)	177 (99.4%)	113 (100%)	221 (100%)	235 (100%)	94 (100%)	169 (100%)	111 (100%)	1362 (100%)
Yes	0 (0%)	1 (0.6%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.0%)
Own generator									
No	242 (100%)	158 (88.8%)	112 (99.1%)	221 (100%)	235 (100%)	71 (75.5%)	168 (99.4%)	108 (97.3%)	1315 (96.5%)
Yes	0 (0%)	20 (11.2%)	1 (0.9%)	0 (0%)	0 (0%)	23 (24.5%)	1 (0.6%)	3 (2.7%)	48 (3.5%)
Solar panels									
No	235 (97.1%)	170 (95.5%)	109 (96.5%)	216 (97.7%)	234 (99.6%)	93 (98.9%)	169 (100%)	110 (99.1%)	1336 (98%)
Yes	7 (2.9%)	8 (4.5%)	4 (3.5%)	5 (2.3%)	1 (.4%)	1 (1.1%)	0 (0%)	1 (0.9%)	27 (2%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Monthly electricity use (kWh)									
Mean (SD)	60 (80)	300 (300)	60 (90)	80 (90)	60 (60)	500 (500)	100 (100)	100 (200)	100 (200)
Median [Min, Max]	30 [0, 600]	200 [0, 3000]	30 [0, 700]	50 [0, 700]	50 [0, 400]	300 [0, 3000]	70 [0, 900]	90 [0, 1000]	70 [0, 3000]
Own air conditioning unit									
No	242 (100%)	106 (59.6%)	113 (100%)	215 (97.3%)	233 (99.1%)	32 (34.0%)	167 (98.8%)	105 (94.6%)	1213 (89.0%)
Yes	0 (0%)	72 (40.4%)	0 (0%)	6 (2.7%)	2 (0.9%)	62 (66.0%)	2 (1.2%)	6 (5.4%)	150 (11.0%)
Own electric stove									
No	238 (98.3%)	115 (64.6%)	111 (98.2%)	208 (94.1%)	233 (99.1%)	41 (43.6%)	161 (95.3%)	95 (85.6%)	1202 (88.2%)
Yes	4 (1.7%)	63 (35.4%)	2 (1.8%)	13 (5.9%)	2 (0.9%)	53 (56.4%)	8 (4.7%)	16 (14.4%)	161 (11.8%)
Own refrigerator									
No	156 (64.5%)	16 (9.0%)	72 (63.7%)	74 (33.5%)	129 (54.9%)	3 (3.2%)	68 (40.2%)	23 (20.7%)	541 (39.7%)
Yes	86 (35.5%)	162 (91.0%)	41 (36.3%)	147 (66.5%)	106 (45.1%)	91 (96.8%)	101 (59.8%)	88 (79.3%)	822 (60.3%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Own television									
No	88 (36.4%)	12 (6.7%)	31 (27.4%)	36 (16.3%)	69 (29.4%)	5 (5.3%)	29 (17.2%)	8 (7.2%)	278 (20.4%)
Yes	154 (63.6%)	166 (93.3%)	82 (72.6%)	185 (83.7%)	166 (70.6%)	89 (94.7%)	140 (82.8%)	103 (92.8%)	1085 (79.6%)
Own washing Machine									
No	241 (99.6%)	125 (70.2%)	112 (99.1%)	213 (96.4%)	234 (99.6%)	45 (47.9%)	167 (98.8%)	108 (97.3%)	1245 (91.3%)
Yes	1 (0.4%)	53 (29.8%)	1 (0.9%)	8 (3.6%)	1 (0.4%)	49 (52.1%)	2 (1.2%)	3 (2.7%)	118 (8.7%)
Own private vehicle									
No	224 (92.6%)	43 (24.2%)	105 (92.9%)	173 (78.3%)	223 (94.9%)	13 (13.8%)	153 (90.5%)	83 (74.8%)	1017 (74.6%)
Yes	18 (7.4%)	135 (75.8%)	8 (7.1%)	48 (21.7%)	12 (5.1%)	81 (86.2%)	16 (9.5%)	28 (25.2%)	346 (25.4%)
Fuel stack when cooking									
No	109 (45.0%)	58 (32.6%)	53 (46.9%)	69 (31.2%)	98 (41.7%)	38 (40.4%)	53 (31.4%)	39 (35.1%)	517 (37.9%)
Yes	133 (55.0%)	120 (67.4%)	60 (53.1%)	152 (68.8%)	137 (58.3%)	56 (59.6%)	116 (68.6%)	72 (64.9%)	846 (62.1%)
Only gas (for cooking)									
No	231 (95.5%)	126 (70.8%)	106 (93.8%)	186 (84.2%)	220 (93.6%)	57 (60.6%)	154 (91.1%)	90 (81.1%)	1170 (85.8%)
Yes	11 (4.5%)	52 (29.2%)	7 (6.2%)	35 (15.8%)	15 (6.4%)	37 (39.4%)	15 (8.9%)	21 (18.9%)	193 (14.2%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Only electricity (for cooking)									
No	242 (100%)	178 (100%)	113 (100%)	221 (100%)	235 (100%)	94 (100%)	169 (100%)	110 (99.1%)	1362 (99.9%)
Yes	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.9%)	1 (0.1%)
Only charcoal (for cooking)									
No	151 (62.4%)	172 (96.6%)	69 (61.1%)	191 (86.4%)	155 (66.0%)	93 (98.9%)	133 (78.7%)	94 (84.7%)	1058 (77.6%)
Yes	91 (37.6%)	6 (3.4%)	44 (38.9%)	30 (13.6%)	80 (34.0%)	1 (1.1%)	36 (21.3%)	17 (15.3%)	305 (22.4%)
Only other (i.e., kerosene or gas) (for cooking)									
No	235 (97.1%)	178 (100%)	111 (98.2%)	217 (98.2%)	232 (98.7%)	94 (100%)	169 (100%)	111 (100%)	1347 (98.8%)
Yes	7 (2.9%)	0 (0%)	2 (1.8%)	4 (1.8%)	3 (1.3%)	0 (0%)	0 (0%)	0 (0%)	16 (1.2%)
Gas and electricity (for cooking)									
No	242 (100%)	147 (82.6%)	113 (100%)	220 (99.5%)	235 (100%)	60 (63.8%)	167 (98.8%)	106 (95.5%)	1290 (94.6%)
Yes	0 (0%)	31 (17.4%)	0 (0%)	1 (0.5%)	0 (0%)	34 (36.2%)	2 (1.2%)	5 (4.5%)	73 (5.4%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Gas and charcoal (for cooking)									
No	145 (59.9%)	119 (66.9%)	79 (69.9%)	104 (47.1%)	146 (62.1%)	83 (88.3%)	80 (47.3%)	58 (52.3%)	814 (59.7%)
Yes	97 (40.1%)	59 (33.1%)	34 (30.1%)	117 (52.9%)	89 (37.9%)	11 (11.7%)	89 (52.7%)	53 (47.7%)	549 (40.3%)
Charcoal and electricity (for cooking)									
No	241 (99.6%)	178 (100%)	113 (100%)	221 (100%)	235 (100%)	94 (100%)	169 (100%)	110 (99.1%)	1361 (99.9%)
Yes	1 (0.4%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.9%)	2 (0.1%)
Gas and other (i.e., kerosene or gas) (for cooking)									
No	240 (99.2%)	176 (98.9%)	113 (100%)	218 (98.6%)	234 (99.6%)	94 (100%)	169 (100%)	111 (100%)	1355 (99.4%)
Yes	2 (0.8%)	2 (1.1%)	0 (0%)	3 (1.4%)	1 (0.4%)	0 (0%)	0 (0%)	0 (0%)	8 (0.6%)
Charcoal and other (i.e., kerosene or gas) (for cooking)									
No	213 (88.0%)	173 (97.2%)	92 (81.4%)	207 (93.7%)	193 (82.1%)	94 (100%)	150 (88.8%)	108 (97.3%)	1230 (90.2%)
Yes	29 (12.0%)	5 (2.8%)	21 (18.6%)	14 (6.3%)	42 (17.9%)	0 (0%)	19 (11.2%)	3 (2.7%)	133 (9.8%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Electricity, gas and charcoal (for cooking)									
No	241 (99.6%)	161 (90.4%)	112 (99.1%)	215 (97.3%)	235 (100%)	84 (89.4%)	168 (99.4%)	104 (93.7%)	1320 (96.8%)
Yes	1 (0.4%)	17 (9.6%)	1 (0.9%)	6 (2.7%)	0 (0%)	10 (10.6%)	1 (0.6%)	7 (6.3%)	43 (3.2%)
Gas, charcoal and other (for cooking)									
No	239 (98.8%)	172 (96.6%)	109 (96.5%)	211 (95.5%)	230 (97.9%)	93 (98.9%)	164 (97.0%)	108 (97.3%)	1326 (97.3%)
Yes	3 (1.2%)	6 (3.4%)	4 (3.5%)	10 (4.5%)	5 (2.1%)	1 (1.1%)	5 (3.0%)	3 (2.7%)	37 (2.7%)
Electricity, charcoal and other (i.e., kerosene or gas) (for cooking)									
No	242 (100%)	178 (100%)	113 (100%)	220 (99.5%)	235 (100%)	94 (100%)	169 (100%)	111 (100%)	1362 (99.9%)
Yes	0 (0%)	0 (0%)	0 (0%)	1 (0.5%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.1%)
Use public transport									
No	18 (7.4%)	47 (26.4%)	14 (12.4%)	23 (10.4%)	37 (15.7%)	31 (33.0%)	18 (10.7%)	14 (12.6%)	202 (14.8%)
Yes	224 (92.6%)	131 (73.6%)	99 (87.6%)	198 (89.6%)	198 (84.3%)	63 (67.0%)	151 (89.3%)	97 (87.4%)	1161 (85.2%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Use bajaji (tricycle)									
No	241 (99.6%)	168 (94.4%)	113 (100%)	194 (87.8%)	233 (99.1%)	92 (97.9%)	165 (97.6%)	109 (98.2%)	1315 (96.5%)
Yes	1 (0.4%)	10 (5.6%)	0 (0%)	27 (12.2%)	2 (0.9%)	2 (2.1%)	4 (2.4%)	2 (1.8%)	48 (3.5%)
Use boda-boda (motorcycle)									
No	215 (88.8%)	172 (96.6%)	110 (97.3%)	201 (91.0%)	214 (91.1%)	94 (100%)	155 (91.7%)	97 (87.4%)	1258 (92.3%)
Yes	27 (11.2%)	6 (3.4%)	3 (2.7%)	20 (9.0%)	21 (8.9%)	0 (0%)	14 (8.3%)	14 (12.6%)	105 (7.7%)
Use BRT									
No	240 (99.2%)	176 (98.9%)	112 (99.1%)	142 (64.3%)	192 (81.7%)	94 (100%)	156 (92.3%)	100 (90.1%)	1212 (88.9%)
Yes	2 (0.8%)	2 (1.1%)	1 (0.9%)	79 (35.7%)	43 (18.3%)	0 (0%)	13 (7.7%)	11 (9.9%)	151 (11.1%)
Use dala-dala (local bus)									
No	86 (35.5%)	138 (77.5%)	51 (45.1%)	153 (69.2%)	123 (52.3%)	85 (90.4%)	79 (46.7%)	59 (53.2%)	774 (56.8%)
Yes	156 (64.5%)	40 (22.5%)	62 (54.9%)	68 (30.8%)	112 (47.7%)	9 (9.6%)	90 (53.3%)	52 (46.8%)	589 (43.2%)
Use private vehicle									
No	233 (96.3%)	85 (47.8%)	107 (94.7%)	196 (88.7%)	223 (94.9%)	37 (39.4%)	159 (94.1%)	93 (83.8%)	1133 (83.1%)
Yes	9 (3.7%)	93 (52.2%)	6 (5.3%)	25 (11.3%)	12 (5.1%)	57 (60.6%)	10 (5.9%)	18 (16.2%)	230 (16.9%)

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Total trips per household per day									
Mean (SD)	3 (2)	2 (2)	3 (2)	2 (2)	3 (2)	2 (2)	2 (2)	3 (2)	2 (2)
Median [Min, Max]	2 [0, 10]	2 [0, 10]	2 [0, 10]	2 [0, 10]	2 [0, 10]	2 [0, 10]	2 [0, 8]	2 [0, 10]	2 [0, 10]
Distance to Kivukoni (city center)									
Mean (SD)	6 (0.5)	10 (0.4)	3 (0.2)	10 (2)	8 (0.3)	7 (0.4)	6 (0.2)	9 (0.5)	9 (4)
Median [Min, Max]	6 [5, 7]	10 [10, 20]	3 [3, 6]	10 [10, 20]	8 [7, 8]	7 [7, 8]	6 [5, 6]	9 [8, 10]	8 [3, 20]
Distance to the closest BRT stop (Phase 1 line)									
Mean (SD)	3 (0.5)	8 (0.5)	2 (0.2)	2 (1)	0.2 (0.1)	3 (0.3)	0.6 (0.2)	1 (0.6)	2 (2)
Median [Min, Max]	3 [2, 4]	8 [7, 9]	2 [1, 3]	1 [0.05, 4]	0.2 [0.01, 0.5]	3 [3, 4]	0.7 [0.2, 1]	1 [0.3, 2]	2 [0.01, 9]
Electric use (GJ/HH/Year)									
Mean (SD)	3 (3)	10 (10)	3 (4)	4 (4)	3 (3)	20 (20)	4 (5)	6 (7)	6 (10)
Median [Min, Max]	1 [0, 30]	7 [0, 100]	1 [0, 30]	2 [0, 30]	2 [0, 20]	10 [0, 100]	3 [0, 40]	4 [0, 40]	3 [0, 100]

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Charcoal use (GJ/HH/Year)									
Mean (SD)	30 (20)	10 (20)	30 (50)	20 (20)	30 (20)	6 (20)	30 (20)	20 (20)	20 (20)
Median [Min, Max]	20 [0, 200]	0 [0, 80]	20 [0, 200]	20 [0, 100]	20 [0, 200]	0 [0, 200]	20 [0, 200]	20 [0, 200]	20 [0, 200]
Gas use (GJ/HH/Year)									
Mean (SD)	3 (4)	8 (5)	4 (10)	4 (4)	2 (4)	10 (10)	4 (8)	7 (10)	5 (7)
Median [Min, Max]	0 [0, 30]	9 [0, 40]	0 [0, 70]	3 [0, 30]	0 [0, 30]	9 [0, 100]	3 [0, 100]	3 [0, 100]	3 [0, 100]
Total household-related energy (GJ/HH/year)									
Mean (SD)	30 (20)	30 (20)	40 (50)	20 (20)	30 (20)	40 (30)	30 (20)	30 (30)	30 (30)
Median [Min, Max]	20 [0, 200]	20 [1, 200]	20 [0, 200]	20 [0, 100]	30 [0, 200]	30 [2, 200]	30 [0, 200]	30 [1, 200]	30 [0, 200]
Total transport-related energy (GJ/HH/Year)									
Mean (SD)	2 (7)	20 (30)	5 (30)	4 (10)	1 (3)	10 (10)	2 (4)	3 (7)	6 (20)
Median [Min, Max]	0.5 [0, 70]	10 [0, 300]	0.3 [0, 300]	0.4 [0, 100]	0.3 [0, 30]	10 [0, 60]	0.5 [0, 30]	0.8 [0, 30]	0.5 [0, 300]

	Buguruni (N=242)	Kawe (N=178)	Keko (N=113)	Kimara (N=221)	Manzese (N=235)	Msasani (N=94)	Mwananyamala (N=169)	Sinza (N=111)	Overall (N=1363)
Total residential energy (GJ/HH/Year)									
Mean (SD)	30 (30)	50 (40)	50 (50)	30 (20)	30 (20)	50 (40)	40 (20)	30 (30)	40 (30)
Median [Min, Max]	30 [0, 200]	40 [3, 300]	30 [0, 300]	30 [0, 100]	30 [0.5, 200]	40 [2, 200]	30 [0.3, 200]	30 [2, 200]	30 [0, 300]

4.11.7 Correlation Matrix Showing Variable Relationships

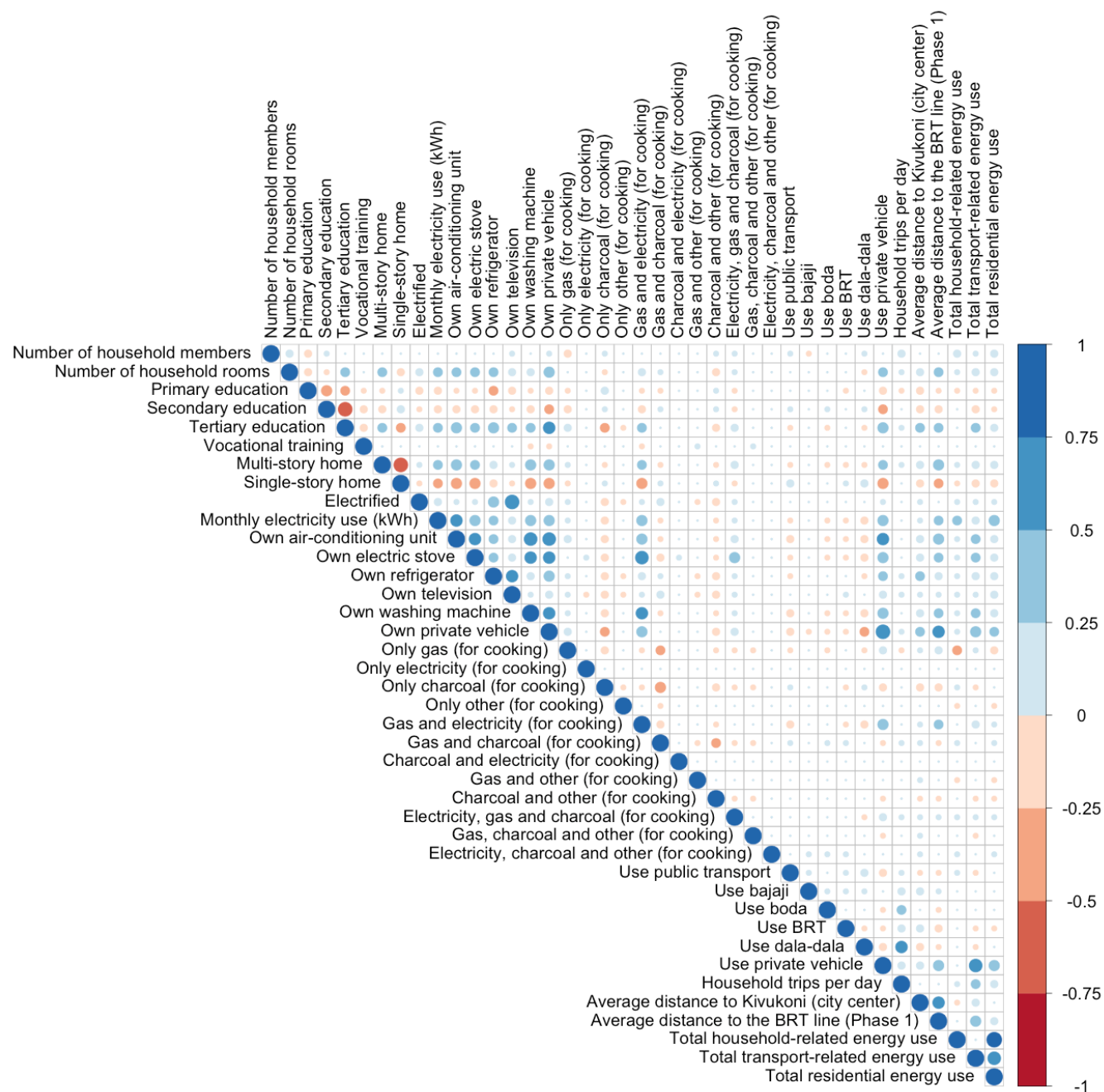


Figure S 4.3. Correlation matrix showing the linear relationships between paired variables. Positive coefficients of correlation are shown in a graduating blue color (where darkest blue color indicates the strongest positive correlation) on the color ramp. Negative coefficients of correlation are indicated in a graduating red color (where the darkest red indicates the strongest negative correlation) on the color ramp.

4.11.8 Effect of Cooking Fuel Choice on Household Energy Use

Table S 4.4. Results from LSD test results showing differences in mean household-related energy use according to household cooking choice among the surveyed wards in the Dar es Salaam region. Groups within each ward are identified by letters and ordered according to their household-related energy use. Note that Group ‘A’, which indicates the highest energy using households within each ward, may be different between wards. Households grouped with the same letter indicate no statistically significant difference between their group means.

Dependent variable (disaggregated by ward): Log (Household Energy)² (Group indicated by letter(s), ordered within each ward³)								
Note: NRO refers to No Recorded Observation.								
Independent variable: Cooking choice¹	Msasani	Sinza	Buguruni	Keko	Manzese	Mwananyamala	Kawe	Kimara
Electricity, charcoal, and other ⁴	NRO	NRO	NRO	NRO	NRO	NRO	NRO	4.7 A
Electricity, gas and charcoal	3.7 AB	3.5 A	3.6 A	3.7 A	NRO	3.8 A	3.7 A	3.4 BC
Charcoal and electricity	NRO	3.2 AB	3.9 A	NRO	NRO	NRO	NRO	NRO
Gas, charcoal and other ⁴	4.1 AB	3.7 A	3.6 A	3.6 A	3.4 A	3.4 A	3.1 B	3.5 AB

Dependent variable (disaggregated by ward): Log (Household Energy) ² (Group indicated by lettering shown below ³)								
Note: NRO refers to No Recorded Observation.								
Independent variable: Cooking choice ¹	Msasani	Sinza	Buguruni	Keko	Manzese	Mwananyamala	Kawe	Kimara
Only charcoal	5.02	3.5	3.3	3.5	3.5	3.5	3.3	3.1
	A	A	A	A	A	A	AB	BCD
Gas and charcoal	3.6	3.4	3.4	NRO	3.4	3.4	3.5	3.3
	AB	A	A		A	A	A	BC
Gas and electricity	3.5	3.1	NRO	NRO	NRO	3.1	3.2	3.2
	AB	AB				A	B	BCD
Charcoal and other ⁴	NRO	2.7	3.2	3.4	3.5	NRO	2.7	3
		ABC	A	A	A		BC	CD
Only gas	2.8	2.9	2.2	2.4	2	2	2.9	2.2
	B	BC	B	A	A	B	BC	DE
Gas and other ⁴	NRO	NRO	3.2	NRO	1.4	3.5	2.2	1.9
			AB		BC	A	C	EF

Dependent variable (disaggregated by ward): Log (Household Energy) ² (Group indicated by lettering shown below ³)								
Note: NRO refers to No Recorded Observation.								
Independent variable: Cooking choice ¹	Msasani	Sinza	Buguruni	Keko	Manzese	Mwananyamala	Kawe	Kimara
Only electricity	NRO	1.4 C	NRO	NRO	NRO	NRO	NRO	NRO
Only other ⁴	NRO	NRO	0.5 C	0.7 B	0.3 C	NRO	NRO	1.1 F
Significance (ANOVA)	p<0.01 *** in all cases							
Note:	*p<0.1; **p<0.05; ***p<0.01							

Table Notes:

¹ My dataset included no household reporting of the following cooking fuel choice combinations: (1) electricity, gas and other (i.e., either kerosene or wood), (2) charcoal and other, and (3) electricity and other.

² Household-related energy use estimates were log transformed to reduce skewness. Note, household-related energy use is estimated based on the sum of energy uses from household charcoal, gas and electricity use.

³ Variables with one or more of the same letters are not statistically different from each other.

⁴ “Other” refers to either kerosene or fuelwood. Note, these fuels are not included in my final estimate for household-related energy use.

4.11.9 Full set of results from OLS and Tobit Regressions

Table S 4.5. Multivariate OLS and Tobit regression results showing the statistical relationship between cooking fuel choice, selected spatial and socio-economic variables, and household-related energy use across surveyed households in the Dar es Salaam region. We compare results when Principal Component (PC) scores and original proxy variables for household wealth are applied as predictors. Coefficients for PC1 and PC2 are statistically significant in all cases ($p < 0.01$). While, in some cases, there was no discernable relationship between some of the original variables (e.g., electric stove and tertiary education) and household-related energy use.

		Dependent variable:							
		Log(Household Energy)							
Independent variable:		OLS (1)	OLS (2)	OLS (3)	OLS (4)	Tobit (5)	Tobit (6)	Tobit (7)	Tobit (8)
<u>Spatial</u>									
	Informal settlement	-0.023	0.058	-0.033	0.060	-0.025	0.057	-0.034	0.059
		(0.065)	(0.076)	(0.066)	(0.076)	(0.065)	(0.076)	(0.066)	(0.077)
	Mixed settlement	-0.089*	-0.034	-0.085	-0.037	-0.090*	-0.034	-0.085*	-0.038
		(0.051)	(0.059)	(0.051)	(0.060)	(0.051)	(0.060)	(0.051)	(0.060)
	Log(Ward density)	0.086**	0.149***	0.093**	0.143***	0.086**	0.150***	0.093**	0.143***
		(0.036)	(0.042)	(0.037)	(0.043)	(0.036)	(0.042)	(0.037)	(0.043)

		Dependent variable:							
		Log(Household Energy)							
Independent variable:		OLS (1)	OLS (2)	OLS (3)	OLS (4)	Tobit (5)	Tobit (6)	Tobit (7)	Tobit (8)
<u>Socio-economic</u>									
Log(Household members)		0.214***	0.362***	0.219***	0.366***	0.215***	0.363***	0.219***	0.367***
		(0.031)	(0.035)	(0.032)	(0.036)	(0.032)	(0.036)	(0.032)	(0.036)
PC1		0.018***	0.010***			0.018***	0.010***		
		(0.002)	(0.002)			(0.002)	(0.002)		
PC2		0.007***	0.013***			0.007***	0.013***		
		(0.002)	(0.003)			(0.002)	(0.003)		
Tertiary education				0.060	-0.063			0.062	-0.062
				(0.042)	(0.049)			(0.042)	(0.049)
Own vehicle				0.155***	0.050			0.156***	0.049
				(0.054)	(0.063)			(0.054)	(0.063)
Own air-conditioning unit				0.246***	0.180**			0.245***	0.180**
				(0.071)	(0.083)			(0.071)	(0.083)
Own electric stove				0.124	0.033			0.126	0.033

		Dependent variable:							
		Log(Household Energy)							
Independent variable:		OLS (1)	OLS (2)	OLS (3)	OLS (4)	Tobit (5)	Tobit (6)	Tobit (7)	Tobit (8)
				(0.082)	(0.074)			(0.082)	(0.075)
	Own television			0.176***	0.257***			0.181***	0.264***
				(0.047)	(0.054)			(0.047)	(0.054)
	Own washing machine			0.143*	0.103			0.142*	0.103
				(0.075)	(0.086)			(0.075)	(0.087)
	Own refrigerator			0.130***	0.133***			0.128***	0.134***
				(0.042)	(0.048)			(0.042)	(0.048)
<u>Cooking fuel choice</u>									
	Only charcoal	2.512***		2.505***		2.669***		2.662***	
		(0.142)		(0.143)		(0.150)		(0.151)	
	Only gas	1.508***		1.502***		1.666***		1.658***	
		(0.147)		(0.148)		(0.155)		(0.155)	
	Gas and electricity	1.904***		1.881***		2.058***		2.035***	
		(0.169)		(0.175)		(0.175)		(0.181)	

		Dependent variable:							
		Log(Household Energy)							
Independent variable:		OLS (1)	OLS (2)	OLS (3)	OLS (4)	Tobit (5)	Tobit (6)	Tobit (7)	Tobit (8)
Gas and charcoal		2.413*** (0.141)	2.413*** (0.143)	2.571*** (0.149)	2.570*** (0.150)				
Charcoal and electricity		2.243*** (0.448)	2.276*** (0.452)	2.398*** (0.451)	2.430*** (0.455)				
Gas and other		1.475*** (0.251)	1.481*** (0.252)	1.635*** (0.256)	1.640*** (0.257)				
Charcoal and other		2.416*** (0.147)	2.407*** (0.148)	2.574*** (0.155)	2.564*** (0.155)				
Electricity, gas and charcoal		2.241*** (0.174)	2.249*** (0.180)	2.396*** (0.181)	2.402*** (0.186)				
Gas, charcoal and other		2.504*** (0.170)	2.506*** (0.171)	2.662*** (0.177)	2.663*** (0.178)				
Electricity, charcoal and other		3.961*** (0.612)	3.913*** (0.613)	4.119*** (0.616)	4.069*** (0.615)				

		Dependent variable:							
		Log(Household Energy)							
Independent variable:		OLS (1)	OLS (2)	OLS (3)	OLS (4)	Tobit (5)	Tobit (6)	Tobit (7)	Tobit (8)
Fuel stack when cooking			0.326*** (0.040)		0.332*** (0.040)		0.328*** (0.040)		0.334*** (0.040)
Constant		0.067 (0.357)	1.129*** (0.378)	-0.496 (0.371)	0.808** (0.402)	-0.094 (0.362)	1.116*** (0.381)	-0.660* (0.375)	0.792** (0.404)
Observations		1,363	1,363	1,363	1,363	1,363	1,363	1,363	1,363
							(Censored: 14)		
Pseudo R ² (McFadden)						0.805	0.769	0.805	0.805
R ²		0.418	0.193	0.420	0.198				
Adjusted R ²		0.411	0.188	0.411	0.191				

Table Notes:

- *p<0.1; **p<0.05; ***p<0.01.
- Standard error denoted by values in parentheses.
- Reference variables not included in table: Formal settlement; Only electricity (for cooking).
- “Other” refers to either kerosene or fuelwood. Note that these fuels are not included in my final estimate for household-related energy use.
- S8, includes a supporting correlation matrix of all data variables.

Table S 4.6. Multivariate OLS and Tobit regression results showing the statistical relationship between public transport use, selected spatial and socio-economic variables and transport-related energy use across the surveyed households in the Dar es Salaam region. Coefficients for PC1 and PC2 are statistically significant in all cases ($p < 0.01$). While, in some cases, there was no discernable relationship between some of the original variables (e.g., washing machine and refrigerator) and transport-related energy use.

Dependent variable:				
Log(Transport Energy Use)				
Independent variable:	OLS (1)	OLS (2)	Tobit (3)	Tobit (4)
<u>Spatial</u>				
Informal settlement	0.316*** (0.103)	0.200** (0.101)	0.402*** (0.145)	0.282** (0.141)
Mixed settlement	0.141* (0.085)	0.118 (0.082)	0.148 (0.118)	0.123 (0.114)
Log(Ward density)	-0.197*** (0.064)	-0.101 (0.063)	-0.177* (0.091)	-0.079 (0.089)
Log(Distance to City Center)	0.028 (0.088)	0.075 (0.085)	0.078 (0.124)	0.126 (0.120)

Dependent variable:				
Log(Transport Energy Use)				
Independent variable:	OLS (1)	OLS (2)	Tobit (3)	Tobit (4)
<u>Socio-economic</u>				
Log(Household members)	0.119**	0.106**	0.184***	0.158**
	(0.048)	(0.047)	(0.069)	(0.066)
PC1	0.043***		0.053***	
	(0.003)		(0.004)	
PC2	-0.007*		-0.005	
	(0.004)		(0.005)	
Tertiary education		0.174***		0.241***
		(0.064)		(0.089)
Own private vehicle		1.100***		1.266***
		(0.083)		(0.114)
Own air-conditioning unit		0.534***		0.618***
		(0.109)		(0.148)
Own washing machine		-0.149		-0.168

Dependent variable: Log(Transport Energy Use)				
Independent variable:	OLS (1)	OLS (2)	Tobit (3)	Tobit (4)
		(0.114)		(0.155)
Own refrigerator		0.062		0.093
		(0.063)		(0.089)
Own television		0.070		0.196*
		(0.070)		(0.101)
Own electric stove		0.169*		0.171
		(0.097)		(0.133)
Use public transport	-0.130*	-0.055	0.054	0.128
	(0.074)	(0.072)	(0.107)	(0.104)
Constant	2.851***	0.919	2.053**	-0.151
	(0.682)	(0.693)	(0.961)	(0.978)
Observations	1,363	1,363	1,363	1,363
			(Censored: 441)	
Pseudo R ² (McFadden)			0.702	0.709

Dependent variable:				
Log(Transport Energy Use)				
	OLS (1)	OLS (2)	Tobit (3)	Tobit (4)
R ²	0.326	0.379		
Adjusted R ²	0.322	0.373		

Table Notes:

- *p<0.1; **p<0.05; ***p<0.01
- Standard error denoted by values in parentheses.
- Reference variables not included in table: Formal settlement.
- Supplementary material, S8, includes a supporting correlation matrix of all data variables.

4.11.10 Multi-logit Regressions to Interpret Travel Behavior

Table S 4.7. Results from multi-logit regression models showing the effect of selected socio-economic variables on the travel mode choice of residents in the surveyed wards in the Dar es Salaam region.

		Travel Mode Choice				
		Bicycle/Walk	Boda/Bajaji	BRT	Private vehicle	Taxi/Uber
		Model I	Model II	Model III	Model IV	Model V
<u>Spatial</u>						
	Formal settlement	-0.169 (0.194)	-1.061*** (0.213)	-2.441*** (0.245)	-0.550*** (0.202)	-0.035 (0.501)
	Informal settlement	0.342*** (0.131)	-1.289*** (0.178)	-1.892*** (0.190)	-0.443** (0.184)	-0.712 (0.448)
	Mixed settlement	0.042 (0.168)	-0.959*** (0.239)	-2.085*** (0.285)	-0.565** (0.279)	0.280 (0.545)
	Distance to the city center	-0.337* (0.174)	1.621*** (0.234)	2.981*** (0.274)	1.092*** (0.274)	-0.766 (0.555)
<u>Socio-economic</u>						
	Log(Household members)	-0.071 (0.114)	-0.173 (0.161)	-0.041 (0.159)	0.198 (0.179)	-0.668 (0.431)
	PC1	-0.029*** (0.007)	-0.016** (0.008)	-0.013* (0.007)	0.149*** (0.007)	0.069*** (0.016)
	PC2	0.011 (0.008)	0.030*** (0.010)	0.018* (0.010)	0.022** (0.010)	0.059** (0.026)

Travel Mode Choice					
	Bicycle/Walk	Boda/Bajaji	BRT	Private vehicle	Taxi/Uber
	Model I	Model II	Model III	Model IV	Model V
Use public transport	-1.163***	-0.859***	0.018	-1.632***	0.148
	(0.172)	(0.228)	(0.290)	(0.198)	(0.748)
Constant	0.215	-3.310***	-6.418***	-1.559***	-0.468
	(0.363)	(0.510)	(0.586)	(0.552)	(1.219)

Table Notes:

- ***p < 0.01; **p < 0.05; *p < 0.1
- Standard error denoted by values in parentheses.
- Reference mode not included in the table: Dala-dala

4.11.11 Inequalities in Energy Use

In economics, Lorenz curves are typically used to estimate income inequality (Kammen & Kirubi, 2008), but in this case, we use them to show inequalities in energy use across the surveyed Dar es Salaam households. Other studies have applied similar methods, e.g., (Jacobson et al., 2005; Kammen & Kirubi, 2008; Wu et al., 2017). I used the same method outlined in (Jacobson et al., 2005) and (Kammen & Kirubi, 2008). First, I calculated each household's share of total energy use, i.e., their energy use divided by the total for the entire sample. Second, I sorted households from the lowest to highest energy user, i.e., based on their relative share of total energy use. I then plotted adapted Lorenz curves for residential energy use, household-related and transport-related energy use, separately. Finally, I calculated Gini coefficients for each plot. Citing directly from Jacobson et al. (2005), “the Gini coefficient for energy consumption is calculated as:

$$G_e = 1 - \sum_i (Y_{i+1} + Y_i) \times (X_{i+1} - X_i)$$

where X_i = (number of energy users in population group i)/(total population) and Y_i = (quantity of energy used by population group i)/(total energy use), with Y_i ordered from lowest to highest energy consumption. The Gini coefficient ranges from perfect equity among all members of the population ($G_e = 0$) to complete inequity ($G_e = 1$). Because more than one Lorenz distribution of a resource can lead to the same Gini value, it is often instrumental to view both metrics simultaneously.” ((Jacobson et al., 2005), page 1826)

Adapted Lorenz curves and estimated Gini coefficients for energy use (G_e) across the surveyed households are shown in Figure S 4.4 to Figure S 4.6.

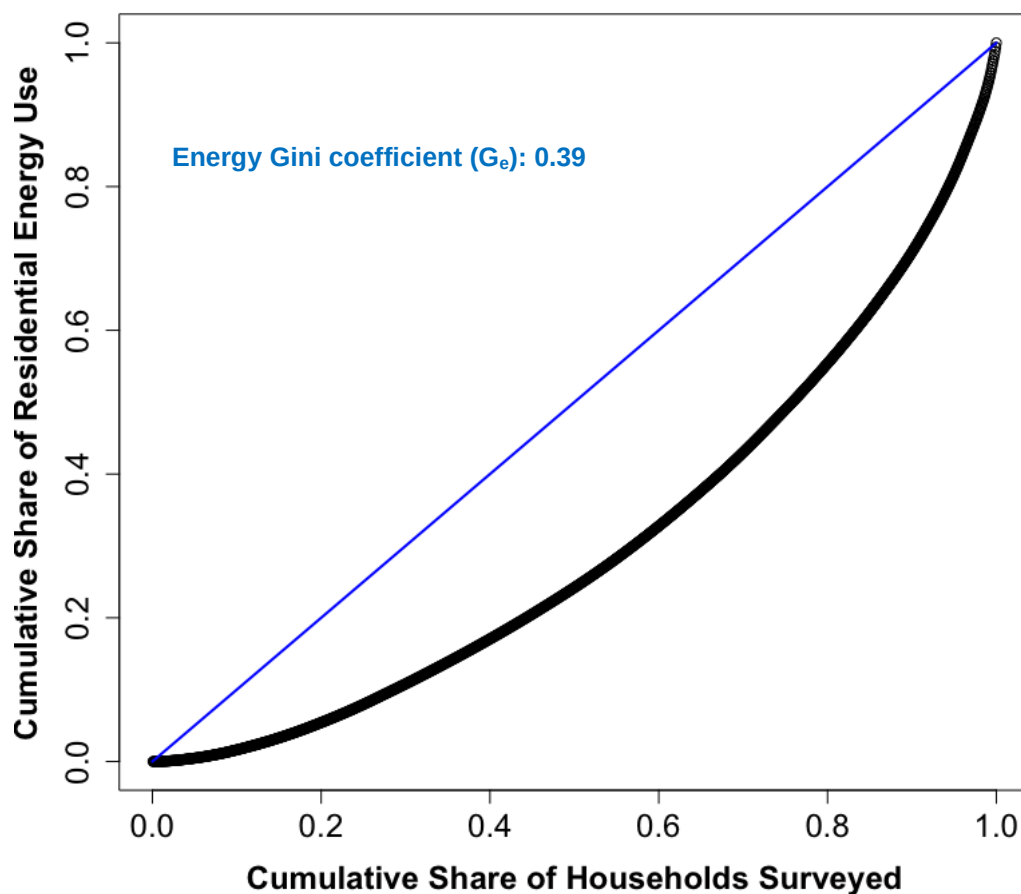


Figure S 4.4. Adapted Lorenz curve showing inequality in residential energy use across the surveyed households in Dar es Salaam. Gini coefficients range from perfect equality ($G_e=0$) to complete inequity ($G_e=1$) (Kammen & Kirubi, 2008). Inequality is indicated as the distance between the curved line (shown in black) and the diagonal straight line (which indicates perfect equality, shown in blue) that extends from the origin to the end point.

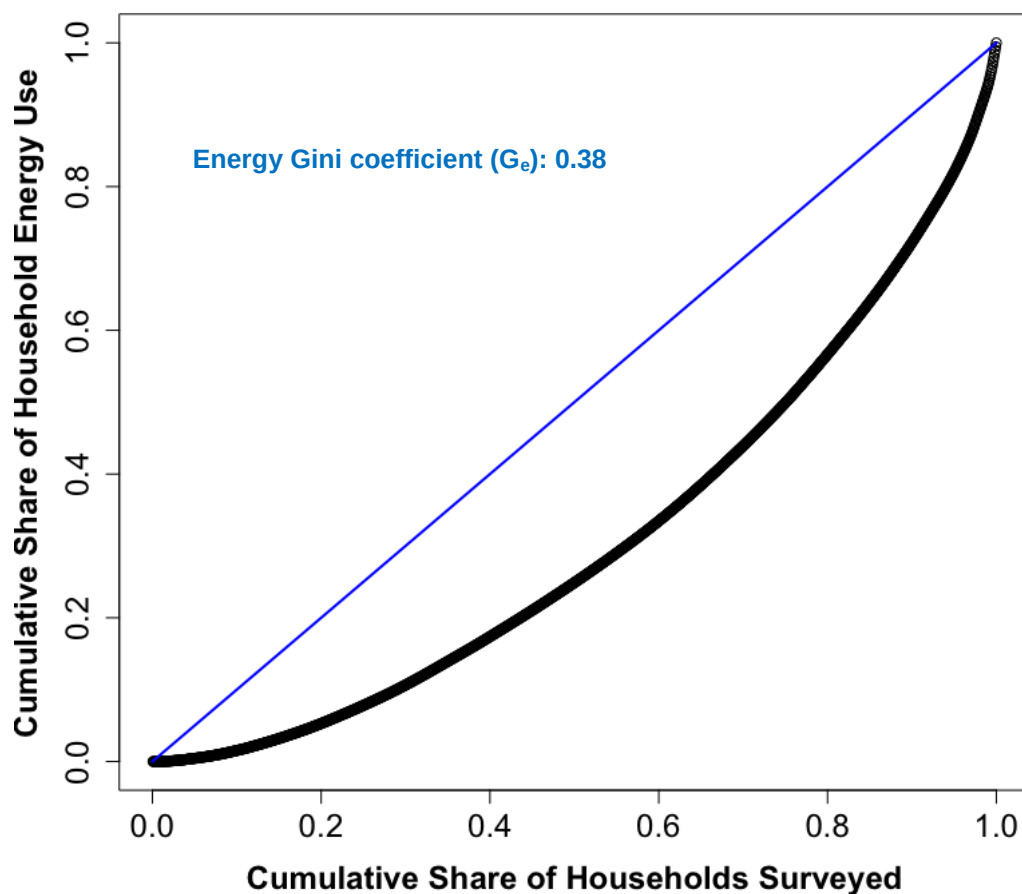


Figure S 4.5. Adapted Lorenz curve showing inequality in household-related energy use across the surveyed households in Dar es Salaam. Gini coefficients range from perfect equality ($G_e=0$) to complete inequity ($G_e=1$) (Kammen & Kirubi, 2008). Inequality is indicated as the distance between the curved line (shown in black) and the diagonal straight line (which indicates perfect equality, shown in blue) that extends from the origin to the end point.

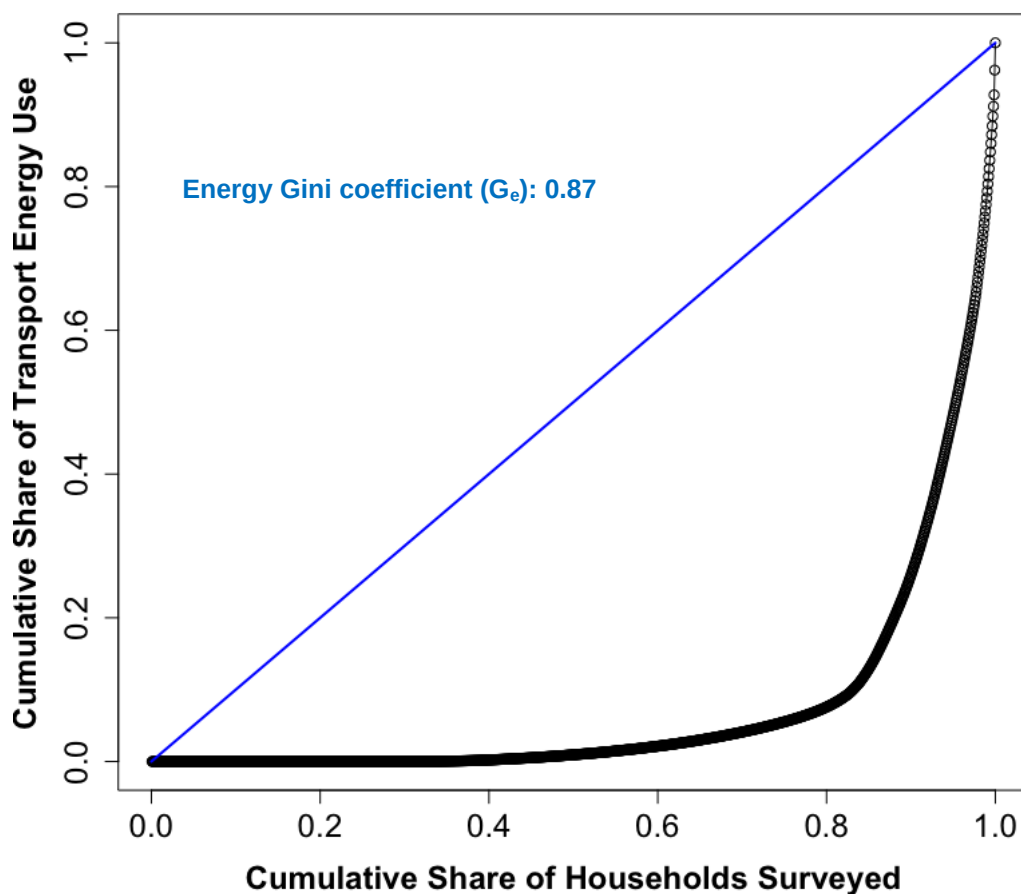


Figure S 4.6. Adapted Lorenz curve showing inequality in transport-related energy use across the surveyed households in Dar es Salaam. Gini coefficients range from perfect equality ($G_e=0$) to complete inequity ($G_e=1$) (Kammen & Kirubi, 2008). Inequality is indicated as the distance between the curved line (shown in black) and the diagonal straight line (which indicates perfect quality, shown in blue) that extends from the origin to the end point.

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Chapter 5

Assessing Institutional and Societal Barriers to Low-Carbon Development in Dar es Salaam

This Chapter is based on a paper being prepared for submission to a peer-reviewed journal, citation below.

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5.1 Abstract

The ongoing and expected future growth of African cities presents an opportunity for stakeholders to promote and adopt low-carbon measures. This study interrogates this discourse and assesses the institutional and societal barriers that may constrain low-carbon development in African cities using the case of Dar es Salaam. Findings from interviews with key informants from national and local government institutions, donor agencies, academia and the private sector suggest that the following institutional factors constrain the implementation of low-carbon initiatives in Dar es Salaam: (1) few ongoing city-level initiatives despite robust policy frameworks for national and local climate change response, (2) limited jurisdiction and capacity of local authorities to spearhead initiatives in key sectors (e.g., transportation and energy), and (3) unawareness of policy processes that engage and sensitize key stakeholders, across multiple sectors, on low-carbon measures. Findings from surveyed households further indicate that the cost and accessibility of low-carbon measures in the residential sector, e.g., electricity use in the home, or the Bus Rapid Transit (BRT) system for commuting, constrain use and uptake among Dar es Salaam residents. To promote higher uptake of low-carbon measures in Dar es Salaam, I recommend for collaborative engagement between a cross-cutting group of stakeholders (e.g.,

local and national government, the private sector, donor groups, and infrastructure services providers) to prioritize and implement low-carbon projects in Dar es Salaam (e.g., electrification initiatives, or planned extensions to the BRT line). Furthermore, local government authorities (e.g., Municipalities and the Dar es Salaam City Council) could take leadership to (1) ensure that low-carbon projects consider the societal context (e.g., household energy use and travel behavior, or the affordability and availability of the BRT service to communities), and (2) engage local communities throughout implementation and planning processes to ensure that overall project objectives align with their socio-economic needs. Finally, while Dar es Salaam was studied in this chapter, these recommendations are expected to be broadly applicable to other cities in the Africa region.

5.2 Introduction

On average, residents of Dar es Salaam use about 17 gigajoules per capita (GJ/capita) of energy annually, based on 2011 estimates of Grubler et al. (2013). To put this in perspective, this is about 27% of the energy used by a resident of Paris (63 GJ/capita), 59% of São Paulo (29 GJ/capita), and 13% of New York (127 GJ/capita) (Kennedy et al., 2015). In the absence of policy measures to support low-carbon development, according to my recent work (Chapter 3), Dar es Salaam's energy use and associated greenhouse gas (GHG) emissions could increase from 1,400 kilotons of carbon dioxide equivalents (ktCO₂e) in 2015 to up to 33,000 ktCO₂e in 2050 (Table 3.4). The latter value is similar to current (2013 to 2015) emissions reported for other major global cities such as New York, San Francisco, and London (Chapter 3). Regionally, van der Zwaan et al. (2018) assert that Africa's CO₂ emissions will become significant at a global level. The authors project Africa could contribute as much as 18% of global CO₂ emissions in 2050 (van der Zwaan et al., 2018), which would be a five-fold increase from Africa's 3.7% contribution in 2018 (IEA, 2019a) and equivalent to the combined 2018 contribution of the United States and Canada (IEA, 2019b). To avoid this possible future, stakeholders – including researchers and development practitioners – have called on African governments to “to act” and prioritize low-carbon development as cities expand, e.g., (AfDB, n.d.; Cartwright et al., 2015; Colenbrander et al., 2019). Most critically, stakeholders highlight the potential for low-carbon

development to spur additional socio-economic gains (e.g., accelerated economic growth, poverty reduction and job creation) across the continent.

Low-carbon development, as defined in this paper, includes city actions to reduce the carbon (or GHG) intensity of urban activities. This may include policies and investments to replace traditional fuels for cooking with modern options (e.g., electricity) or promoting sustainable public transport or non-motorized travel (Kennedy et al., 2019), among other measures. In the context of Africa, regional reports and studies (e.g., African Union (2014), Tait & Euston-Brown (2017), Bawakyillenuo et al. (2018)) have called on policymakers in local and national government to incorporate low-carbon development as part of their urban planning and development goals. Meanwhile, other authors have asserted that low-carbon development could be enabled through initiatives that avoid a “lock-in” to GHG or carbon-intensive urban growth (IPCC, 2014; Cartwright et al., 2015; Kennedy et al., 2014, 2018, 2019). Specifically, the work of Kennedy et al. (2014; 2018; 2019) describes different urban infrastructure strategies to support low-development, including (1) low-carbon electricity generation and supply from renewable sources such as solar, hydro, wind and geothermal; (2) the use of electric appliances, engines and devices (i.e., electrification); and (3) energy efficiency measures (e.g., use of efficient electric appliances or transport modes).

However, the capacity of African cities to effectively implement low-carbon initiatives may be constrained by various institutional and societal factors. Firstly, local governments are often overburdened with competing development agendas (e.g., rising urban poverty, informality and inequality (Lall et al., 2017)), or they are severely restricted within multi-level governance arrangements that allocate more power to national government institutions (Bawakyillenuo et al., 2018; Tait and Euston-Brown, 2017). Secondly, societal factors present additional challenges as community energy consumption behaviors may be influenced by various socio-economic factors, e.g., individual preferences for public or private transportation (Nkurunziza, et al. 2012; Salon & Aligula, 2012) or household energy choices (Bisu et al., 2016; Makonese et al., 2018; Uhunamure et al., 2017; Doggart et al., 2020). Findings from these specific studies are detailed in the literature review (Section 5.3). However, with the exception of the study by Doggart et al. (2020), no other study has simultaneously included both expert judgments of key informants

(i.e., technical experts in national and local government institutions, donor agencies, academia and the private sector, among other groups) and judgments of community members in the authors' assessments and/or policy recommendations. In this paper, I argue that local government authorities (e.g., Municipalities and the Dar es Salaam City Council) – as the most visible institution to urban residents – are uniquely positioned to ensure that low-carbon projects (e.g., wide-scale electrification or public transport projects) align with the socio-economic needs of communities (especially the poor). Therefore, enabling progress on low-carbon development would require two-pronged approach, i.e., addressing barriers to low-carbon development from an institutional perspective, and ensuring that policies account for societal factors (i.e., resident energy choices and travel behaviors) that may enable or constrain the wider uptake of low-carbon investments in communities.

This study aims to address these gaps in the literature. Specifically, I identify possible institutional and societal factors that may constrain progress on low-carbon development in Dar es Salaam. For example, how do institutions of urban governance influence progress on low-carbon development? What societal factors and supply considerations (e.g., availability of fuels for cooking or transport options) contribute to household cooking fuel and travel mode choices? And what are resident perceptions on the environmental and health burdens associated with the use of wood-based fuels (i.e., charcoal)? To answer these questions, I draw on the perspectives of two stakeholder groups in Dar es Salaam: (1) key informants (experts) across national and local government institutions, donor agencies, academia and the private sector (see Figure 5.2), and (2) surveyed households across eight Dar es Salaam wards (Table 5.1). My other work (Chapter 4) is based on the same household survey and presents the differences in residential energy use among the surveyed wards. This current study examines institutional and societal aspects, which were not considered in my previous work, but which constitute an essential part of the overall survey and research objectives (details in Methods, Section 5.5).

I selected Dar es Salaam as a case study as it is the fastest-growing city in East Africa, estimated to grow at over 4% annually between 2020 and 2035 (UN, 2018; Figure 1.1). Dar es Salaam is also the largest city in Tanzania, with a population of 6.1 million people as of 2018, and the most important city for business and economic activity (see Chapter 4). I also focus my analysis on the

residential sector given the sector's high contribution to energy use in Africa. According to the International Energy Agency (IEA), the residential sector accounted for 64% of energy demand in sub-Saharan Africa, compared to 14% from industry, 15% from transportation (including domestic and commercial transportation), and 8% from other energy-using sectors (IEA, 2019a). This current paper extends on my previous work (Chapter 4) and uses the term “residential energy use” to refer to domestic household and transport-related energy use, behaviors, and perspectives, as well as various demographic aspects.

There are four remaining sections of this paper. Section 5.3 reviews the literature on institutional responses to urban climate change (mitigation and adaptation) in African cities, and specifically, in Dar es Salaam. Section 5.5 presents the research methods, i.e., Dar es Salaam key informant interviews, household surveys, and their analyses. Sections 5.6 and 5.7 discuss overall research findings and conclusions.

5.3 Literature Review

5.3.1 The Role of Institutions in Supporting Climate Change (Mitigation and Adaptation) Responses in African Cities

Literature on urban governance in African cities asserts that processes of collaboration between national and local institutions influence progress on climate change adaptation and mitigation (low-carbon development) in African cities (Diep et al., 2016; Herslund et al., 2018; Leck & Simon, 2018; Vedeld et al., 2016, 2015). For example, Leck and Simon (2018) conducted in-depth, semi-structured interviews with government officials in two neighboring municipalities in South Africa: eThekweni and Ugu (the latter is predominantly rural). The authors found that weak “inter-municipal collaboration” across the two city regions significantly impeded progress on local climate change adaptation responses. In Saint Louis (Senegal), Vedeld et al. (2016) conducted interviews with public, private and civil society officials across local and national government. In their policy recommendations, the authors suggested that progress on climate

change adaptation in Saint Louis could be accelerated through collaborative processes that engaged local and national authorities (Vedeld et al., 2016).

The few regional studies focusing on climate change/GHG mitigation have similarly suggested that collaboration between national and local governments (municipalities) would be required to support low-carbon development in cities, e.g., Bawakyillenuo et al. (2018) and Tait and Euston-Brown (2017). In the case of Dar es Salaam, Shemdoe et al. (2015) recommended technical training and capacity building among local officials to improve local responses to climate change mitigation and adaptation in the metropolitan region (Shemdoe et al., 2015). However, Bawakyillenuo et al. (2018), Tait and Euston-Brown (2017), and Shemdoe et al. (2015) did not consider or integrate the community context in their policy recommendations.

5.3.2 Community Energy Use Behaviors in African Cities

There is a body of literature, distinct from the one reviewed in Section 5.3.1, that relies on household surveys to understand the societal factors that influence community energy use or travel behaviors, e.g., the household's choice to use charcoal instead of electricity or gas for cooking (Bisu et al., 2016; Kimemia & Annegarn, 2011; Mensah & Adu, 2015; Salon & Aligula, 2012; Van der Kroon et al., 2014) or private vehicles instead of non-motorized options or public transport (Nkurunziza, Zuidgeest, & Van Maarseveen, 2012; Salon & Aligula, 2012). For example, Bisu et al. (2016) examined factors affecting household energy choices in Bauchi (Nigeria) using data collected from 100 households. The results showed that kerosene was used by most households as a primary cooking fuel (41% of households), followed by wood (35%), liquefied petroleum gas ("gas") (24%), and charcoal (20%) (no household reported use of electricity or solar energy) (Bisu et al., 2016). In the transport sector, Salon and Aligula (2012) used survey data to examine travel behavior in Nairobi (Kenya). In the absence of policies for safe and reliable non-motorized or public transport, the authors projected that sharp increases in car ownership could be expected given anticipated rises in household wealth expected with future economic growth in Nairobi (Salon & Aligula, 2012).

5.3.3 Household Energy Use in Dar es Salaam and Tanzania

There have been a limited number of studies on household energy use or travel choices in Dar es Salaam, or Tanzania, more broadly, e.g., Choumert-Nkolo et al. (2019) and D’Agostino et al. (2015). Choumert-Nkolo et al. (2019) assessed household cooking fuel choices in Tanzania and showed that improvements in household socio-economic status (i.e., household income) was correlated with a larger diversity of fuels purchased and a propensity for “fuel stacking”, i.e., where households use a combination of two or more fuels for cooking (Choumert-Nkolo et al, 2019). In transportation, few studies have assessed resident travel behaviors in Dar es Salaam. Using interview data from 174 households sampled across five peripheral settlements, Andreasen and Møller-Jensen (2017) examined the travel mode choice of communities in Dar es Salaam. The authors showed that 131 of the 174 surveyed participants depended on the local minibuses (“dala-dala”) for commuting as it is the cheapest form of public transport in Dar es Salaam. To encourage greater use of public transport, the authors recommended the scale-up of public transport services in peripheral communities (which remain largely un-serviced) among their key policy recommendations. Other transport studies in Dar es Salaam have also suggested the need for improved non-motorized transport options for residents, e.g., Mkalawa and Haixiao (2014) and Nkurunziza, Zuidgeest, and Van Maarseveen (2012).

The current study offers new insights on possible barriers to low-carbon development in Dar es Salaam from an institutional perspective. For this, I draw on insights from interviews with selected key informants within national and local government institutions, donor agencies, academia and the private sector. Secondly, based on our household surveys of 1,363 households sampled across formal, informal, and mixed settlements in the city, I assess the household energy use and travel behaviors of Dar es Salaam residents, as these activities contribute to total residential energy use in the city, as demonstrated in my previous work (see Chapter 4). The originality of my study results from drawing insights from both the qualitative key informant interviews and the primary household data we (the field team) collected. The study uses insights from a wide range of key informants and findings from household surveys to understand the institutional and societal context that may be essential to informing progress on low-carbon development in Dar es Salaam.

5.4 Institutions of Urban Governance in Dar es Salaam

In this section, I discuss the institutions of urban governance for the Dar es Salaam region and their relative power to coordinate policies and project financing decisions (refer to Sections 2.3, 3.4, and 4.3.1 for additional background information on Dar es Salaam). Outlining these power structures provides critical context for understanding the possible role of institutions (i.e., national, and local government authorities) to coordinate low-carbon projects in Dar es Salaam.

Figure 5.1 maps the institutions of urban governance in Tanzania and for the Dar es Salaam region. At the highest level, the Regional Administration and Local Government – a body within the office of the President (also known as “PORALG”) – coordinates all regional and local government activities in Tanzania and works alongside sector ministries to develop policies and projects in key infrastructure sectors (e.g., transportation or energy). The Regional Administration for the Dar es Salaam region (still at the level of the national or central government) is headed by a Regional Commissioner, who is appointed by the President, and plays an oversight and coordinating role, including oversight on the duties and functions of local government authorities. At the level of local government, the city has a two-tier governance structure, i.e., urban activities are coordinated by both the Dar es Salaam City Council and five autonomous Municipal Councils (or Districts: Kinondoni, Ilala, Ubungo, Temeke and Kigamboni). Day to day urban management mostly occurs at the City and Municipal Council level, where local government authorities control questions of land use, service delivery (e.g., solid waste management) and have the power to levy taxes and fees (e.g., service taxes, or market and bus fees). The Dar es Salaam City Council also coordinates cross-cutting issues between the Municipal Councils. From an administrative perspective, the Dar es Salaam City Council is headed by an elected Mayor, and each Municipal Council has their own budget, management process, and District Commissioner and Executive Director (who report to national government, i.e., the Regional Commissioner and PORALG). Municipal Councils are further sub-divided into divisions (“Wards”), then streets (“Mtaa”), which are the city’s smallest administrative units (Figure 5.1).

Other aspects relevant to this study are Tanzania's ongoing policy processes for charcoal use and public transportation. Firstly, Tanzania's biomass strategy envisions a 50% reduction in urban charcoal use between 2015 and 2050 to be realized through households' use of improved cookstoves. The national energy policy and action agenda envisions that 75% (or more) of Tanzania's households will substitute their charcoal use with modern alternatives, i.e., electricity or liquefied petroleum gas ("gas") by 2030 (Government of Tanzania, 2015). Secondly, transport and urban development policies for the Dar es Salam region (e.g., Government of Tanzania, 2017) aim to improve the efficiency of public transport, mostly through the Bus Rapid Transit (BRT) system. The BRT development is coordinated at the national level (i.e., through PORALG and the Ministry of Local Government, Figure 5.1), though policy processes have engaged local government institutions (e.g., the Dar es Salaam City Council and Municipal Councils) in design and implementation activities. The Phase 1 line of the BRT is currently in operation (as of 2016), and future phase extensions are being planned (additional details on the BRT are provided in Chapter 3 and Chapter 4).

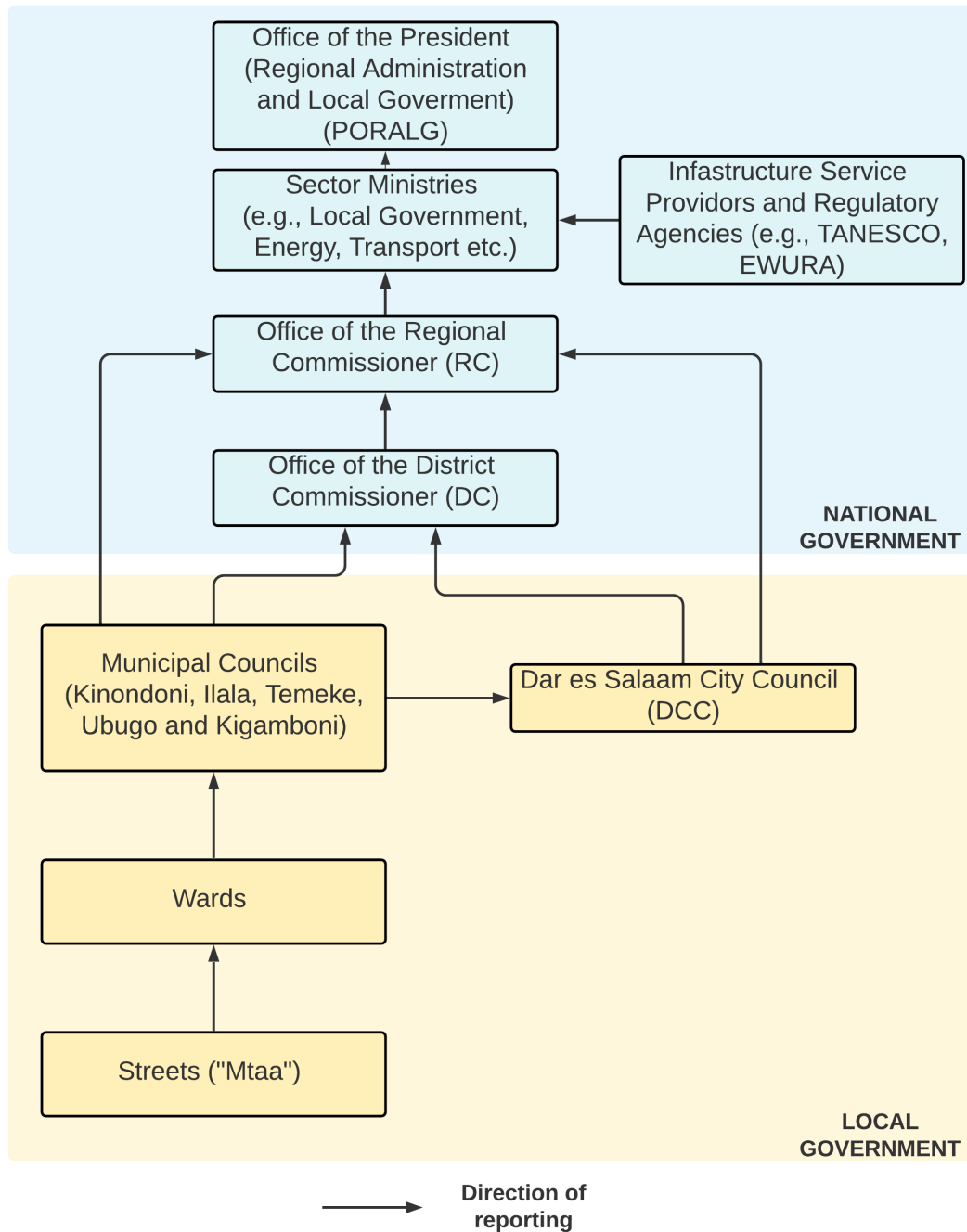


Figure 5.1. Institutions of urban governance in Tanzania and for Dar es Salaam region. Direction of arrow indicates the direction of reporting between institutions in local and national government. This figure was created by the author (Alice Chibulu Luo) based on descriptions provided by key informants during interviews, and other supporting information received from PORALG, with their permission.

5.5 Methods

I employed a phased approach to the study. First, I conducted semi-structured interviews with key informants from government, academia, the private sector, and civil society to examine the key barriers to progress on low-carbon development in Dar es Salaam (further details in Section 5.5.1). I used qualitative methods (thematic analysis) to elucidate key informant perspectives on the institutional factors that constrain the implementation of low-carbon initiatives in Dar es Salaam. Second, I surveyed households, sampled across eight Dar es Salaam wards (Table 5.1), and employed statistical methods (chi-square tests) to test the statistical effects of different settlement types (e.g., informal, formal, or mixed wards) on household energy choices and travel behaviors.

Field activities (i.e., key informant interviews and household surveys) were conducted in Dar es Salaam between August 2018 and November 2018. Earlier visits to Dar es Salaam in the previous year (September 2017) allowed me to establish partnerships with local universities and researchers, pilot early versions of the survey, and receive feedback from community members on the survey's overall interpretation and sensitivity to the local context. The study received research ethics approval through the University of Toronto Ethics Review Board and the Tanzania Commission for Science and Technology (details in Chapter 4, Section 4.11.2).

Subsequent field activities were also conducted in Lusaka (Zambia) between November 2018 and February 2019 (where I also hosted a policy workshop, as discussed in Section 1.5). However, due to resource constraints, I was unable to complete a full study in Lusaka. I broadly summarize preliminary findings from key informant interviews conducted in Lusaka in the supplementary material (Section 5.8.1) but do not incorporate them within the main body of this chapter. Conducting more detailed comparative work and field activities in Lusaka (and possibly, other African cities) would be an important area of future work.

5.5.1 Interviews with Key Informants and Thematic Analysis

I conducted in-person, semi-structured interviews with key informants from local and national government institutions, academia, and the private sector. The key informants were identified based on their prior work, research, expertise, or present position within the institution they represented, e.g., engineers, managers, specialists, professors and urban planners who worked in urban planning, transportation, and energy sectors. I summarize the affiliations and expertise of the interviewed key informants in the supplementary material, Table S 5.1. The first set of experts was identified through early outreach (via email or telephone) using networks established from initial visits in 2017. These initial experts were then invited to suggest additional participants/interviewees who I subsequently contacted for interview. I also conducted introductory discussions (in-person or via telephone) with each participant to describe the overall research objectives, and in some cases, ensured that letters of “research introduction” were provided to their respective institutions (as some experts declined to participate without the approval of existing within their organizations).

I developed open-ended interview questions, and these were designed to cover a range of urban planning, transportation, and energy access topics. The interview format was adapted based on participant expertise or knowledge of any of the topics. For example, transport sector experts had the liberty to address only the transport-related questions, or all questions if they had knowledge of other topics. Interviews lasted approximately one hour and were conducted only after informed consent was obtained. The following three questions informed the main themes presented in Section 5.6.1, although the full set of guiding questions is detailed in the supplementary material (Section 5.8.1).

1. What are your general thoughts on how Dar es Salaam is promoting low-carbon development?
2. In your opinion, what are key institutional factors that enable or constrain the adoption of low-carbon initiatives or measures at the city or national level?
3. Are you aware of any urban sector projects or processes that focus on GHG mitigation or low-carbon development in Dar es Salaam?

Interviews were digitally recorded with participant consent or summarized via hand-written notes. Recorded formats were transcribed using an automated transcription tool (“f4transkript”). Approximately 3 to 4 interviews were completed each week resulting in a final sample of 24 key informants, as shown in Figure 5.2. The final sample size was determined when content saturation was observed during subsequent interviews (i.e., new themes or concepts did not emerge with additional interview participants). The use of content saturation in determining sample size has been widely applied in qualitative research (see Bengtsson (2016), Trotter (2012), and Wolf & Moser (2011)) and therefore was deemed appropriate for this study. Finally, the transcribed interviews were analyzed using thematic analysis (Bengtsson, 2016), i.e., where prevailing themes and ideas that emerged from the interviews were coded manually based on key words and phrases that were commonly used by experts.

5.5.2 Household Surveys

Structured household surveys were conducted in Dar es Salaam for six weeks between August and October 2018 (additional details on the survey are outlined in Section 4.11.1 and 4.11.2). At the start of our survey activities, research permits were obtained from city, district and ward level authorities, i.e., the Regional Administrative Secretary, District Council Directors (for Kinondoni, Temeke, Ubungu and Ilala districts), and Ward Directors in the 8 pre-selected wards. Wards were selected based on their estimated density (according to 2012 census data); residential land-use; assumed average household income, determined based on anecdotal feedback from the field team and local community members; and estimated distance from the closest BRT stop along the current line (Phase 1).

It is important to note that residential land-uses in Dar es Salaam are complex, and there are several nomenclatures to describe the different residential land-use structures in the city, see studies by Kironde (2000; 2006), Kombe (2005), and Lupala (2002). The current study describes settlements as either “formal” or “informal”. Formal wards have land or housing transfers mostly regulated by local or national government, compared to informal wards where there are informal arrangements through individuals (Kironde, 2000; 2006). We also surveyed households from

“mixed” settlements, which are comprised of both formal and informal settlements within the same sub-city boundary. Other common terms in the literature to describe these differences include “unplanned” and “planned” settlements, or “customary” and “non-customary” land (see studies by Kombe (2005) and Lupala (2002)). The socio-economic and spatial characteristics of the surveyed wards are shown in Table 5.1. We interviewed a total of 1,363 households across the 8 wards.

We adopted a stratified random sampling approach in which proportional numbers of households were randomly selected and interviewed in each ward, i.e., estimated based on the ward population density, additional details can also be found in Chapter 4. Our final sample covered a spatially and socio-economically diverse set of wards to ensure that the final sample reflected the range of household and ward typologies present in Dar es Salaam.

Table 5.1. Socio-economic and spatial characteristics of wards surveyed in Dar es Salaam¹

Name	Ward density (people/sq.-km)	Residential land-use	Average household income (assumed) ²	Mean distance of surveyed wards to closest BRT stop (km)	Total number of households surveyed
Msasani	4,402	Formal	High-income	3.4	94
Sinza	12,151	Formal	Middle-income	1.1	111
Buguruni	20,460	Informal	Low-income	3.1	242
Keko	4,336	Informal	Low-income	1.6	113
Manzese	38,496	Informal	Low-income	0.2	235
Kawe	24,179	Mix	High-income	7.9	178
Kimara	5,569	Mix	Middle-income	1.7	221
Mwananyamala	20,409	Mix	Low-income	0.6	169
Total households surveyed					1,363

Table Notes:

¹ Map of wards visited is shown in supplementary file.

² Income categories were determined anecdotally based on insights and local knowledge of the city from residents, researchers, and the field team members.

Surveys were administered in Swahili and later translated into English. The field team participated in a 3-day mandatory training in both English and Swahili prior to administering the surveys. Questions were structured and required participants to respond to a set of pre-determined open-ended and closed-ended questions. In some cases, Likert scale responses were required, i.e., participants were asked to rank their preferences or opinions over a pre-defined range. Questions that were most relevant to this study are detailed in Table 5.2.

Table 5.2. Structured survey questions administered to households in Dar es Salaam^{1,2}**Household energy behaviors**

1. Does the household have access to electricity?³
2. What is your preferred stove for cooking? (i.e., electricity, charcoal, gas or other)?
3. What is the main factor that influences the household's choice to use a non-electric stove for cooking? (i.e., charcoal or gas)⁴
4. Would the household support the eventual phase-out of charcoal use in Dar es Salaam?³
5. Are you aware of the negative health impacts associated with charcoal use for cooking?³

Household travel behaviors and perceptions of the BRT service

1. Does the household own a private car?³
2. Does the household use public transport during the weekday?³
3. What is the household's main mode of travel (i.e., represents the largest travel distance per day)?
4. How accessible is the BRT service to your neighborhood?⁴
5. How affordable is the BRT service relative to the "dala-dala" mini-bus service?⁴

Table Notes:

¹ Only the questions relevant to this study are presented in this table. See Appendix A for the full questionnaire.

² The survey was based on a structured questionnaire.

³ This question was based on a Yes/No response.

⁴ A ranking of options was presented, and participants were required to select one response, scale: 1 = not accessible/affordable at all, 6 = extremely accessible/affordable.

5.5.3 Household Survey Data Cleaning and Statistical Analysis

Household survey response data were cleaned and restructured for analysis using the R programming language. My final data tables (Table 5.3 and Table 5.4) summarize responses based on the proportion of total households that responded to each of the survey questions. Given the categorical nature of the analyzed survey responses, Pearson's chi-square tests of independence (or association) were employed to test the statistical relationship between household responses and ward type (the list of surveyed wards is presented in Table 5.1). Chi-square tests are a widely recognized statistical method for analyzing categorical survey data and

have been used in various Africa region studies examining household energy choices (e.g., electricity, charcoal, firewood, and other fuels), e.g., Nigeria (Ifegbesan et al., 2016; Kiyawa & Yakubu, 2017), South Africa (Uhunamure et al., 2017) and Zimbabwe (Campbell et al., 2003). Chi-square tests are usually performed at a 5% significance level, meaning that the test yields a statistically significant result when p-values are below 0.05. Therefore, in the context of this research, p-values less than 0.05 would indicate that survey responses are statistically dependent on ward type.

Employing chi-square tests to determine whether survey responses were statistically dependent on ward type allowed me to answer questions such as, are household perceptions on charcoal use statistically different at the ward level? Are household perceptions of the BRT relative to the local mini-bus (“dala-dala”) statistically different at the ward level? My previous work (Chapter 4) employed other statistical methods (i.e., Analysis of Variance and multivariate regression models) to show the statistical relationship between estimated energy use among the surveyed wards and ward type but did not examine the societal factors that influence household energy use behaviors (i.e., household perceptions of different fuels or transport options) as in the current chapter.

5.6 Results and Discussion

This section first presents results from the key informant interviews (i.e., the institutional perspective), followed by those from the household surveys (i.e., the societal perspective), and lastly, links the two sets of findings and suggests an approach or “way forward” for stakeholders to support progress on low-carbon development in Dar es Salaam.

5.6.1 Dar es Salaam Key Informant Interviews

I interviewed 24 key informants in Dar es Salaam with demonstrated expertise in the following sectors: urban planning, transport and energy (details in supplementary material, Section 5.8.2). Of the 24 key informants, 6 were from academia (referred to as “Academic”), 7 from national government (“NGOV”), 3 from local government (“Municipality”), 4 from non-governmental organizations or donor agencies (“NGO-Donor”), and 4 from the private sector (“Private Sector”).

As shown in Figure 5.2, key informants expressed the following as factors that may be constraining progress on low-carbon development in Dar es Salaam: (1) Policies exist, but implementation processes are inadequate, (2) Local authorities (Municipal and City Councils, see Figure 5.1) have little jurisdiction, (3) Limited funding and own-source-revenue collection among local authorities constrains their capacity to spearhead their own initiatives, and (4) Interviewees were unaware of processes that enabled stakeholders to engage on low-carbon initiatives in Dar es Salaam. Table S 5.1 in the supplementary material shows a more detailed mapping of the key informant responses.

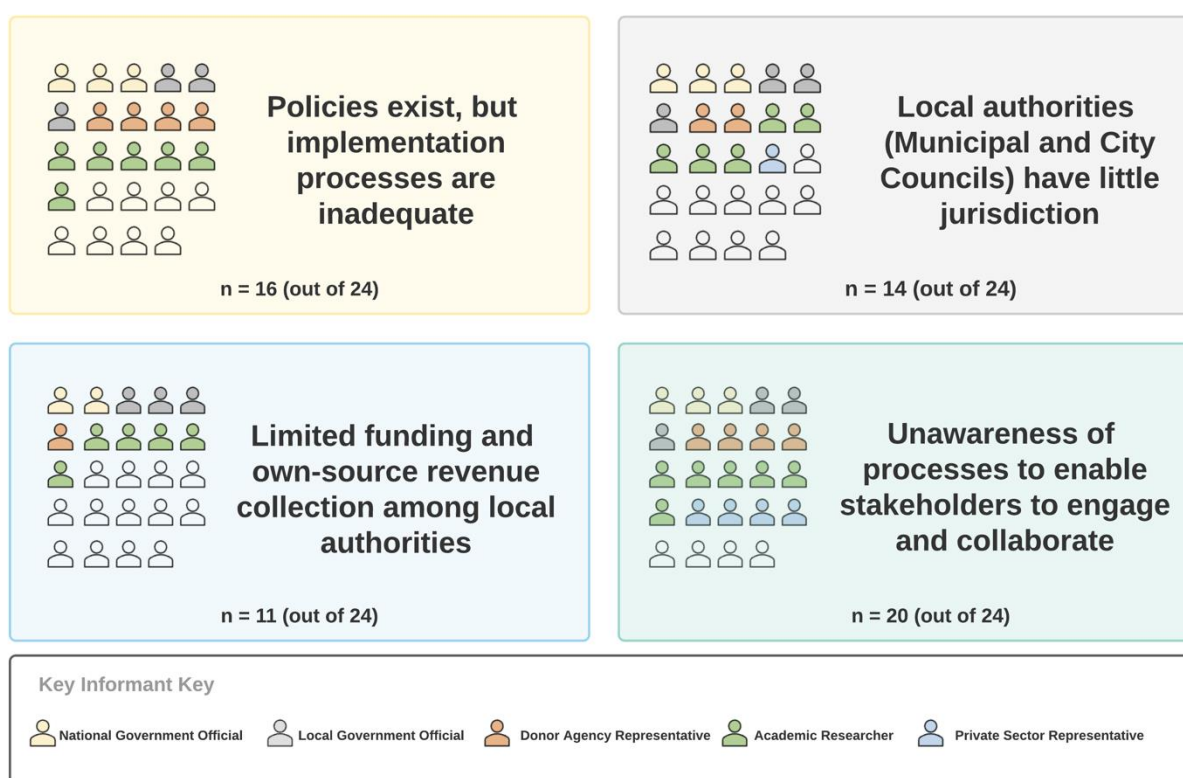


Figure 5.2. Summary of responses from interviews with Dar es Salaam key informants, selected across national and local government institutions, donor agencies, academia and the private sector.

5.6.1.1 Policies exist, but implementation processes at the city-level are inadequate

Sixteen of the 24 key informants interviewed (68%, see Figure 5.2) were of the view that despite several strategies and policies supporting low-carbon development in Tanzania, progress on their implementation at the city level (i.e., for the Dar es Salaam region) is inadequate. “Not much has been done so far,” stated one academic. “There are quite a lot of information on the literature, written documents, but going into the ground – the implementation – not much has been done.” Another donor representative similarly expressed: “You have so many initiatives, and you can

always wonder if they belong to the same country. You know, because there is Sustainable Energy for All, the Action Agenda that Tanzania adopted, you have Scaling-up Renewable Energy, you have Paris Agreements, SDGs, you have so many things and Tanzania participates. But the actual deployment, implementation of it, is usually a question mark.” Another official from national government described Tanzania’s energy sector policies and noted that intended plans are not widely reflected at the city level: *“As I said, the energy sector planning that we have in Tanzania – most of the plans are at the country level (the national level). I am not sure of planning at the city level when it comes to energy.”*

5.6.1.2 Local authorities have little jurisdiction

Historically, the relationship between national and local government in Tanzania has been focused on consolidating power at the national level with little autonomy given to the local authorities (TULab, 2019). After the independence of Tanganyika in 1961, the country inherited the colonial system where local authorities were used as vehicles to provide basic social services (Kombe & Namangaya, 2016). However, by 1974, this initiative was reversed and the national government through other regional bodies (e.g., the Regional Commissioner’s Office and PORALG) became the main provider of services, retaining most, if not all decision-making powers (TULab, 2019). The Tanzanian Local Government Reform Act introduced in 1998 – and based on the principle of decentralization – aimed to empower local authorities with more political, administrative and fiscal control by offering more expenditure responsibility to lower administrative levels and enabling the collection of local government authorities’ own revenues from different sources (e.g., through projects with the private sector or development agencies). The reality, as suggested by 14 of the 24 key informants (58%, Figure 5.2), is that local authorities have limited jurisdiction to influence policy processes and direct investments for the implementation of low-carbon projects in Dar es Salaam. *“Indeed, the decentralization policy exists; but the reality is that the central government is slowly taking over the city”*, stated one donor representative. National government authorities (i.e., PORALG and the sector ministries, see Figure 5.1) retain most decision-making power in key sectors that are important to low-carbon development (e.g., energy and transport). Sector ministries and national utilities

coordinate investments in key infrastructure sectors (e.g., electrification, road transport, and public transportation), which highlights the centralized structures of infrastructure service delivery in Tanzania.

5.6.1.3 Limited funding and own-source revenue collection among local authorities constrains their capacity to spearhead their own initiatives

The existing funding constraints among local authorities was a critical issue highlighted by almost half of the key informants (11 out of 24, i.e., 46%, see Figure 5.2). The views expressed by these key informants are consistent with other local studies showing that 80% to 90% of current local government budgets are disbursed through intergovernmental transfers from national government institutions (e.g., PORALG) (Government of Tanzania, 2016a). These funding flows underscore the sizable dependence of local governments on national government funds, which limits their autonomy to implement low-carbon projects in their jurisdictions. Relatedly, two key informants from government and academia also highlighted that local authorities were limited in their capacity to coordinate budgets and funding flows from donor groups (e.g., the World Bank). Therefore, these key informants expressed that funding flows needed to be coordinated at the national level where financial management systems were more robust but noted that implementation of low-carbon projects should engage both local authorities and communities in a collaborative process. According to one official in national government, *“Even though the implementation of low-carbon investments will be done at the sector level, it needs to include all stakeholders. But currently, every sector is planning on its own.”* Similarly, other studies (e.g., (Diep et al., 2016; Leck & Simon, 2018; Tait & Euston-Brown, 2017; Vedeld et al., 2016)) have highlighted the benefits of multiple stakeholder collaboration in enabling progress on local climate change adaptation or mitigation responses in African cities.

5.6.1.4 Interviewees were unaware of processes that enabled stakeholders to engage on low-carbon initiatives in Dar es Salaam

Twenty of the 24 key informants (83%, see Figure 5.2) were unaware of specific policy processes or initiatives that sought to engage all relevant stakeholders at the national and local level. According to one official from national government, *“there is no clear policy direction for low-carbon development at the city level in a manner that engages all stakeholders. The issue is rarely discussed among institutions, and less so in the urban context.”* One expert in the private sector referred to individual activities focused on off-grid renewable electrification (e.g., solar home systems for residential application), though their implementation was limited to specific communities or neighborhoods; the expert was unable to determine whether such processes could encourage eventual scale-up at the city level. Some key informants referred to the BRT project as an example of a project that has realized significant scale at the city level, through partnership between stakeholders at the city and national level (including the private sector). For example, one NGO (donor) representative highlighted that, *“the initial idea to develop a BRT in Dar es Salaam all started with the City Council. Since the solution has been successfully adopted in Latin America, the idea was that it could be adapted to the Dar es Salaam context. ...So, although the project is now overseen by national government, the initial idea came from local government.”* Another representative from national government also noted that the BRT was not initially implemented from a “GHG mitigation” perspective, but rather to address socio-economic needs and urban mobility challenges in Dar es Salaam. Therefore, the BRT exemplifies how stakeholders at different levels can collaborate on a “development project” that also supports low-carbon development as a “positive co-benefit”.

5.6.2 Dar es Salaam Household Surveys

I present the results from statistical analysis (chi-square tests) that was conducted using data from household responses to questions shown in Table 5.2. Household findings are disaggregated into the following sections: (1) energy access, (2) cooking behavior, (3) perceptions on charcoal use, (4) travel behavior, and (5) perceptions on the affordability and availability of the BRT service.

5.6.2.1 Household Electricity Access

Chi-square tests show a statistically significant effect of ward type on reported electricity access levels among surveyed households ($p < 0.01$), i.e., high and middle-income wards (Msasani, Sinza, Kawe, Mwananyamala and Kimara) are more likely to report “having access to electricity” compared to low-income wards (Buguruni, Manzese and Keko), details in Table 5.3. Overall, between 75% (Buguruni) and 100% (Msasani and Sinza) of surveyed households reported having access to an electricity connection. The lower-bound estimate (75%) is consistent with electricity access levels reported for the Dar es Salaam region in 2017 (i.e., 75% in 2017, see Government of Tanzania (2017)). However, the higher upper-bound estimate (100%) may suggest two things: (1) my survey sample is not entirely representative of the city-wide (Dar es Salaam) average and additional data collection may be required to confirm my estimates, or (2) the city-wide data is under-estimated as it does not consider the heterogeneity in electricity access at the ward level (i.e., as shown in this study and my previous work in Chapter 4). Nonetheless, my findings demonstrate the higher electricity access levels in Dar es Salaam compared to the national (Tanzania) average of 33% and rural average of 17% in 2016 (Government of Tanzania, 2017). The heterogeneity in the city’s electricity access is also similar to levels reported for other African cities, e.g., 95% in Kampala (Uganda) (2018/2019 data), 87% in Nairobi (2014 data), and 76% in Lusaka (Zambia) (2018 data) (DHS Program, 2020).

5.6.2.2 Household Cooking Behavior

Despite the generally high electricity access levels, I find that between 51% (Msasani) and 99% (Keko) of households reported that they *did not* use electricity for cooking. Chi-square tests show a statistically significant effect of ward type on household cooking behavior ($p < 0.01$), i.e., high-income wards (Msasani and Kawe) are more likely to report using electricity for cooking (49% and 29% of households) compared to low-income wards (less than 1% of households in Keko and Manzese).

My previous study (Chapter 4) showed that charcoal is a majority contributor to household-related energy use among low-income wards (Buguruni, Keko, and Manzese). Similarly, chi-square tests show a statistically significant effect of ward type on charcoal use, i.e., low-income wards are more likely to use charcoal as a cooking fuel (95% of households). This includes households that “fuel stack”, i.e., they use charcoal in combination with one or more other fuels for cooking. In Chapter 4 I also disaggregated the different fuel combinations among households that fuel stack and examines the statistical relationship between fuel stacking and household-related energy use. Findings from this previous work show that charcoal use is typically present in fuel stacking and is significantly correlated with higher household-related energy use, while cooking with only electricity or only gas is correlated with lower household-related energy use (Chapter 4).

Charcoal and gas are widely perceived as “more affordable” cooking fuels. Between 74% (Sinza) and 85% (Keko) of surveyed households perceived non-electric fuels (i.e., charcoal and gas) as “more affordable” fuel sources. The high p-value (i.e., $p = 0.254$, Table 5.3) indicates that households’ perceptions of charcoal, wood or gas stoves as “more affordable” are common among all wards, i.e., these perceptions are independent of ward type.

5.6.2.3 Household Perceptions on Charcoal Use

Between 44% (Mwananyamala) and 51% (Buguruni) of surveyed households “disagreed” with policies promoting a charcoal phase out, while between 24% (Keko) and 42% (Manzese)

“agreed”s with such policies. And between 15% (Manzese) and 27% (Keko) were “neutral” details in Table 5.3). The high p-value ($p = 0.141$) indicates that variations in household opinions are common among all wards, i.e., independent of ward type.

Households in low-income and informal wards are also less aware of the “health effects” associated with charcoal use. On household sensitivity to the “negative consequences” of charcoal use, which refers to the health burdens from indoor air pollution associated with charcoal use, I found a statistically significant relationship between household responses and ward type ($p < 0.01$). Respondents from Buguruni and Keko, in particular, have less awareness of these consequences: 54% and 56% of households reported that they were not aware of the “negative health consequences associated with using a charcoal stove for cooking”, compared to 35% and 32% of households in high-income wards (Msasani and Kawe) (Table 5.3). Finally, based on anecdotal observations from our field interactions, even though some households stated knowledge or awareness of the negative effects of charcoal use, i.e, between 44% (Keko) and 68% (Kawe), existing economic challenges and the perceived high associated with electricity or gas use (e.g., due to the high upfront cost of purchasing an electric or gas stove) were among the key reasons for continued charcoal use in general.

5.6.2.4 Household Travel Behavior

Households in low and middle-income wards (e.g., Keko and Kimara) are more likely to use public transport – i.e., local mini-buses (“dala-dalas”), motorcycles (“bodas”), tricycles (“bajajis”) or the BRT – during the working week (Monday to Friday, which are the days we surveyed) relative to high-income wards (e.g., Msasani). For example, between 67% (Msasani) and 93% (Buguruni) of households stated that they used public transport during the working week (see Table 5.4).

Households in low-income wards are more likely to travel by dala-dala as their main travel mode (between 43% and 63% of households in Keko, Buguruni and Manzese, Table 5.4), while high-income wards are more likely use private vehicles as their main mode (between 49% and 59% of

households in Msasani and Kawe). Based on anecdotal evidence from the fieldwork, I observed that some households in Msasani and Kawe were among Dar es Salaam's elite class, and during the course of interviewing, expressed an overall preference for private vehicles due to safety concerns (the issue of public transport safety has also been noted by Andreasen and Møller-Jensen (2017)). Respondents also cited wanting to avoid long-commuting times (and possible route transfers) associated with dala-dala use.

5.6.2.5 Household Perceptions on the Accessibility and Affordability of the BRT Service

Household perceptions on the accessibility and affordability of the BRT service (i.e., the Phase 1 line) were correlated with ward type ($p < 0.01$). Between 29% (Kimara) and 58% (Keko) of surveyed households stated that the BRT was “not accessible” within their community (Table 5.4). Relatedly, some households believed the BRT was “not affordable” compared to the local mini-bus (dala-dala) service. While, between 30% (Mwananyamala) and 55% (Keko) of households perceived the BRT as “not affordable” relative to the existing dala-dala service. Other households were “neutral” on the question of BRT affordability (between 12% (Keko) and 26% (Mwananyamala)) or agreed that it was “affordable” (between 31% (Kawe) and 44% (Mwananyamala)). These findings suggest that ongoing efforts to expand the BRT service, which was referenced among key informants (Section 5.6.1), are needed to deliver a competitive service, i.e., in terms of affordability and accessibility that is equivalent or better than other public transport modes. This would improve perceptions of the BRT service and encourage higher BRT use within communities. Therefore, processes to enable multi-stakeholder collaboration on low-carbon development as recommended in Section 5.6.1 could engage residents in mapping of preferred routes, locations or stops to be serviced by future BRT lines.

Table 5.3. Results from household surveys and chi-square tests showing the effect of ward type on household survey responses to questions on household electricity access, cooking behavior and perceptions on charcoal use.

	Msasani (N=94)	Kawe (N=178)	Mwanayamala (N=169)	Kimara (N=221)	Sinza (N=111)	Buguruni (N=242)	Keko (N=113)	Manzese (N=235)	Significance
Income bracket ¹	High	High	Middle	Middle	Middle	Low	Low	Low	
1. Does the household have access to electricity?									
No	0%	4%	6%	8%	0%	25%	22%	14%	p<0.01***
Yes	100%	96%	94%	92%	100%	75%	78%	86%	
2. What type of fuel(s) does the household use for cooking?									
<u>Charcoal</u>									
No	80%	46%	11%	19%	24%	8%	8%	8%	p<0.01***
Yes	20%	55%	89%	81%	76%	92%	92%	92%	
<u>Electricity</u>									
No	51%	71%	98%	97%	87%	99%	99%	100%	p<0.01***
Yes	49%	29%	2%	3%	13%	1%	1%	0%	
<u>Gas</u>									
No	1%	6%	34%	23%	20%	53%	59%	53%	p<0.01***
Yes	99%	94%	66%	77%	80%	47%	41%	47%	
<u>Other</u>									
No	100%	100%	91%	99%	97%	90%	91%	82%	p<0.01***
Yes	0%	0%	9%	1%	3%	10%	9%	18%	
3. What is the main factor that influences the household's choice to use a non-electric stove (i.e., charcoal or gas) for cooking?									
It is more affordable	80%	83%	82%	82%	74%	79%	85%	84%	p = 0.254
It is widely accessible	9%	6%	5%	6%	11%	10%	11%	5%	
Other	11%	12%	8%	12%	12%	10%	4%	11%	
4. Would the household support the eventual phase out of charcoal use in Dar es Salaam?									
Agree	36%	32%	38%	33%	41%	30%	24%	42%	p=0.141
Neutral	18%	19%	18%	16%	16%	19%	27%	16%	
Disagree	46%	49%	44%	50%	43%	51%	50%	42%	
5. Are you aware of the negative health consequences associated with using charcoal for cooking?									
No	35%	32%	40%	35%	38%	54%	56%	43%	p<0.01***
Yes	65%	68%	60%	65%	62%	46%	44%	57%	

Table Notes:

¹Income categories were determined based on the field team's knowledge and expertise of neighborhood types in the Dar es Salaam region.

- Totals do not always add to 100% due to rounding.
- *p<0.1; **p<0.05; ***p<0.01.

Table 5.4. Results from household surveys and chi-square tests showing the effect of ward type on household survey responses to questions on household travel behavior and perceptions on the affordability and accessibility of the BRT service (i.e., relative to the “dala-dala” minibus service).

	Msasani	Kawe	Mwanayamala	Kimara	Sinza	Buguruni	Keko	Manzese	Significance
Income bracket ¹	High	High	Middle	Middle	Middle	Low	Low	Low	
1. Does your household use a private car during the working week?									
No	40%	48%	6%	11%	84%	96%	95%	95%	p<0.01***
Yes	60%	52%	94%	89%	16%	4%	5%	5%	
2. Does your household use public transport during the working week?									
No	33%	27%	13%	10%	13%	7%	12%	16%	p<0.01***
Yes	67%	73%	87%	90%	87%	93%	88%	84%	
3. What is your household's main mode of travel during the working week?									
<u>BRT</u>									
No	100%	99%	95%	69%	90%	99%	100%	84%	p<0.01***
Yes	0%	1%	5%	31%	10%	1%	0%	16%	
<u>Dala-dala</u>									
No	93%	81%	49%	78%	60%	37%	51%	57%	p<0.01***
Yes	8%	19%	51%	22%	41%	63%	49%	43%	
<u>Private vehicle</u>									
No	41%	51%	96%	91%	87%	96%	96%	97%	p<0.01***
Yes	59%	49%	4%	9%	13%	4%	4%	3%	
4. How accessible is the BRT service in your neighborhood?									
Accessible	51%	68%	51%	71%	47%	58%	43%	60%	p<0.01***
Not accessible	49%	32%	49%	29%	53%	42%	58%	40%	
5. How affordable is the BRT relative to the existing dala-dala service?									
Affordable	40%	31%	44%	42%	42%	35%	34%	40%	p<0.01***
Neutral	16%	18%	26%	12%	13%	18%	12%	20%	
Not affordable	44%	51%	30%	46%	45%	48%	55%	40%	

Table Notes:

¹Income categories were determined based on the field team’s knowledge and expertise of neighborhood types in the Dar es Salaam region.

- Totals do not always add to 100% due to rounding.
- *p<0.1; **p<0.05; ***p<0.01.
- Due to the small sample size, the use of the boda/bajaji was not considered in the statistical analysis.

5.6.3 Linking Findings from Key Informant Interviews and Household Surveys in Dar es Salaam

Findings from the key informant interviews suggest a call to action to stakeholders to reflect on the following constraints to low-carbon development from an institutional perspective: (1) Policies exist, but implementation processes are inadequate; (2) Local government authorities have little jurisdiction; (3) Funding constraints and insufficient own-source-revenue collection limit the capacity of local government authorities to spearhead their own initiatives; and (4) Policy processes to support collaboration among key stakeholders are inadequate.

Based on the current structure of urban governance for the metropolitan region (Figure 5.1), key informants suggested that national government authorities (PORALG and sector ministries) hold the most power to initiate climate change policies and projects Dar es Salaam, including low-carbon projects (e.g., public transportation initiatives, or renewable electrification). Although Tanzania's decentralization policy supports own source revenue collection among local government authorities, urban governance in the metropolitan region is highly centralized and coordinated at the national level, i.e., between 80% to 90% of local government funds are transferred through PORALG and the sector ministries (Government of Tanzania, 2016a).

Therefore, investments to support low-carbon development would need to engage national and local government authorities on specific mandates. For example, City Council and Municipal Councils could take leadership in engaging national government authorities around a priority set of project/investments that would be eventually funded through national government. However, local governments could also play a key role in ensuring that investments account for the societal context, i.e., community needs and their socio-economic challenges. Results from household surveys (Section 5.6.2) show that (1) Dar es Salaam's electrification levels are generally high, (2) most households do not use electricity for cooking, (3) fuels other than electricity are widely perceived as more affordable and close to half of households (between 44% and 51%) disagreed with policies promoting a charcoal phase out, (4) households in low-income and informal wards are less aware of the "health effects" associated with charcoal use, and (5) there are significant differences in travel choices among surveyed wards. Therefore, these societal factors (e.g.,

community preference for non-electric cooking, or sensitivity around charcoal use and its negative health effects) should be considered in the implementation of low-carbon projects. This is likely to require leadership from multiple stakeholders through a collaborative process that leverages the power and mandate of different institutions. I conclude by outlining a possible structure for such a process in the following Section (5.7).

5.7 Conclusions – Which Institution(s) should “Take the Lead” to Enable Low-Carbon Development in Dar es Salaam?

This study shows the institutional and societal barriers that may constrain low-carbon development in Dar es Salaam. Findings from interviews with key informants and surveyed households suggest that leadership on low-carbon development in Dar es Salaam will require multi-stakeholder processes of collaboration that engage stakeholders across both local and national government, donor agencies, infrastructure service providers, and local communities. “Champions” within each stakeholder group could be identified and invited as official members of a task force or steering committee (as suggested by some key informants), tasked with the responsibility to (1) establish a pipeline of low-carbon projects to be implemented in Dar es Salaam over a specified time period (e.g., the next 5 years); (2) define roles and responsibilities for different institutions based on their power and capacity to lead in specific activities or policy processes; and (3) ensure that selected institutions deliver on their roles/responsibilities through monitoring and evaluation of progress towards envisioned goals and objectives.

As an example, steering committee members could include representatives of PORALG and sector ministries (e.g., Ministry of Local Government, Energy and Transport), City and Municipal Councils, donor institutions, and infrastructure service providers (including the private sector). Roles and responsibilities for each institution could include the following.

- **PORALG and sector ministries:** coordinate funding and investment towards low-carbon projects in Dar es Salaam, while also ensuring that projects align with broader urban planning and energy sector goals. PORALG and the sector ministries would also

lead policy development and financing of low-carbon projects – although developing a priority set of pipeline low-carbon projects would require partnership and engagement of other members of the steering committee, i.e., City and Municipal Councils, funding institutions, and infrastructure service providers, among other stakeholders.

- **Dar es Salaam City and Municipal Councils:** ensure that the “voices” of lower-level jurisdictions, e.g., wards and streets (“mtaas”) (Figure 5.1), are considered as pipeline projects are developed. Their roles/responsibilities could also include (1) engaging with community members via focus group discussions, or community-specific activities, and (2) collecting community data and information on energy use and travel patterns (data presented in this study could inform these secondary processes).
- **Multilateral Development Banks (MDBs) (e.g., the World Bank, Africa Development Bank) and the private sector:** develop additional financing options for pipeline projects (e.g., grants, concessional loans or private equity) to accelerate their implementation in Dar es Salaam. Institutions could also lead mapping efforts to identify example projects, business models, or technologies that would be considered “bankable” and therefore attractive to MDB or private sector finance.
- **Energy utilities, transport authorities, and regulators:** improve existing approaches to infrastructure service delivery, e.g., supporting electrification efforts among low-income wards where access levels are lowest, or introducing subsidies to encourage electricity and gas use for cooking among low-income wards.

To conclude, effective collaboration among the above institutions would ensure progress toward low-carbon development in Dar es Salaam, where leadership from City and Municipal councils would ensure the societal context is adequately considered in all stages of project development and implementation.

5.8 Supplementary Material

The supplementary material includes supporting methods, calculations and background data that are important in the context of this chapter (Chapter 5). Tables and figures are presented with the letter “S” in their caption title, and are referenced throughout the main body of this chapter, where relevant.

5.8.1 Interview Guide for Key Informant Interviews



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

KEY INFORMANT INTERVIEWS – QUESTION GUIDE

- What are your general observations on the urban development process in your city?
- Can you comment on the institutional framework for urban governance in your city with respect to:
 - Key institutions and organizations involved in urban governance?
 - How urban governance is evolving i.e. what are the main opportunities and challenges?
 - Any policy processes responding to the challenges of rapid urbanization?
 - Key research bodies coordinating and collecting data on urban infrastructure e.g., electricity access and use, transportation demand, or other aspects? And how does this data feed into the existing urban development process?
 - Any innovations or changes taking place that are relevant to mention?
- What are your general thoughts on how the city is promoting low-carbon development? Please provide examples.
- In your opinion, what are key institutional factors that enable or constrain the adoption of low-carbon measures at the city or national level?
- Are you aware of any urban sector projects or processes that focus on climate change mitigation or low-carbon development in the city?
- Are you aware of any institutions at the local or national level have the power to structure policies and/or interventions to enable low-carbon development? For example, policies or interventions to support public transport use? Or a transition to modern fuels such as electricity or gas? Or reducing the domestic use of charcoal?
- Is there a working or active master plan for the city?
 - If so, what are the main elements of the plan?
 - What are its limitations?
- Are you aware of any policies or processes that examine then nexus between energy use, greenhouse gas (GHG) emissions and spatial planning in the city?
 - If so, what are they?
- What are the existing policy processes to manage and coordinate processes for energy use and GHG emissions management at the city or national level?
 - What are the main priorities in this regard?
 - What are the infrastructure sectors being prioritized?
 - Are these processes linked to national level targets?
- What are your general observations on energy use in key sectors e.g., transportation or housing, in the city?
- Are there any other aspects you would like to comment on?

5.8.2 Mapping of Dar es Salaam Key Informant Responses

Table S 5.1. Mapping of key informant responses based on main themes that emerged from interviews. Key informants who stated opinions that were generally consistent with one or more of the four themes are identified with a check mark (√).

#	Code	Affiliation and Expertise	1: Policies exist, but implementation processes at the city-level are inadequate	2: Local authorities have little jurisdiction	3: Limited funding and own-source revenue collection among local authorities constrains their capacity to spearhead their own initiatives	4: Interviewees were unaware of processes that enabled stakeholders to engage on low-carbon initiatives in Dar es Salaam
<u>Urban Planning</u>						
1.	Academic1	Professor, Ardhi University	√	√	√	√
2.	Academic2	Professor, Ardhi University	√	√	√	√
3.	Academic3	Professor, Ardhi University	√	√	√	√
4.	Academic4	Professor, Ardhi University	√	√	√	√
5.	Academic5	Professor, University of Dar es Salaam	√		√	√
6.	Municipality1	Director, Dar es Salaam City Council	√	√	√	√
7.	Municipality2	Urban Planner, Dar es Salaam City Council	√	√	√	√

#	Code	Affiliation and Expertise	1: Policies exist, but implementation processes at the city-level are inadequate	2: Local authorities have little jurisdiction	3: Limited funding and own-source revenue collection among local authorities constrains their capacity to spearhead their own initiatives	4: Interviewees were unaware of processes that enabled stakeholders to engage on low-carbon initiatives in Dar es Salaam
8.	Municipality3	Urban Planner, Dar es Salaam City Council	√	√	√	√
9.	NGO1	Specialist, Donor Agency (Multilateral)	√	√		√
<u>Transport</u>						
10.	NGOV1	Engineer, Dar es Salaam Rapid Transit Agency		√		
11.	NGOV2	Engineer, Dar es Salaam Rapid Transit Agency				
12.	NGOV3	Engineer, Dar es Salaam Rapid Transit Agency				
13.	NGOV4	Engineer, Tanzania National Roads Agency		√	√	
14.	Academic6	Researcher, University of Dar es Salaam	√	√		√
15.	NGO2	Specialist, Donor Agency (Multilateral)	√	√		√

#	Code	Affiliation and Expertise	1: Policies exist, but implementation processes at the city-level are inadequate	2: Local authorities have little jurisdiction	3: Limited funding and own-source revenue collection among local authorities constrains their capacity to spearhead their own initiatives	4: Interviewees were unaware of processes that enabled stakeholders to engage on low-carbon initiatives in Dar es Salaam
<u>Energy</u>						
16.	NGOV5	Engineer, Tanzania Electric Supply Company Limited	√	√		√
17.	NGOV6	Manager, Energy and Water Utilities Regulatory Authority	√			√
18.	NGOV7	Director, President's Office – Regional and Local Government	√		√	√
19.	NGO3	Specialist, Donor Agency (Multilateral)	√		√	√
20.	NGO4	Specialist, Donor Agency (Multilateral)	√			√
21.	PS1	Manager, Private Sector				√
22.	PS2	Manager, Private Sector				√
23.	PS3	Manager, Private Sector		√		√
24.	PS4	Manager, Private Sector				√
Total			16	14	11	20

5.8.3 Preliminary Work in Lusaka

This section presents findings from subsequent interviews with key informants in Lusaka. I conducted interviews between November and February 2018 to broadly compare insights with those from Dar es Salaam. Insights from Lusaka serve as a subordinate case – my analysis of key informant responses is generalized, and I do not complete a detailed mapping of key informant responses. Rather, I broadly reflect on common and distinct themes and findings. This is a limitation of my study, and future work could incorporate more detailed engagement with key informants in Lusaka, and other cities in Africa.

In total, I interviewed 17 key informants in Lusaka who were professors, managers and specialists in urban planning, energy, climate change, and transport sectors. Table S 5.2 summarizes the affiliations and expertise of the Lusaka key informants. In summary, of the 17 interviewed key informants, 2 were from academia (also coded as “Academic”), 9 from national government (“NGOV”), 1 from local government (“Municipality”), 3 from non-governmental organizations or donor agencies (“NGO-Donor”), and 2 from the private sector (“Private Sector”).

Table S 5.2. Affiliation and expertise of key informants interviewed in Lusaka.

	<u>Urban Planning</u>	
1	Academic ¹	Professor Department of Natural Resources and Environment, University of Zambia
2	NGOV ¹	Manager, Lusaka South Multi-Facility Economic Zone
3	NGOV ²	Manager, Ministry of Local Government and Housing
4	Municipality ¹	Director, Lusaka City Council
	<u>Energy</u>	
5	Academic ²	Professor, Department of Mechanical Engineering, University of Zambia
6	NGOV ³	Manager, Energy Regulation Board
7	NGOV ⁴	Manager, Energy Regulation Board
8	PrivateSector ¹	Manager, Private Sector (Energy)
9	PrivateSector ²	Advisor, Private Sector (Energy)
	<u>Climate Change</u>	
10	NGOV ⁵	Advisor, Interim Climate Change Secretariat, Ministry of Finance and National Planning
11	NGOV ⁶	Specialist, Ministry of Lands, Natural Resources and Environment
12	NGOV ⁷	Director, Ministry of Lands, Natural Resources and Environment
13	NGOV ⁸	Manager, Ministry of Lands, Natural Resources and Environment
14	NGO ¹	Specialist, Donor Agency (Multilateral)
15	NGO ²	Director, Donor Agency (Multilateral)
16	NGO ³	Specialist, Donor Agency (Multilateral)
	<u>Transport</u>	
17	NGOV ⁹	Director, Ministry of Transport and Communications
¹ NGOV – National Government ² Municipality – Municipal Government (city-level) ³ NGO – Non-Governmental Organization		

5.8.3.1 Lusaka Key Informant Interviews

Urban governance in Zambia is a mix of top-down and bottom-up processes. Zambia's long-standing commitment to decentralization has allowed local authorities to retain some responsibility over urban activities. However, similar to Dar es Salaam, key informants in Lusaka expressed that local authorities have remained constrained in their capacity to coordinate climate responses in key infrastructure sectors, i.e., transportation and energy (which fall outside the mandate of local authorities) (PMRC Zambia, 2016). Specifically, some key informants expressed that national institutions (e.g., ministries) coordinated most infrastructure initiatives, though local governments are engaged in the design, preparation, and implementation activities, where relevant. Others in national government and the private sector highlighted the handful of initiatives at the city level that may contribute to low-carbon development in Lusaka, e.g., solar electrification initiatives through the Lusaka South Multi-Facility Economic Zone and feasibility studies for a Light Rail Tram (LRT) system.

Finally, like Dar es Salaam, key informants in Lusaka were unaware of specific policy processes that sought to engage stakeholders around low-carbon initiatives in Lusaka. One key informant from national government highlighted that urban climate change responses (e.g., mitigation or adaptation) were often planned in "ad-hoc" manner, despite a robust policy framework for low-carbon development at the national and city level. Relatedly, others from national government and NGO (donor) groups noted that donor funding for infrastructure in Zambia is largely directed towards "rural development" given lower levels of socio-economic development in these regions. As a result, investment in urban areas often lag those in rural areas and fail to engage a diverse group of stakeholders in a collaborative process to promote low-carbon development in Lusaka.

5.9 References for Chapter 5

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Chapter 6

Conclusions

This thesis addresses four main objectives: Objective 1: Estimate current and future residential energy use and GHG emissions in Dar es Salaam; Objective 2: Identify key drivers of residential energy use and GHG emissions in Dar es Salaam; Objective 3: Assess differences in residential energy use that may exist at the Dar es Salaam sub-city (ward) level; and Objective 4: Assess the institutional and societal factors that may constrain low-carbon development in Dar es Salaam (also stated in Section 1.2). This chapter describes the main conclusions for each research objective and question, and discusses research limitations and areas of future work.

6.1 Objective 1: Estimate current and future residential energy use and GHG emissions in Dar es Salaam

Question 1: What are current levels of residential energy use and GHG emissions in Dar es Salaam, and how might they evolve from 2015 to 2050?

In Chapter 3, I estimate residential energy use for the average Dar es Salaam household at 46GJ/HH/year in 2015. This level of consumption is associated with residential greenhouse gas (GHG) emissions of 1,400 ktCO₂e (or 0.2 tCO₂e/capita). Looking to 2050, I project that Dar es Salaam may see a 4 to 24-fold increase in residential GHG emissions between 2015 and 2050, which corresponds with a range of between 5,700 ktCO₂e (0.5 tCO₂e/capita) and 33,000 ktCO₂e (2 tCO₂e/capita). These findings indicate that Dar es Salaam's absolute residential emissions could catch up to (2012 to 2015) levels reported in cities such as New York, San Francisco and London (as shown in Section 6.8). Although, emissions changes on a *per capita basis* are less sizeable, i.e., increasing 1 to 10-fold between 2015 and 2050. At the highest level (2 tCO₂e/capita), per capita emissions still remain at 3-times less than those observed in New York in 2015 (5.7 tCO₂e/capita, see Section 3.9.4), for example.

My work in Chapter 3 was inspired by the Shared Socio-Economic Pathways (SSPs), which I discuss in detail in Chapter 2 and Chapter 3 (Section 2.4.3 and 3.5.1). I used the SSPs as a basis for projecting annual residential energy use and GHG emissions in Dar es Salaam between 2015 and 2050. Although, some key limitations of this early work were that (1) I did not validate my model assumptions and estimates (shown in Table 3.2) with household survey data, and (2) I later collected household data in as part of work for Chapter 4 and Chapter 5, although data were not incorporated into the LEAP model to compare differences in findings. These data comparisons would be an essential part of my future work. Furthermore, for studies interested to apply the LEAP model approach to other cities in Africa, I would suggest that: (1) researchers develop models that incorporate and compare model results based on field data collected at the local level (i.e., household surveys initiated in participating cities), and (2) conduct structured workshops and interviews with infrastructure experts, policymakers and community members (i.e., via expert elicitation) to refine the different SSP narratives and develop city-specific “futures” that account for the unique political, societal and infrastructure contexts of cities.

As mentioned, my work in Chapter 4 and Chapter 5 employed household energy use data that was collected as part of field work activities in Dar es Salaam in 2018. I initiated these activities to account for differences in energy use at the ward or settlement level. Results from this fieldwork show that mean residential energy use of surveyed households was approximately 38GJ/HH/year – this value incorporated a range of 30GJ/HH/year for low-consuming wards (e.g., Kimara) and 50GJ/HH/year for high-consuming wards (e.g., Kawe). Comparing my results with those in Chapter 3, I found that the estimated value from my previous work (46GJ/HH/year) falls within this range but also 21% higher than the mean value estimated from my field data, i.e., due to the higher electricity and gas use assumed in the SSP model, which may have resulted in a slightly over-estimated energy use and GHG emissions values.

Question 2: What modelling approaches can be employed to estimate and project residential energy use or GHG emissions when data availability is limited?

I use the Long-Range Energy Alternatives Planning (LEAP) modelling tool (Heaps, 2016) to estimate Dar es Salaam's current and future residential energy use and GHG emissions to 2050. My assumptions for each SSP narrative (see Table 3.1, Chapter 3) inform my approach to modelling three distinct urban growth and GHG emissions pathways in LEAP. I chose LEAP as a modelling tool given its wide use in modelling energy/GHG emissions futures in the Africa region, e.g., see studies of Kemausuor et al. (2015), Mahumane & Mulder (2016), and SEA (2015), and other Global South cities where data availability may be limited, e.g., São Paulo (Collaço et al., 2019), Panama (McPherson and Karney, 2014), Bangkok (Phdungsilp, 2010), and several Chinese cities (Zhou et al., 2016; Fan et al., 2017; Yang et al., 2017; Lin et al., 2018). At the national level, 37 developing countries (as of 2018) have used LEAP to develop their national emissions scenarios as part of their Intended Nationally Determined Contributions (INDCs) to the United Nations Framework Convention on Climate Change (SEI, 2018). However, to my knowledge, LEAP has not been used to model energy use and/or GHG emissions futures in Dar es Salaam or Tanzania.

My estimates for Dar es Salaam's energy use/GHG emissions for 2015 and 2050 are informed by various input indicators and assumptions. These include population levels, household size, the electric grid mix, household activities (e.g., fuel uses and travel behaviors). See Table 3.2 and Table 3.3, for key parameters and underlying assumptions that I integrated within the LEAP platform. Finally, LEAP's Technology and Energy Database includes GHG emissions data for a range of fuels based on the Intergovernmental Panel on Climate Change (IPCC) guidelines. The supplementary material appended to Chapter 3 (Section 3.9.9) details the GHG emissions calculation structure within LEAP.

6.2 Objective 2: Identify key drivers of residential energy use and GHG emissions in Dar es Salaam

Question 3: What activities drive residential energy use and GHG emissions in Dar es Salaam in 2015 and in 2050?

Throughout the thesis, I use the term “residential energy use” to refer to energy uses from domestic household and transport-related activities. At the aggregate city level (Chapter 3), household-related energy use, mostly from wood fuel use (charcoal and firewood), contributes the majority share of residential energy use in Dar es Salaam. Charcoal and firewood account for 78% of household-related energy use, compared to 11% from electricity, 9% from LPG “gas” and 2% from kerosene (Table 3.3, Chapter 3). In my later work (Chapter 4), I likewise show that charcoal use, which is typically present in with fuel stacking, is a majority contributor to residential energy use among low-income wards especially, e.g., over 80% of residential energy use in Buguruni and Keko.

However, it is clear that the conversion from energy use (i.e., GJ of energy) to GHG emissions – i.e., the global warming potential of different fuels represented in tons of carbon dioxide equivalents (CO₂e) – is not one-to-one. There is need to consider different metrics (i.e., energy use vs. GHG emissions) when assessing drivers of energy use in cities. For example, even though charcoal accounts for most household-related energy use in Dar es Salaam, it does not contribute most to GHG emissions. For example, I show that charcoal accounts for 44% of household-related energy use at the aggregated city level (Chapter 3), and over 80% among surveyed low-income wards (Buguruni, Keko and Manzese) (Chapter 4). But when I convert these energy values to GHG emissions in my LEAP model, GHG emissions from charcoal are carbon neutral given the assumption that these emissions are biogenic and have limited global warming potential. This assumption in the model is consistent with the 2006 IPCC guidelines which does not count biogenic emissions from wood fuel burning in the energy sector (as these values are assumed to be counted in the Agriculture, Forestry, and Land-Use (AFOLU) sector (IPCC, 2006)). However, it is clear that charcoal is an important energy driver, and policy measures for aggressive climate change mitigation in Dar es Salaam should consider reducing

charcoal use, alongside efforts to decarbonize electrification and transportation, which were my recommended areas of policy action in Chapter 3.

Therefore, future extensions of my work in Chapter 3 could explore the implications of continued charcoal use in Dar es Salaam, especially under an SSP3 scenario, which does not assume a phase-out of charcoal use by 2050 (see Table 3.3). It is also clear from my findings in Chapter 4 and Chapter 5 that while charcoal use may be less-GHG intensive from a global warming perspective, its continued use among households could lead to other environmental and health burdens (e.g., from indoor air pollution) – although, I did not consider broader environmental and health burdens as part of the thesis work. Therefore, while my findings in Chapter 3 provide an overall (city-wide) picture of GHG emissions in Dar es Salaam, it is important to consider other environmental implications of wood fuel use as energy sector and GHG-mitigation policies and actions are developed in Dar es Salaam, and Tanzania more broadly.

Question 4: What is the influence of household cooking fuel choice, travel mode choice and household wealth on energy use among households in Dar es Salaam?

In Chapter 3, findings show a statistically significant effect of cooking fuel choice on mean household energy use ($p < 0.01$). I also find that fuel stacking has a significant effect on household-related energy use, and in particular, households that fuel stack with three fuels (6% of households surveyed) generally appear to be the highest energy users (Table 4.3). Based on results from OLS and Tobit regressions, fuel stacking is correlated with ~35% higher household-related energy use among surveyed wards (i.e., relative to households that do not fuel stack). Households that fuel stack with electricity, charcoal, and kerosene/firewood (3% of households surveyed) are correlated with the highest household-related energy (11 and 13 times more) relative to households that use only electricity (reference case). Finally, among other socio-economic indicators, I find a significant effect of household wealth – which I assume based on other indicators (proxies) for wealth (Section 4.6.2, Chapter 4) – and household-related energy use. These findings are consistent with other studies that likewise found positive and significant

correlations relating both income and fuel stacking to household energy use in Tanzania, e.g., Choumert-Nkolo et al. (2019) and D’Agostino et al. (2015).

In transportation, OLS regression outputs show that using public transport as part of a household’s commuting activities is correlated with 13% lower transport-related energy use. Public transport use (particularly use of the local minibuses, “dala-dala”) is higher in informal wards, e.g., 93% of households in Buguruni reported using public transport as part of their weekly commuting, compared to 67% of households in Msasani (Table S 4.3). I also find a positive and significant correlation between transport-related energy use and proxies for household wealth (Section 4.6.2), i.e., high-income wards (Msasani and Kawe) contribute the highest transport-related energy use, largely due to higher levels of private vehicle use in these wards (see Figure 4.5).

6.3 Objective 3: Assess differences in residential energy use that may exist at the Dar es Salaam sub-city (ward) level

Question 5: Is there a statistically significant difference in residential energy use between informal, formal, and mixed ward types in Dar es Salaam?

Using Analysis of Variance (ANOVA) in Chapter 4, I find a statistically significant difference in mean residential energy use between the different surveyed wards in Dar es Salaam ($p < 0.01$). I also conduct additional post-hoc tests (i.e., Fisher’s Least Squared Differences) map where these differences existed for the different ward types. Results from post-hoc tests indicate 4 distinct categories of wards based on their energy use, i.e., “high”, “medium-high”, “medium”, and “medium-low” energy users (see Figure 4.3).

As shown in Figure 4.3, in some cases I find no statistically significant difference in mean residential energy use between formal and informal wards, or high-income and low-income wards, e.g., Msasani (high-income, formal ward) and Keko (low-income, informal ward) both have a “medium-high” level of energy use relative to other wards. In other cases, there is a

statistically significant difference in mean residential energy use between wards e.g., between Kawe and Keko, or Keko and Kimara (Figure 4.3).

6.4 Objective 4: Assess the institutional and societal factors that may constrain low-carbon development in Dar es Salaam

Question 6: How do key informants perceive the enabling environment for low-carbon development Dar es Salaam?

In Chapter 5, I draw on the perspectives of key informants, i.e., selected experts across various local and national government, academia, civil society and the private sector, to ascertain possible institutional barriers to low-carbon development in Dar es Salaam. Findings from interviews with key informants suggest the following institutional constraints to low-carbon development in Dar es Salaam: (1) Policies exist, but implementation processes are inadequate; (2) Local government authorities have little jurisdiction; (3) Funding constraints and insufficient own-source-revenue collection limit the capacity of local government authorities to spearhead their own initiatives; and (4) Policy processes to support collaboration among key stakeholders are inadequate. Based on these findings, I assert that the implementation of low-carbon projects would need to engage multiple stakeholders in a collaborative process to leverage the power and mandate of different institutions. This collaborative process may be enabled through a dedicated “steering committee” tasked with the mandate to develop a roadmap for implementing low-carbon projects in the city. I also highlight that the steering committee should consist a mixed group of stakeholders from local and national government (i.e., PORALG, sector ministries, Dar es Salaam City and Municipal Councils, donor agencies, and infrastructure service providers etc., see Figure 6.1). At the same time, local government authorities (i.e., the Dar es Salaam City and Municipal Councils), as the closest government institution to urban residents, could take leadership in ensuring that the societal/community context is always considered in the implementation of low-carbon projects.

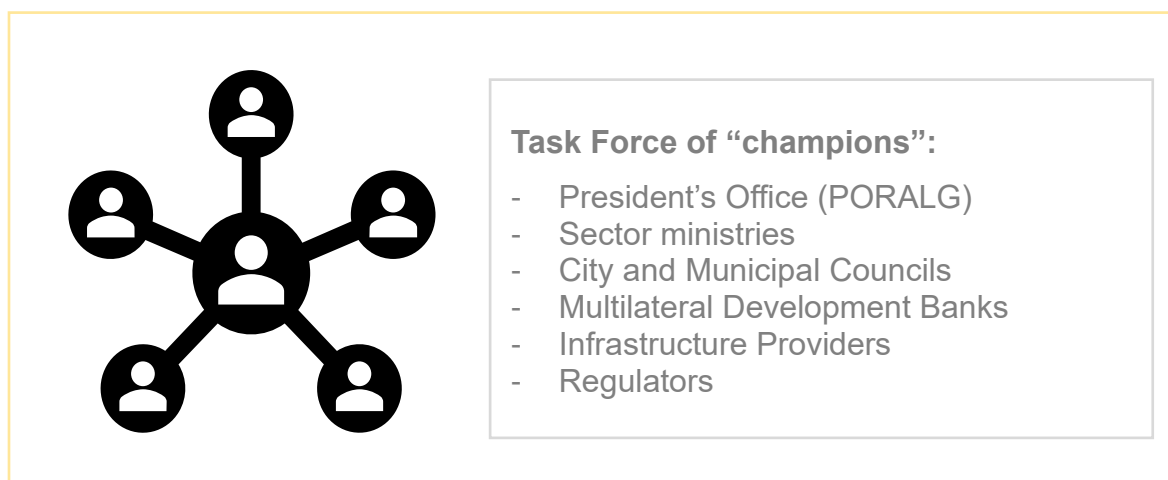


Figure 6.1. Proposed Steering Committee of local champions from PORALG, sector ministries, city and municipal councils, multilateral development banks (MDBs), service providers, and regulators.

Question 7: What are possible societal factors that influence household energy and travel choices in Dar es Salaam?

In addition to understanding the institutional context (i.e., Chapter 5), my policy recommendations in Chapter 5 call on policymakers to account for the societal context, i.e., the key factors that may influence household energy use and travel choices. Insights from interviews with surveyed households show: (1) a high preference for non-electric stoves, i.e., wood-fuels (i.e., charcoal) and gas stoves; (2) the perception that charcoal is an important cooking fuel and should not be “phased out”, and (3) limited awareness (particularly among informal wards) of the negative “health effects” (i.e., due to indoor air pollution) associated with charcoal use (Chapter 5). In transportation, the perceived accessibility and affordability of the “dala-dala” minibus relative to the BRT service (i.e., the Phase 1, see Chapter 3) influences household preferences to use the dala-dala. Therefore, as stakeholders (i.e., the steering committee recommended in Chapter 5) develop low-carbon projects (e.g., electrification initiatives or expansion of the BRT services) these societal factors need to be considered in the design and implementation of projects.

6.5 Concluding Remarks Linking Thesis Objectives and Questions

My work in Chapter 3 shows that GHG emissions in Dar es Salaam may cover a wide range of future scenarios. However, as demonstrated in my later work in Chapter 4, it is important to couple long-term energy models and GHG assessments with local data that consider differences in energy use at the settlement (or ward level). Findings from my surveys in Dar es Salaam show a high level of inequality and difference in the way households' access and use energy sources and transport infrastructure. Therefore, most future increases in GHG emissions may be attributed to high-consuming and wealthier households (i.e., in wards such as Kawe and Masani). While my work in Chapter 4 did not quantify differences in GHG emissions among surveyed households, future work would need to consider these socio-economic differences among households as part of GHG emissions projections.

Differences in my findings across thesis chapters (i.e., Chapters 3 to 5) show that it is important to consider a range of metrics (e.g., GHG emissions versus total energy use) when assessing drivers of energy use and GHG emissions in African cities. This is clearly demonstrated in the case of charcoal, which is an important driver of energy use (i.e., from a GJ per household perspective), but does not contribute substantially to GHG emissions (i.e., from a tCO_{2e} per household perspective). Given these differences, it is important that the results from my LEAP model – which do not consider biogenic emissions from charcoal – should be expanded to consider other environmental impacts (e.g., land use changes or air pollution). Therefore, an important area of future work would be to consider a larger set of impact indicators for each household fuel (i.e., charcoal, gas, electricity or kerosene) to assess the broader environmental implications of their use.

Finally, it would be important to engage a diverse set of actors when developing and implementing strategies. In the case of Dar es Salaam, overall success would rely on political leadership at the highest level (i.e., leadership from the Office of the President) to ensure that envisioned strategies translate to meaningful policy actions.

6.6 Future Work

I have identified four key areas of future work that would make for interesting post-doctoral work or research within an international development context: for example, with the World Bank Group, World Resources Institute, or C40 Cities, or other multilateral agencies working on urban sustainability issues. These include (1) examining the linkages between urban sustainability and resilience, (2) piloting energy use/GHG emission studies in other African cities, (3) accounting for (Scope 3) GHG emissions, (4) developing a road map and financing structures, including private sector finance, to support GHG mitigation efforts.

6.6.1 Linkages Between Urban Sustainability and Resilience

Urban resilience is not explicitly considered in the work presented in this thesis, but the concept is an important dimension of urban sustainability. The injustice in climate change is that African cities, though contribute least to climate change, are most burdened by climate-induced risks and vulnerabilities. Current discourse on urban resilience recognizes that urban growth should occur in tandem with efforts to protect cities from climate-induced disasters (Fankhauser, 2017). The concept is often also used interchangeably with “climate change adaptation” to describe processes of adjusting to climate change or alleviating its negative impacts (Fankhauser, 2017). In my previous work with the Global Environment Facility (GEF) and the World Bank (i.e., prior to starting my doctoral studies), I supported country work to “mainstream” climate change adaptation with sector-specific plans and strategies in developing regions (including Africa). Programs focused on adaptation actions in key economic sectors e.g., urban planning, natural resources management, energy, and infrastructure, among others. Arguably, if developing countries can “climate-proof” their development sectors, i.e., through climate change adaptation actions, they can achieve both economic and adaptation benefits. The notion of “climate-resilient development” is therefore critical to the region’s development.

I was previously interested to continue work on urban resilience within the scope of my doctoral research (and focus on African cities). At the time, I conducted broad literature reviews on

resilience as to identify key research gaps, and later developed a conceptual framework to diagnose resilience in African cities from an engineering perspective. This work was based the notion of infrastructure “carrying capacity”, which describes the optimum material or energy demands that can be “carried” by infrastructure systems in changing urban environments. I hypothesized that crossing carrying capacities could induce significant changes in infrastructure performance and impact community resilience. Therefore, the high population growth in African cities would put pressures on urban infrastructure beyond their carrying capacities and compromise infrastructure resilience and the ability for systems withstand climate-related disasters. For communities, compromised infrastructure resilience would also limit the availability of services for the poor during these disaster events.

Table 6.1 provides an illustrative example on the data requirements related to my early work on urban resilience. For example, the “current resource use” would show the net material and energy use of the end user (i.e., building, neighborhood, or city). The “available capacity” would estimate the carrying capacity of the infrastructure system (e.g., estimated from infrastructure service providers, building owners and/or utility companies). Where capacity estimates are unknown, expert elicitation could be employed to specify a range of values. Infrastructure resilience would therefore be quantified based on the difference between the available capacity and the current resource use.

Table 6.1. Carrying capacity analysis for different urban infrastructures

Urban Infrastructure	Estimated Use (per day)	Available Capacity
Water	▪ m ³ per day.	Determined from infrastructure service providers, utility companies, or expert elicitation
Energy (electricity)	▪ kWh of per day.	
Solid waste	▪ Kilograms per day	
Public transport (e.g., BRT bus)	▪ Passengers/day	

Carrying capacity data (i.e., “available capacity” estimates shown in Table 6.1) are unavailable for most African cities and especially for informal settlements. These constraints limited my progress on this work. Contingent on funding availability for data collection and additional fieldwork, this would be an important area of future research. Some of the data collected during my fieldwork in Dar es Salaam could support energy sector inputs, while data collection for other sectors would need to engage a range of stakeholders (e.g., government agencies, infrastructure service providers and the private sector).

6.6.2 Piloting Energy Use/GHG emission Studies in other African Cities

Future work could also employ similar studies in other African cities and enable comparative reviews of energy use. Disaggregated studies (i.e., those employed at the sub-city level) could also estimate inequalities in energy use and access for different energy use sectors. For example, in Chapter 4, I use Gini coefficients (G_e) to estimate the inequality in energy use among the surveyed wards. Findings showed that inequality was highest for transport related energy use (indicated by the high G_e of 0.87) i.e., only a few households account for the majority share of energy use in the transport sector. It would be important to consider if similar patterns exist for other African cities. Research approaches could also integrate qualitative methods (e.g., observational, or unstructured interviews) to assess community way of life and the cultural or local factors that influence energy use. While some of these aspects are highlighted through the structured surveys, we conducted in Dar es Salaam, this format did not allow for informal interactions with communities. Unstructured methods could capture additional cultural aspects that have not previously been considered in this thesis or urban energy research more broadly.

6.6.3 Accounting for Upstream (Scope 3) GHG Emissions

Quantifying emissions from upstream (Scope 3) processes would support more holistic accounting of urban GHG emissions. In Chapter 3, I only consider for Scope 1 and 2 emissions attributed to residential activities in Dar es Salaam. Unavailable data on upstream and

downstream GHG emissions (that are attributed to energy use in the Dar es Salaam) limited the research scope and accounting methods applied in my research. Some authors have already shown that scope 3 GHG emissions account for a majority share of total emissions for individual fuel uses (e.g., charcoal use, see (Ekeh et al., 2014; FAO, 2016; Fisher et al., 2011)). A study by (Ekeh et al., 2014) shows that direct emissions from charcoal use accounted for 47% percent of total emission in Kampala (Uganda), compared to 53% from upstream production processes. Similarly, the Food and Agricultural Organization (FAO) estimate that GHG emissions from charcoal use and use are mostly associated with upstream wood sourcing (~ 21 to 69 percent) and burning (~28 to 61 percent), compared to end use in the home (approximately 9 to 18 percent) (FAO, 2016). These studies show that accounting for scope 3 GHG emissions can better reflect the full extent of environmental impacts associated with urban activities. Accounting methods can be a data-intensive process, and future work would need to engage a diverse set of stakeholders (including the private sector) around data collection efforts for different scope 3 processes (e.g., GHG emissions from upstream transportation or downstream waste processes).

6.6.4 Developing Roadmaps and Financing Structures to Support Low-Carbon Initiatives in Cities

Policy workshops hosted as part of the thesis research engaged stakeholders in a dialogue on strategies to enable low-carbon development in Dar es Salaam and Lusaka. Discussions also engaged stakeholders around the following question: “Which governing body or institutions are best equipped to take the lead in implementing low-carbon measures in Dar es Salaam/Lusaka?” (see agenda in Appendix). Participants shared several examples of key institutions, e.g., local government authorities, designated line ministries or designated units within the President’s or Vice President’s Office. Drawing on these insights, future work could employ structured or unstructured surveys with selected institutions at the city or national level to develop roadmaps or approaches for “leading” institutions to collaborate on low-carbon initiatives.

Roadmaps could also outline mechanisms to scale up financing for low-carbon measures. For example, governments could consider a dedicated budget allocation for low-carbon measures at

the city or national level. For example, governments could set up an “infrastructure” fund for low-carbon investment. Similar funds are available in other sectors, for example, road development is funded through the National Roads Agency (TANROADS) in Tanzania and the Road Development Agency (RDA) in Zambia. Research could also assess options for scaling-up own-source revenues among local government authorities. For example, through may be through different financing structures such as concessional loans or grants provided by the World Bank and other entities, climate change finance provided by the Global Environment Facility or the Climate Investment Funds.

The private sector could also help countries close fulfill infrastructure funding gaps, estimated at \$130 and \$170 billion a year, according to the African Development Bank (AfDB, 2018). While new loans and grants to help countries respond to COVID-19 pandemic, e.g., (World Bank, 2020), could present opportunities to channel funds towards low-carbon measures. For example, the World Bank is including resources for off-grid solar electricity and battery power systems as part of COVID-19 response strategies in the energy. Established contracts (e.g., with ministries of energy, utility companies or the private sector) could serve as basis for larger electrification projects (e.g., solar powered microgrids or integrating renewable supply within the on-grid network) and provide incentive for private sector investment in technologies.

6.7 Final Thesis Remarks

Africa's current and ongoing urbanization means that future energy use and GHG emissions, both regionally and globally, will concentrate in African cities. In the case of Dar es Salaam, rising energy and population levels could increase the city's GHG emissions to reach as high as 24 times 2015 levels in 2050. The anticipated rise in Dar es Salaam's GHG emissions, as shown in Chapter 3, could provide some insight to the GHG emissions pathways of other fast-growing cities in the Africa region. Therefore, if investments for low-carbon (or low-GHG) development are prioritized in this current and early stage of Africa's urban growth (e.g., through investments to decarbonize electricity generation and transportation sectors), cities could leapfrog towards low-GHG and resilient communities (that also benefit the poor).

At the same time, implementing low-carbon projects would require policymakers to account for broader urban planning and socio-economic goals, and the intra-community differences in energy use that may exist at the sub-city (settlement or ward) level (see Chapter 4). Related to this, it would be important to recognize the unique institutional and societal context in which African cities are growing, and the critical role of local government authorities in ensuring that societal context (e.g., local community behaviours) are considered in project design and implementation processes (see Chapter 5).

Finally, this thesis has presented several exciting opportunities for future work that I look forward to exploring over the course of my career.

6.8 Supplementary Material

6.8.1 LEAP Model Results (Presentation Format)

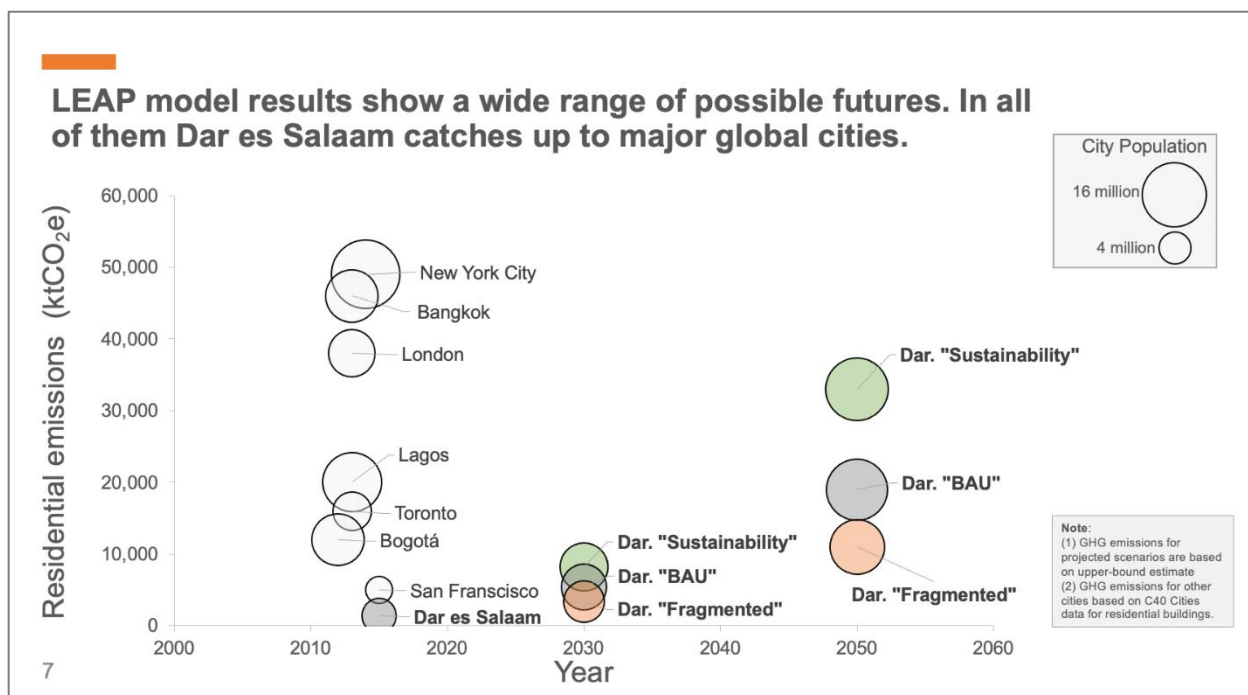


Figure S 6.1. Leap Model Results, shown as presentation slide. GHG emissions for projected scenarios are based on the upper-bound estimate. GHG emissions for other cities are based on C40 cities data for residential buildings.

6.9 References for Chapter 6

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Appendix A: Dar es Salaam Household Survey

RESIDENTIAL ENERGY USE SURVEY

SECTION 1: INTRODUCTION

Note: Some questions have been omitted, where relevant, to ensure privacy for the research team and interview participants.

Name of Region

☐ Dar es Salaam

Name of District

☐ Kinondoni

☐ Ilala

☐ Temeke

☐ Ubungo

Name of Ward

Street Name

Date of Interview

yyyy-mm-dd

Interview Start Time

Time in 24 hours

hh:mm

SECTION 2: INTERVIEW CONSENT FORM
Please read this consent loud to respondent

Introduction

My name is _ , I am working with a research firm called Ideas In Action (IAct) and I'm here to collect household data for the study, "Diagnosing energy-efficient and resilient urban development in Dar es Salaam". The research is being led by researchers at the Department of Civil Engineering at the University of Toronto and the Institute for Human Settlement Studies (IHSS) at Ardhi University.

The aim of the study is to assess your household energy use and travel patterns. Given the rapid urbanization that is expected in Dar es Salaam in the coming years, the responses you provide to these questions are intended to inform future urban planning and energy use in the city.

Participation is voluntary

You don't have to participate in the survey, but we hope you will agree to answer our questions since your views and experiences are very important. If I ask you any question you don't want to answer, just let me know and I will go on to the next question or you can stop the interview at any time. Refusing to participate or stopping your participation will involve no penalty to you or your household. The respondent must be an adult (18 +)

Confidentiality

We will take measures to keep all personal information collected confidential. Your name(s) will not be listed in any report that comes out of the survey. Your answers may be anonymously combined with the answers of other people in your community to calculate the average opinions in your community.

2.1 Are you willing to be interviewed?

- ☐ Yes
- ☐ No

Section 3: HOUSEHOLD SOCIO-ECONOMICS

3.0 Household ID

Use unique number provided by supervisor

3.4 Including yourself, how many members are in your household?

3.5. Number of adults in household (i.e. aged 18 or older)?

3.6 Number of children in household? (i.e. aged below 18)

3.7. Education level

Highest in household

- ☐ No education
- ☐ Primary education level
- ☐ Secondary education level
- ☐ Vocational training level
- ☐ University education level
- ☐ Other. Please specify

3.7.1 Please specify

Section 4: GPS COORDINATES AND PHYSICAL STRUCTURE OF HOME

4.1. What is the general structure or design of the household?

- ☐ House, single-level
- ☐ House, multi-level
- ☐ Flat or apartment, multi-level
- ☐ Flat or apartment, single-level
- ☐ Other. Please specify

4.1.1 Please specify

4.2. What are the general materials used for your roof construction?

- ☐ Steel or metal sheet
- ☐ Concrete
- ☐ Straw
- ☐ Tiles
- ☐ Other. Please specify

4.2.1 Please specify

Section 5: BUILDING PROFILE**5.1. How long have you lived in this current home?**

- ☐ Less than 1 or 2 years
- ☐ 2 to 5 years
- ☐ 5 to 10 years
- ☐ More than 10 years

5.2. Do you own or rent your home?

- ☐ Own
- ☐ Rent

5.3. If you own, how did you purchase your home?

- ☐ Customary land purchase (through an individual)
- ☐ Formal land purchase (Through the ward or city council)
- ☐ Don't know
- ☐ Other. Please specify

5.3.1 Please specify

5.5. What are the total number of rooms in your household?

5.6. What is your building living arrangement?

- ☐ Shared (i.e. building is shared with one or more families)
- ☐ Single (i.e. building is not shared with one or more families)

5.7 If living in a shared building, approximately how many other families occupy the building?

5.8. How do you get rid of solid waste in your home?

- ☐ Formal collection service from city council, local ward or other service provider
- ☐ Informal collection services e.g. organized by residents within the neighborhood
- ☐ Open pit/burning
- ☐ Other. Please specify:

5.8.1 Please specify

5.9. How do you get water in your home?

- ☐ Piped water supply
- ☐ Borehole
- ☐ Community tap/kiosk
- ☐ Other. Please specify:

5.9.1 Please specify

5.10. Do you use the internet?

- ☐ Yes
- ☐ No

6.6. On average, how much does your household pay for electricity per month? TSH/month
Enter 99 if Don't know

6.7. Do you have the most recent bill that I can take a picture?

- ☐ Yes
☐ No

6.7.1 Take image of the bill

[Click here to upload file. \(< 10MB\)](#)

5.11. If you have used internet, how do you access it ?*Check all that apply*

- ☐ Wi-Fi in the home
- ☐ Cellphone service
- ☐ Other. Please specify:

5.11.1 Please specify

Section 6: BUILDING ENERGY USE**6.1. Is your building electrified?**

- ☐ Yes
- ☐ No

6.2. How does your building receive electricity?*Check all that apply*

- ☐ Grid connection through a utility company
- ☐ Community generator
- ☐ Own generator
- ☐ Solar panels
- ☐ Don't Know
- ☐ Other. Please specify

6.2.1 Please specify

6.9.3 How much charcoal do you use on average?*specify in KG if possible*

6.9.3.1 Specify time of usage for the charcoal☐ Per Month☐ Per Week☐ Per Day**6.9.4 How much gas do you use on average?***specify in KG if possible*

6.9.4.1 Specify time of usage for the gas☐ Per Month☐ Per Week☐ Per Day**6.10.1 On average, how many times a day does your household use the electric stove?**

6.10.1.1 On average , approximately how many minutes are you using the electric stove per time of use?

6.10.2 On average, how many times a day does your household use the gas stove?

6.10.2.1 On average , approximately how many minutes are you using the gas stove per time of use?

6.10.3 On average, how many times a day does your household use the charcoal stove?

6.10.3.1 On average , approximately how many minutes are you using the charcoal stove per time of use?

6.8. What electric appliances do you use in your household?*Check all that apply*

- ☐ Air-conditioning/space heating unit
- ☐ Electric stove/cooker
- ☐ Electric water heater
- ☐ Microwave
- ☐ Fan
- ☐ Water heater
- ☐ Refrigerator
- ☐ Television
- ☐ Computer/Laptop
- ☐ Cellphones
- ☐ Light bulbs
- ☐ Radio
- ☐ Washing machine/dryer
- ☐ Blender
- ☐ Rice cooker
- ☐ Iron
- ☐ Water dispenser
- ☐ Other. Please specify:
- ☐ Not applicable. I don't own any electric appliances

6.8.1. Please specify

6.9 What type of stove does your household use for cooking?*Check all that apply*

- ☐ Electric stove
- ☐ Gas stove
- ☐ Charcoal stove
- ☐ Wood stove
- ☐ Other. Please specify:

6.9.1 Please Specify

6.18. Gas stoves are a good alternative for households that cannot only afford electric stoves.

- ☐ Strongly disagree
☐ Disagree
☐ Neutral
☐ Agree
☐ Strongly agree
☐ Don't know

6.19. Strict energy efficiency standards should be required for all new buildings and appliances in Dar es Salaam.

- ☐ Strongly disagree
☐ Disagree
☐ Neutral
☐ Agree
☐ Strongly agree
☐ Don't know

» 6.20 What is your preferred stove for cooking?

Rank 1

- | | | |
|--------------------------------------|---------------------------------|--------------------------------------|
| ☐ Electric stove | ☐ Gas stove | ☐ Charcoal stove |
| ☐ Wood stove | | |

Rank 2

- | | | |
|--------------------------------------|---------------------------------|--------------------------------------|
| ☐ Electric stove | ☐ Gas stove | ☐ Charcoal stove |
| ☐ Wood stove | | |

Rank 3

- | | | |
|--------------------------------------|---------------------------------|--------------------------------------|
| ☐ Electric stove | ☐ Gas stove | ☐ Charcoal stove |
| ☐ Wood stove | | |

Rank 4

- | | | |
|--------------------------------------|---------------------------------|--------------------------------------|
| ☐ Electric stove | ☐ Gas stove | ☐ Charcoal stove |
| ☐ Wood stove | | |

6.14. Are you aware of any negative health impacts associated with:

	Yes	No
Wood stove	☐	☐
Charcoal stove	☐	☐
Gas stove	☐	☐
Electricity stove	☐	☐

6.15. Policymakers should promote more subsidizing of electricity supply to our neighborhood.

- ☐ Strongly disagree
☐ Disagree
☐ Neutral
☐ Agree
☐ Strongly agree
☐ Don't know

6.16. What is most important to you when choosing a stove for cooking:

Scale: 1 = Not Important, 5 = Very important

	1	2	3	4	5
Health impacts	☐	☐	☐	☐	☐
Ease of use	☐	☐	☐	☐	☐
Affordability	☐	☐	☐	☐	☐
Impact on food taste	☐	☐	☐	☐	☐

6.17. Charcoal use should be phased out as Dar es Salaam continues to develop.

- ☐ Strongly disagree
☐ Disagree
☐ Neutral
☐ Agree
☐ Strongly agree
☐ Don't know

6.10.4 On average, how many times a day does your household use the wood stove?

6.10.4.1 On average, approximately how many minutes are you using the wood stove per time of use?

6.10.5 On average, how many times a day does your household use the other stove?

6.10.5.1 On average, approximately how many minutes are you using the other stove per time of use?

6.11. We use non-electric stoves for cooking (charcoal, wood or gas) in our household because:

- ☐ Charcoal, wood and gas are cheaper than electricity services
- ☐ We cannot afford an electric stove
- ☐ Charcoal, wood and gas are more available and accessible within our neighborhood
- ☐ Charcoal, wood and gas make the food taste better
- ☐ This question does not apply. We use electricity for cooking.
- ☐ Other. Please specify

6.11.1 Please specify

6.12. What reasons would make you switch to electricity for cooking

- ☐ If it was cheaper
- ☐ If the reliability of the service was better (few power disruptions)
- ☐ If I could afford an electric stove
- ☐ This question does not apply. We use electricity for cooking.
- ☐ Other. Please specify

6.12.1 Please specify

6.13. What is the maximum amount of money you would be willing to pay for electricity (per month) so that you only use electricity for cooking?

7.8 How far is the closest public transit stop to your home

- ☐ Less than 5 minutes walk
- ☐ 5 to 10 minutes walk
- ☐ 10 to 20 minutes walk
- ☐ 20 minutes walk or more

7.9 On average, how much do you pay for public transport per trip? TSH/trip

7.10 Does your neighborhood have foot or bicycle paths for pedestrian walking and cycling?

- ☐ Yes
- ☐ No

7.11. We would like to understand the general travel behavior for each working member of the household during a typical week day. Please can you describe your general movements to us. For example, you can recall to us how you travelled yesterday.

* Acknowledge that you are ready to describe a family member general movements

- ☐ OK

» Household working member

7.12. What are the factors you consider when choosing a public transport?

Scale: 1 = Not Important, 5 = Very important

	1	2	3	4	5
Safety	☐	☐	☐	☐	☐
Convenience	☐	☐	☐	☐	☐
Speed	☐	☐	☐	☐	☐
Comfort	☐	☐	☐	☐	☐
Affordability	☐	☐	☐	☐	☐

7.13. How safe would you say using the public transit in Dar is?

- ☐ Not safe at all
- ☐ Somewhat safe
- ☐ Safe
- ☐ Very safe
- ☐ Extremely safe

7.4 Does your household own any "boda" or "bajaj" (i.e. motorcycles or 3-wheeler)?

- ☐ Yes
☐ No

7.4.1 How many

7.5 Are any workers in the household a "boda" or "bajaj" driver?

- ☐ Yes
☐ No

7.5.1 How many

7.7. Do you use public transport (dala-dala, BRT, bajaji or boda) during the weekday?

- ☐ Yes
☐ No

Section 7: TRANSPORTATION

7.1 How many workers are in your household (i.e. generate income for the home)?

7.2 Does your household own any private vehicles?

☐ Yes

☐ No

7.2.1 How many

7.14. How comfortable is it to travel using public transport?

- ☐ Not comfortable at all
- ☐ Somewhat comfortable
- ☐ Comfortable
- ☐ Very comfortable
- ☐ Extremely comfortable

7.15. How reliable is it to travel using public transport?

- ☐ Not reliable at all
- ☐ Somewhat reliable
- ☐ Reliable
- ☐ Very reliable
- ☐ Extremely reliable

7.16. How accessible is the Bus Rapid Transit (BRT) to your neighborhood.

- ☐ Not accessible at all
- ☐ Somewhat accessible
- ☐ Accessible
- ☐ Very accessible
- ☐ Extremely accessible

7.17. What are your thoughts on the affordability of the BRT?

- ☐ It is not affordable at all (compared to dala-dala)
- ☐ It is somewhat affordable (compared to dala-dala)
- ☐ It is the same price as dala-dala
- ☐ It is slightly more affordable than dala-dala
- ☐ It is very much more affordable than dala-dala

Interview results

- ☐ Completed
- ☐ Not at home
- ☐ Refused
- ☐ Other. Please specify:

Please specify

Interview End Time

hh:mm

End of Interview

Appendix B: Policy Workshop in Dar es Salaam

STEERING SUSTAINABLE ENERGY USE AND LOW-CARBON GROWTH IN DAR ES SALAAM

BANK OF TANZANIA
CONFERENCE CENTRE
MT. MERU ROOM (1ST FLOOR)
MIRAMBO STREET, DAR ES SALAAM

WEDNESDAY
NOV. 7TH 2018
8:30AM – 4:30PM

AN INTERACTIVE WORKSHOP
LINKING URBAN GOVERNANCE,
FINANCING AND IMPLEMENTATION
ACTIONS IN KEY INFRASTRUCTURE
SECTORS



UNIVERSITY OF
TORONTO

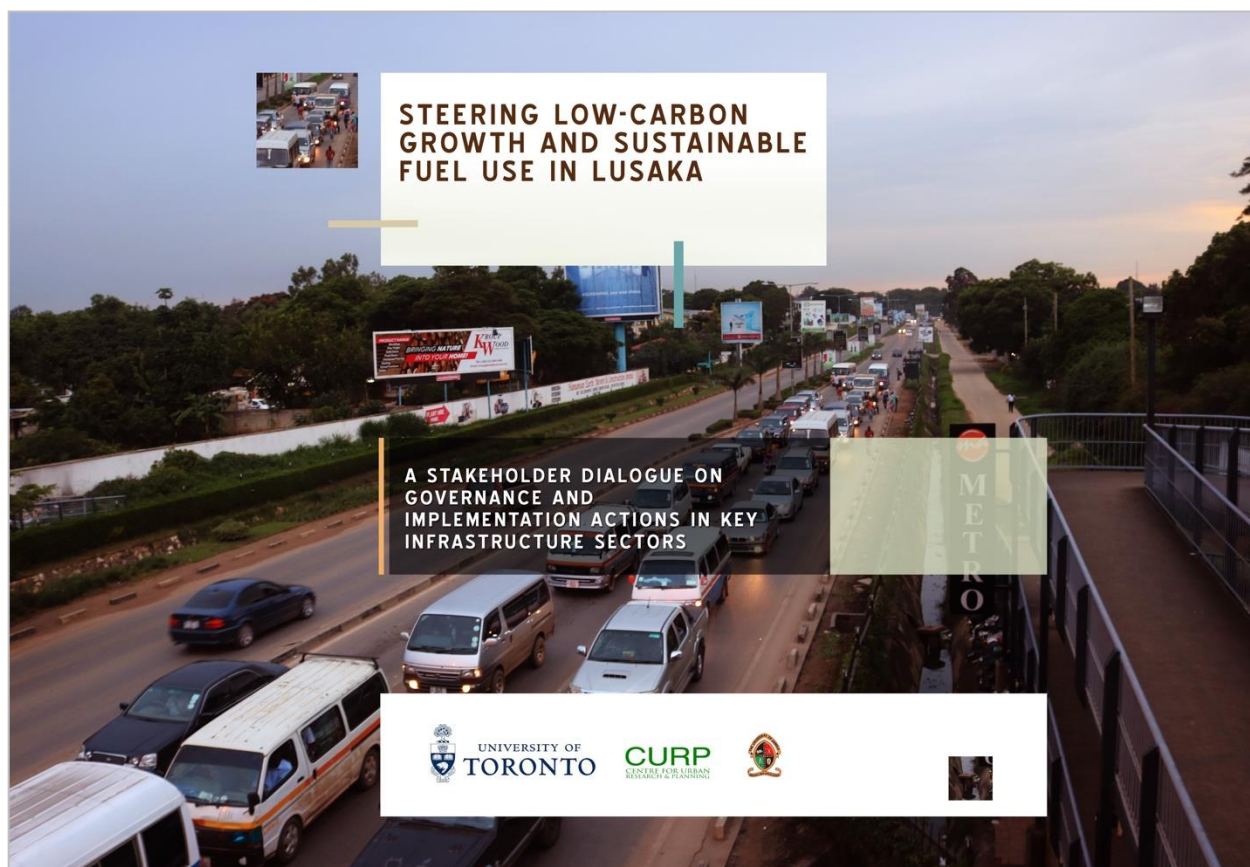
IGC
International
Growth Centre





Photos by Victor Faustine,
KababahGraphics,
Dar es Salaam

Appendix C: Policy Workshop in Lusaka



Photos by Alice Chibulu Luo

